

EEEE Codes

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
825-7691	1	EEEE (MLB_BOT_ICE, JP, 639-84872)	EEEE_18XR	CRITICAL	JP
825-7691	1	EEEE (MLB_BOT_ICE, ROW, 639-84346)	EEEE_HYFC	CRITICAL	ROW

Global Capacitors

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00148	138S00149	BOM_TABLE_ALTS	ALL	0402-3T, 10.5uF@1V, Kyocera
138S00150	138S00149	BOM_TABLE_ALTS	ALL	0402-3T, 10.5uF@1V, SEMCO
138S00151	138S00149	BOM_TABLE_ALTS	ALL	0402-3T, 10.5uF@1V, TY

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00143	138S00144	BOM_TABLE_ALTS	ALL	0402, 16uF@1V, Kyocera
138S00163	138S00144	BOM_TABLE_ALTS	ALL	0402, 16uF@1V, Taiyo

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00138	138S00139	BOM_TABLE_ALTS	ALL	0201, 3uF@1V, Kyocera
138S00164	138S00139	BOM_TABLE_ALTS	ALL	0201, 3uF@1V, Taiyo

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00221	138S00146	BOM_TABLE_ALTS	ALL	0402, 5.1uF@3V, Kyocera

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00140	138S00141	BOM_TABLE_ALTS	ALL	0201, 1.1uF@3V, Kyocera
138S00142	138S00141	BOM_TABLE_ALTS	ALL	0201, 1.1uF@3V, SEMCO
138S00166	138S00141	BOM_TABLE_ALTS	ALL	0201, 1.1uF@3V, Taiyo

Global Ferrites

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
152S00558	152S00557	BOM_TABLE_ALTS	ALL	IND, ML2, 0.47UH, 20%, 2.5A, 80MD, 1600
155S00194	155S00610	BOM_TABLE_ALTS	ALL	FERR BD, 150 OHM, 25%, 200MA, 0.7 DCR, 01005
155S00200	155S00610	BOM_TABLE_ALTS	ALL	FERR BD, 150 OHM, 25%, 200MA, 0.7 DCR, 01005
155S00338	155S00661	BOM_TABLE_ALTS	ALL	FERR BD, 33 OHM, 25%, 1.5A, 55MDHM DCR, 0201

CRITICAL PART#	COMMENT
138S00149	0402-3T, 10.5uF@1V

CRITICAL PART#	COMMENT
138S00144	0402, 16uF@1V

CRITICAL PART#	COMMENT
138S00139	0201, 3uF@1V

CRITICAL PART#	COMMENT
138S00146	0402, 5.1uF@3V

CRITICAL PART#	COMMENT
138S00141	0201, 1.1uF@3V

CRITICAL PART#	COMMENT
152S00557	IND, ML2, 0.47UH, 20%, 2.5A, 80MD, 1600
155S00610	FERR BD, 150 OHM, 25%, 200MA, 0.7 DCR, 01005

155S00661	FERR BD, 33 OHM, 25%, 1.5A, 55MDHM DCR, 0201
-----------	--

Global R/C Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00648	138S00652	BOM_TABLE_ALTS	ALL	CAP, XSR, 4.7UF, 6.3V, 0.65MM, 0402, TAIYO
138S00024	138S00986	BOM_TABLE_ALTS	ALL	CAP, CER, 3-TERM, 1.0UF, 50V, 0.65MM, 0402, TAIYO/TDK
138S0706	138S0739	BOM_TABLE_ALTS	ALL	CAP, CER, 2UF, 25V, 10V, XSR, 0201, MURATA
138S0945	138S0739	BOM_TABLE_ALTS	ALL	CAP, CER, 2UF, 25V, 10V, XSR, 0201, KYOCERA
138S0739	138S0706	BOM_TABLE_ALTS	ALL	CAP, CER, XSR, 0.22UF, 20%, 6.3V, 20%
132S0436	132S0400	BOM_TABLE_ALTS	ALL	CAP, CER, XSR, 0.22UF, 20%, 6.3V, 01005
138S00049	138S00831	BOM_TABLE_ALTS	ALL	CAP, CER, XSR, 2.2UF, 20%, 6.3V, 0201
138S00133	138S00128	BOM_TABLE_ALTS	ALL	CAP, XSR, 0.47UF, 20%, 6.3V, KYO, 01005
138S00116	138S00071	BOM_TABLE_ALTS	ALL	CAP, XSR, 4UF, 20%, 6.3V, 0201, T10-50MM
138S00117	138S00071	BOM_TABLE_ALTS	ALL	CAP, XSR, 4UF, 20%, 6.3V, 0201, T10-50MM
138S00048	138S00003	BOM_TABLE_ALTS	ALL	CAP, XSR, 100F, 20%, 6.3V, 0.65MM, RECTTL, 0402
131S00172	131S00164	BOM_TABLE_ALTS	ALL	CAP, CER, C0G, 220PF, 5%, 25V, 01005
131S00173	131S00164	BOM_TABLE_ALTS	ALL	CAP, CER, C0G, 220PF, 5%, 25V, 01005
155S00338	155S00402	BOM_TABLE_ALTS	ALL	FERR BD, 33 OHM, 25%, 1.5A, TY
155S00437	155S00402	BOM_TABLE_ALTS	ALL	FERR BD, 33 OHM, 25%, 1.5A, TDK

CRITICAL PART#	COMMENT
138S00652	CAP, XSR, 4.7UF, 6.3V, 0.65MM, 0402, TAIYO
138S00986	CAP, CER, 3-TERM, 1.0UF, 50V, 0.65MM, 0402, TAIYO/TDK
138S0739	CAP, CER, 1UF, 20%, 10V, XSR, 0201, MURATA

138S0706	CAP, CER, XSR, 0.22UF, 20%, 6.3V, 20%
132S0400	CAP, CER, XSR, 0.22UF, 20%, 6.3V, 01005
138S0031	CAP, CER, XSR, 2.2UF, 20%, 6.3V, 0201
138S00128	CAP, XSR, 0.47UF, 20%, 6.3V, KYO, 01005
138S00071	CAP, XSR, 4UF, 20%, 6.3V, 0201, T=0.55MM

138S00003	CAP, XSR, 15UF, 20%, 6.3V, 0.65MM, HRZTL, 0402
131S00164	CAP, CER, C0G, 220PF, 5%, 16V, 01005

Global Inductors

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
152S00653	152S00651	BOM_TABLE_ALTS	ALL	IND, 1.2UH, 3A, 2016, 0.65Z
152S00654	152S00652	BOM_TABLE_ALTS	ALL	IND, 1.2UH, 3A, 2016, 0.8Z

CRITICAL PART#	COMMENT
152S00651	IND, 1.2UH, 3A, 2016, 0.65Z
152S00652	IND, 1.2UH, 3A, 2016, 0.8Z

Iktara

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
116S00002	6	RES, High Current 0 Ohm	C3401, C3402, C3403, C3404, C3405, C3406	CRITICAL	UNBALANCED CAPS
132S00186	1	CAP, 0.33uF, 10%, 100V, X6S	C3451	CRITICAL	UNBALANCED CAPS
132S00187	3	CAP, 0.047uF, 10%, 100V, X6S	C3452, C3453, C3456	CRITICAL	UNBALANCED CAPS

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
685-00183	685-00182	BOM_TABLE_ALTS	SUBBOM_DS	DI000A, MLB_BOT, DI000ES, DI000ES, 1001

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
685-00182	1	SUBBOM, MLB, BOT, DI000ES, DI000ES, X891	SUBBOM_DS	CRITICAL	COMMON
371S00133	4	DI000ES, SHOTTKY DIODE, 30V, 2A, 0603	D3400, D3401, D3402, D3403	CRITICAL	DIODES_DS
371S00132	4	ONSEMI, SHOTTKY DIODE, 30V, 2A, 0603	D3400, D3401, D3402, D3403	CRITICAL	ONSEMI_DS

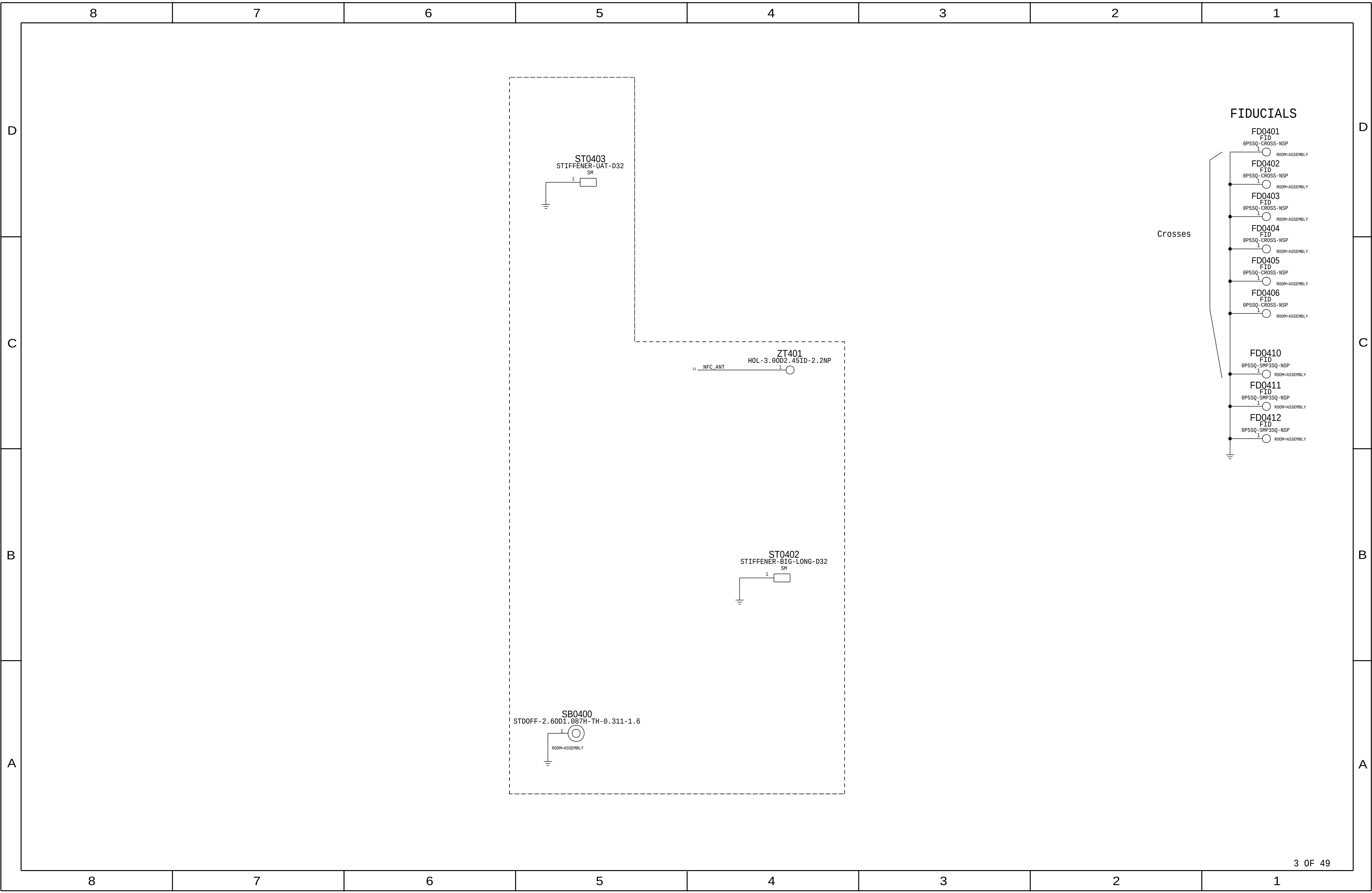
Hall Effect Alts

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
353S01327	353S3697	BOM_TABLE_ALTS	ALL	ROHM, RES

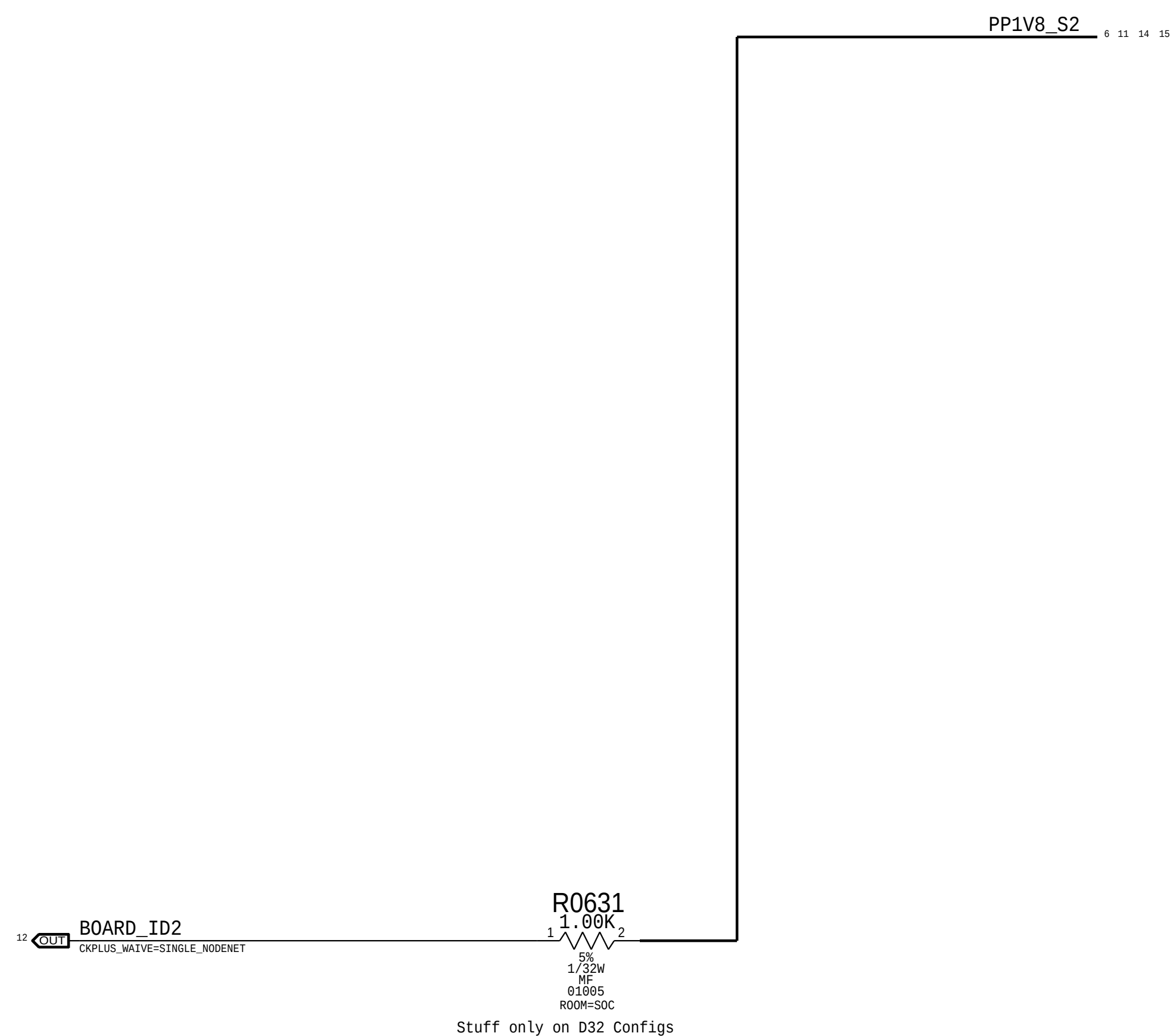
Multi-Vendor Criticals

CRITICAL PART#	COMMENT
138S00979	CAP, CER, XSR, 18UF, 20%, 10V, 0402, H=0.65MM
138S00683	CAP, CER, XSR, 1UF, 10%, 25V, 0402
132S0663	CAP, CER, XSR, 1UF, 10%, 25V, 0402
132S0288	CAP, CER, XSR, 0.1UF, 10%, 10V, 0201
132S0275	CAP, CER, XSR, 470PF, 10%, 10V, 01005
132S0245	CAP, CER, XSR, 0.01UF, 10%, 6.3V, 01005
138S00692	CAP, CER, XSR, 1UF, 20%, 6.3V, 0201

CRITICAL PART#	COMMENT
132S00008	CAP, CER, 0.1UF, 10%, 50V, XTR, 0402
131S00004	CAP, CER, 27PF, 5%, C0G, 25V, 0201
131S0307	CAP, CER, NP0/C0G, 100PF, 5%, 16V, 01005
131S00053	CAP, CER, C0G, 220PF, 5%, 10V, 01005
117S00055	RES, MF, 1/20W, 2% OHM, 5, 0201, SMD
107S0257	THERMISTOR, NTC, 10K OHM, 1%, B=3435, 01005
131S00170	CAP, CER, C0G, 220PF, 5%, 25V, 01005
131S00643	CAP, CER, NP0/C0G, 50PF, 5%, 25V, 01005
131S0316	CAP, CER, XSR, 0.1UF, 20%, 6.3V, 01005




```
BOOTSTRAPPING:BOARD ID[3]
```



Board ID[4:0] Float = Low PU = High					
	4	3	2	1	0
	Denali = 0, Imola = 1	Mav = 0, Ice = 1	00=Open 01=D33 10=N84 11=D32		MLB = 0, Dev = 1
D33 MLB Mav	0	0	0	1	0
D33p MLB Mav	1	0	0	1	0
D33 Dev Mav	0	0	0	1	1
D33p Dev Mav	1	0	0	1	1
D33 MLB Ice	0	1	0	1	0
D33p MLB Ice	1	1	0	1	0
D33 Dev Ice	0	1	0	1	1
D33p Dev Ice	1	1	0	1	1
D32 MLB Mav	0	0	1	1	0
D32 Dev Mav	0	0	1	1	1
D32 MLB Ice	0	1	1	1	0
D32 Dev Ice	0	1	1	1	1

SELECTED - ->

Test Points

Probe Points

D

C

B

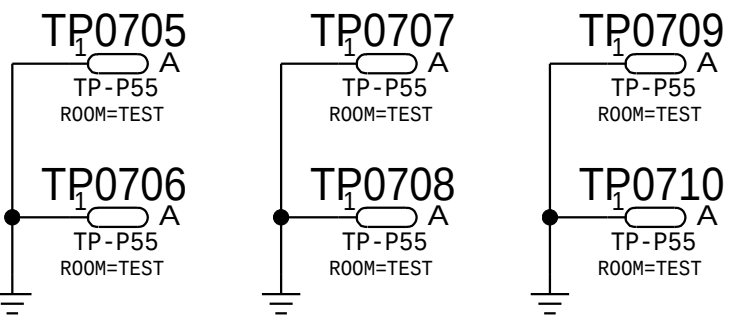
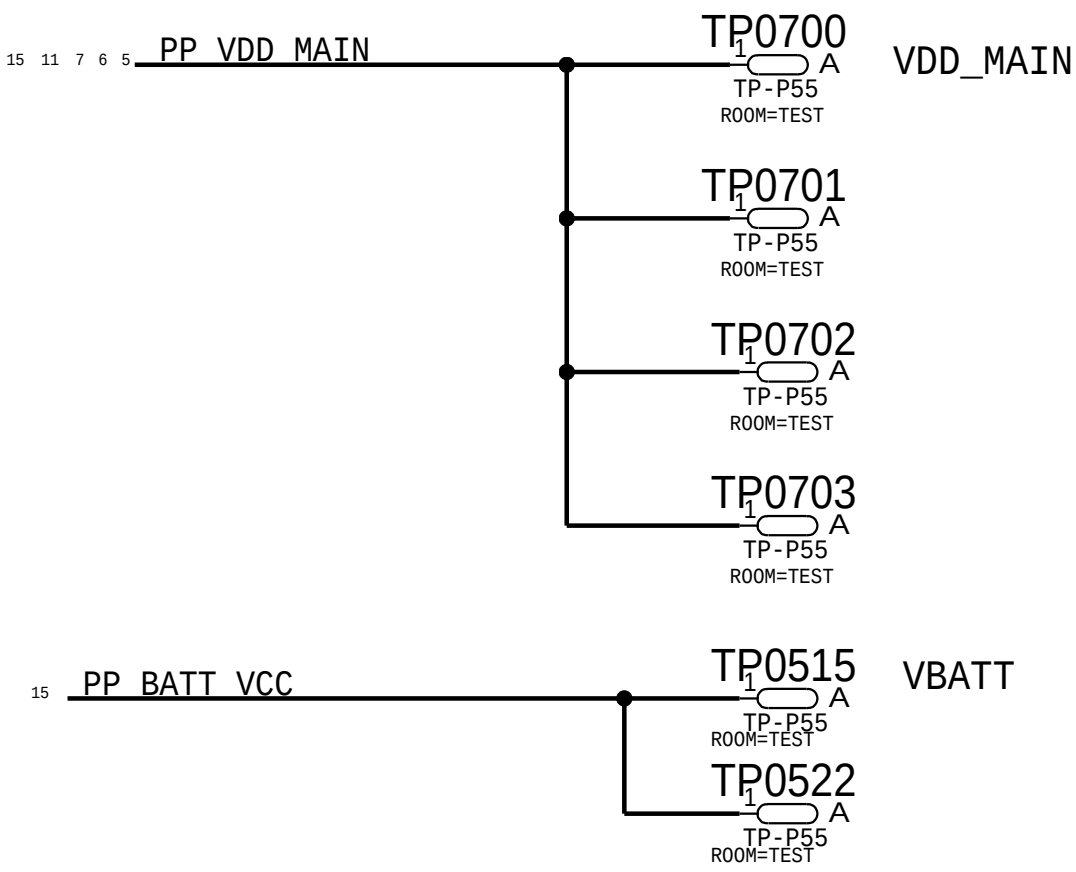
A

D

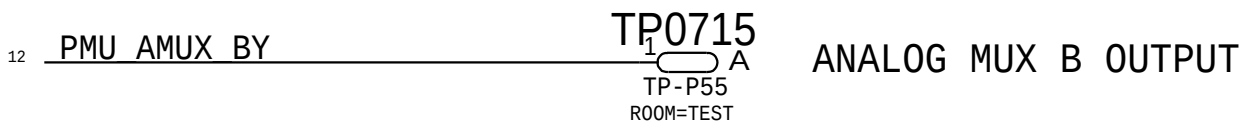
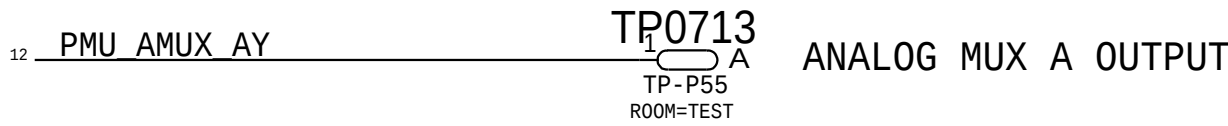
C

B

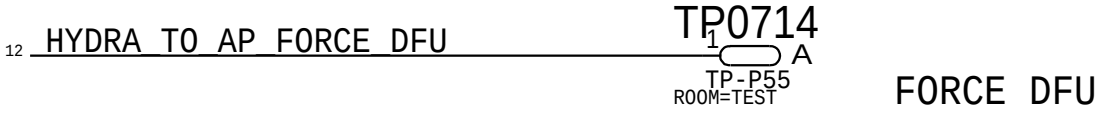
A



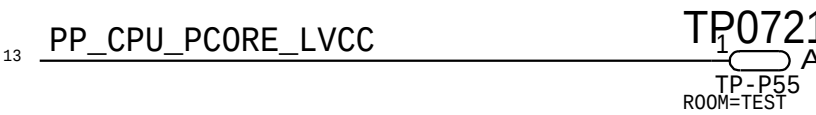
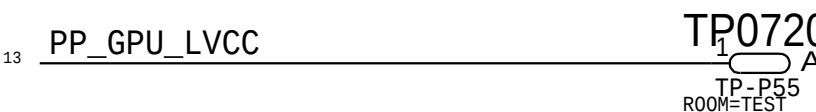
AMUX



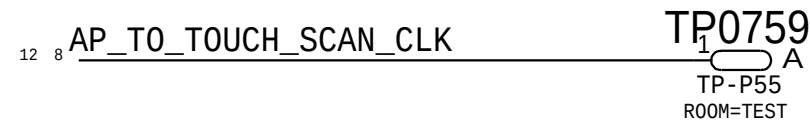
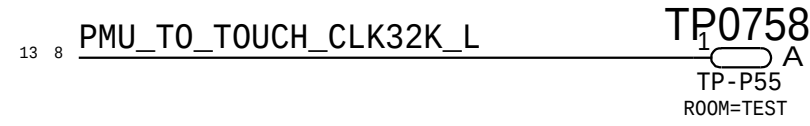
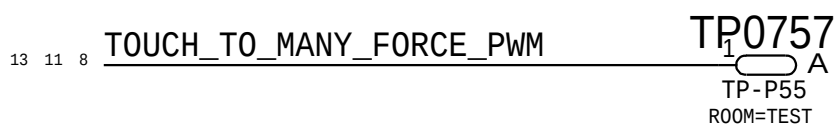
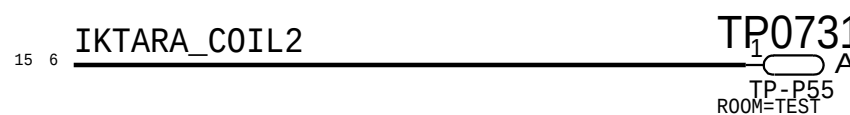
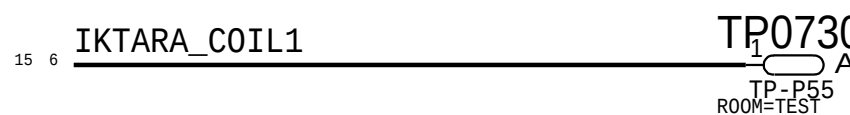
DFU



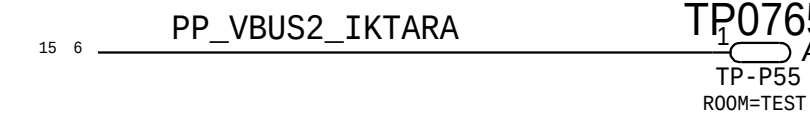
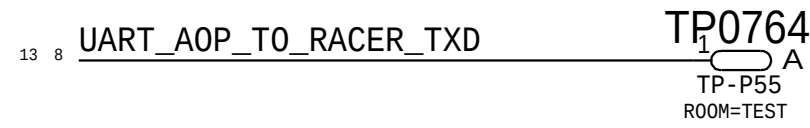
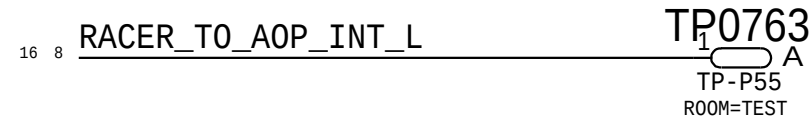
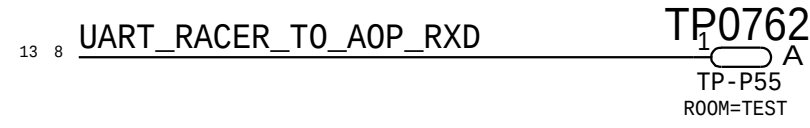
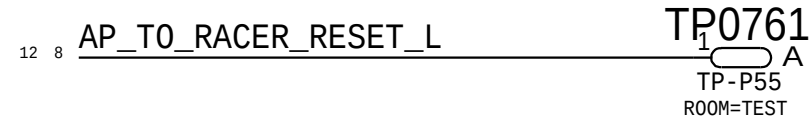
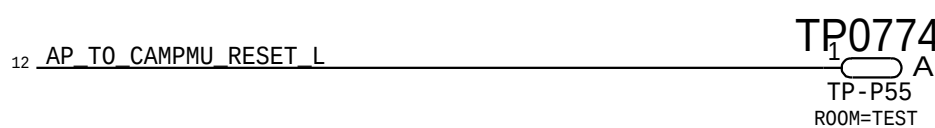
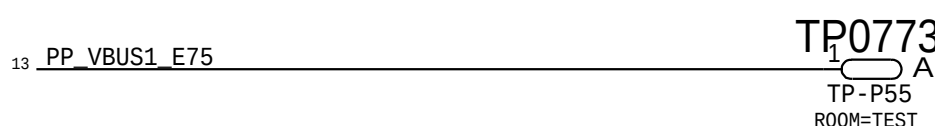
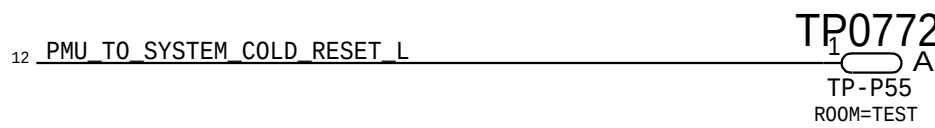
LVCC



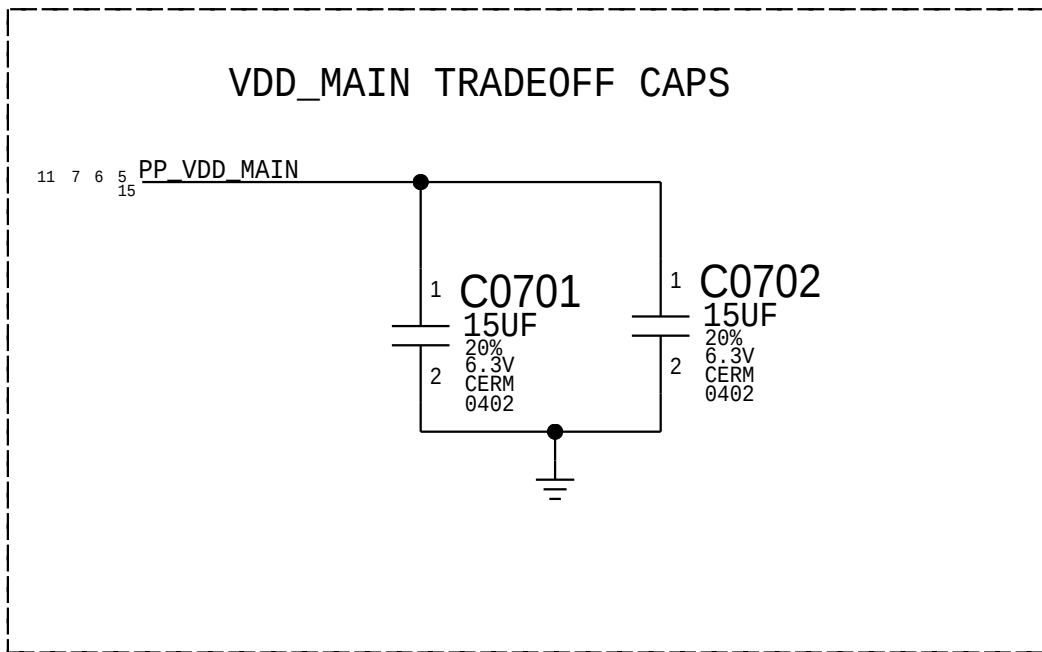
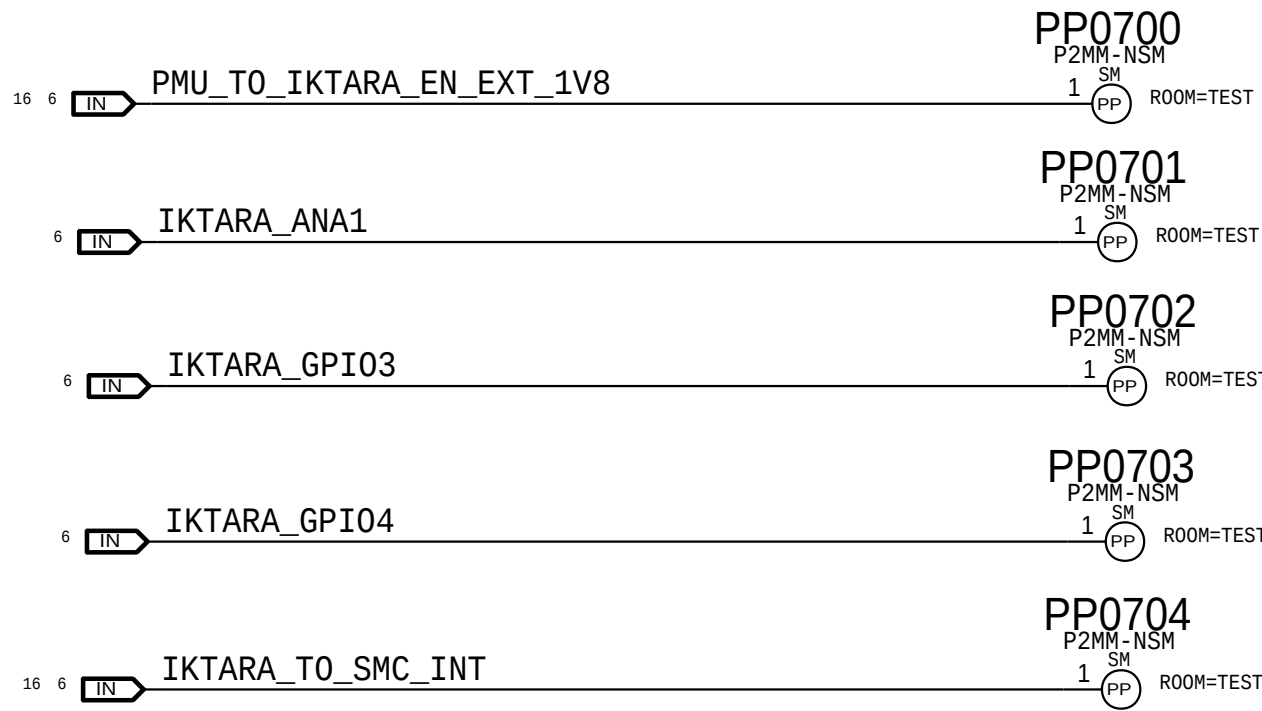
COIL



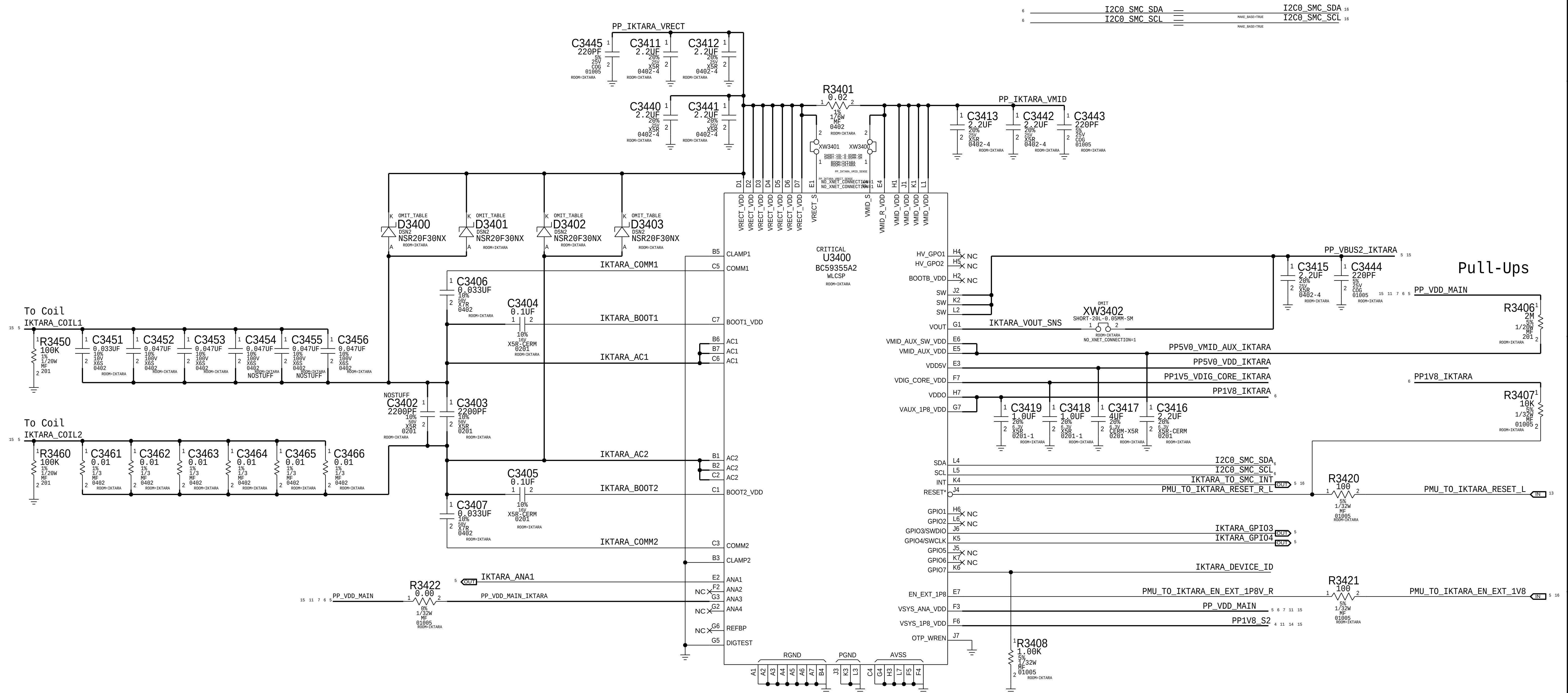
REL BIAS



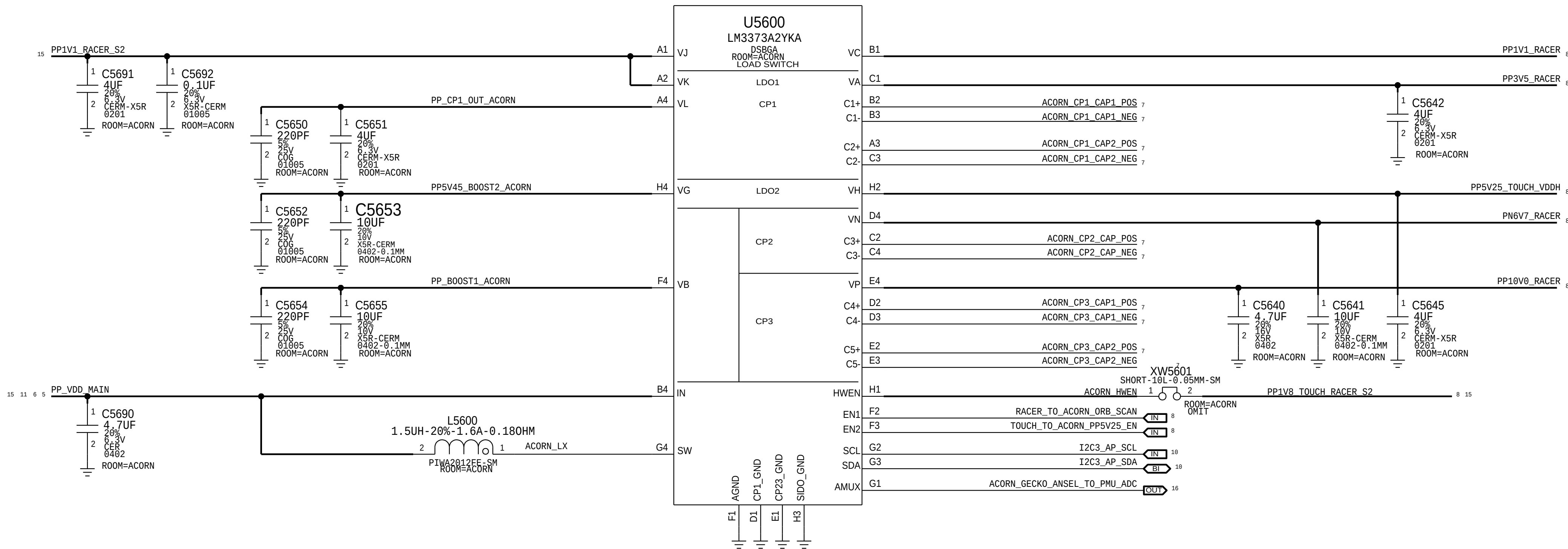
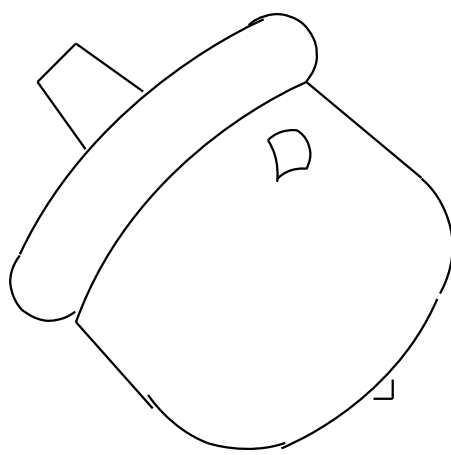
Iktara Debug



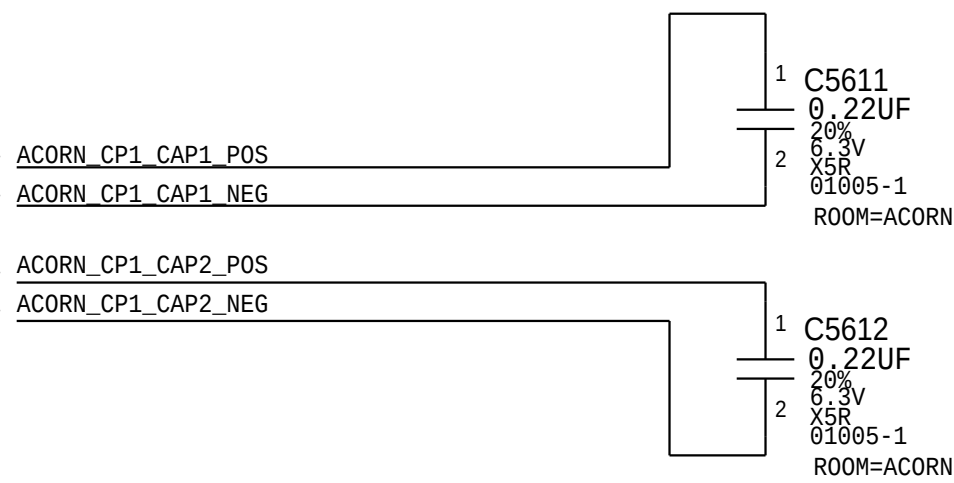
Iktara



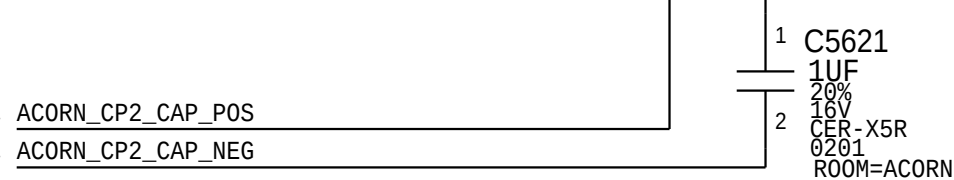
Acorn PMU



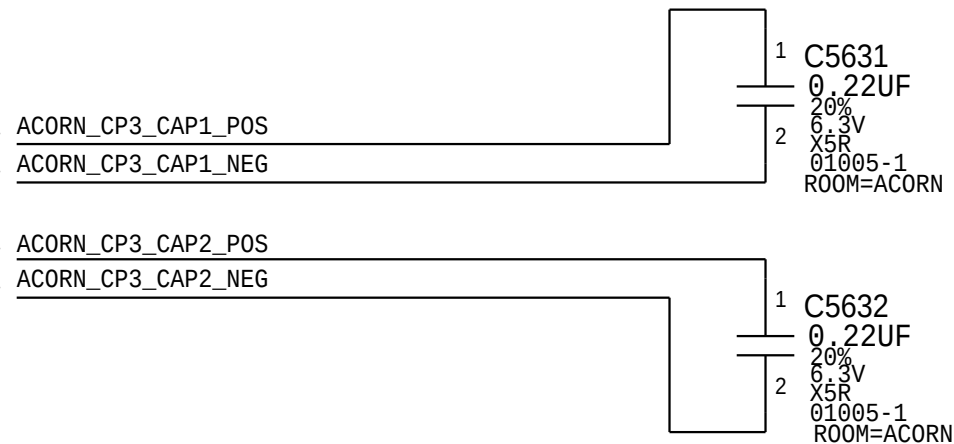
Charge Pump 1 Caps



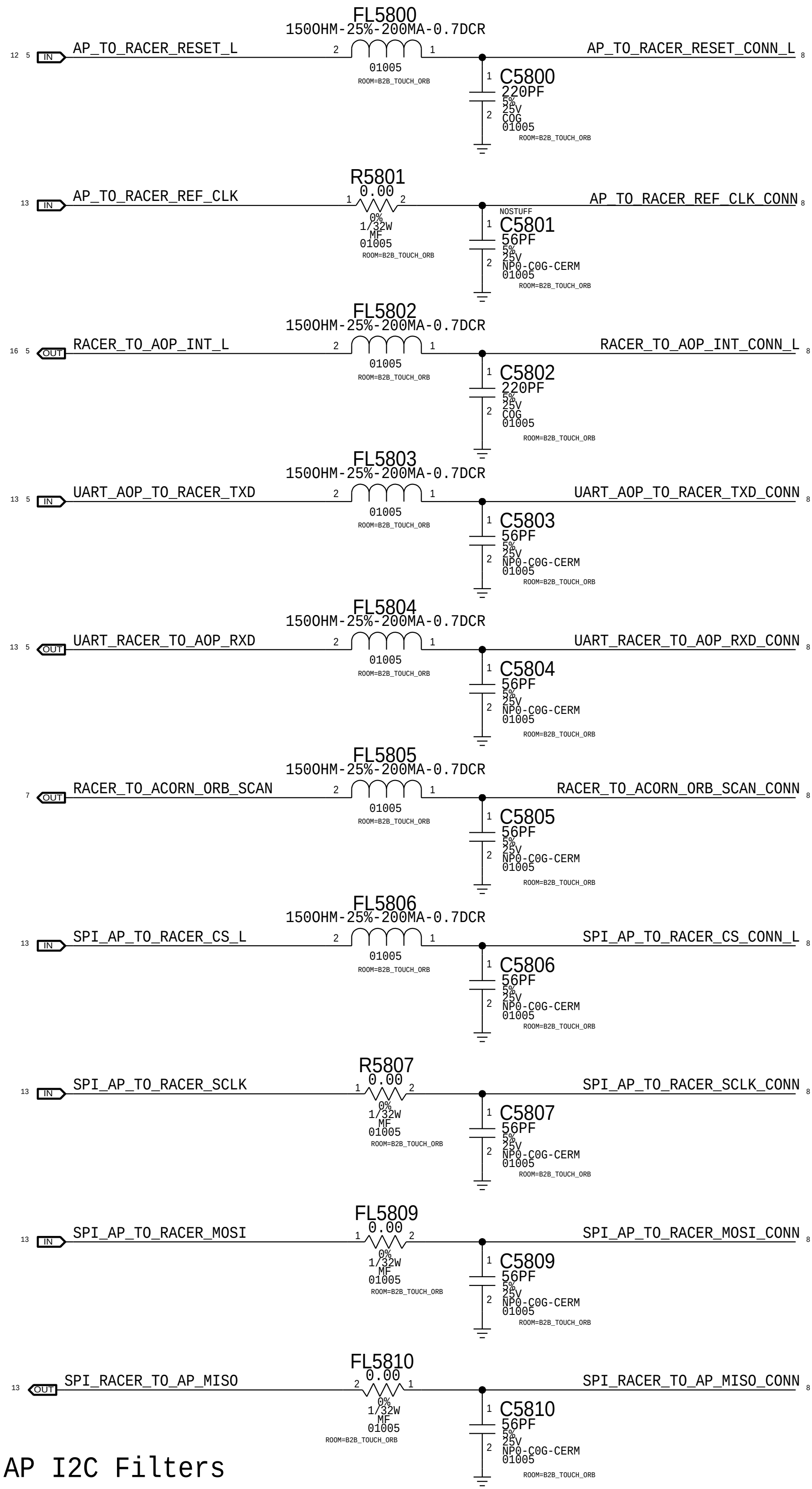
Charge Pump 2 Caps



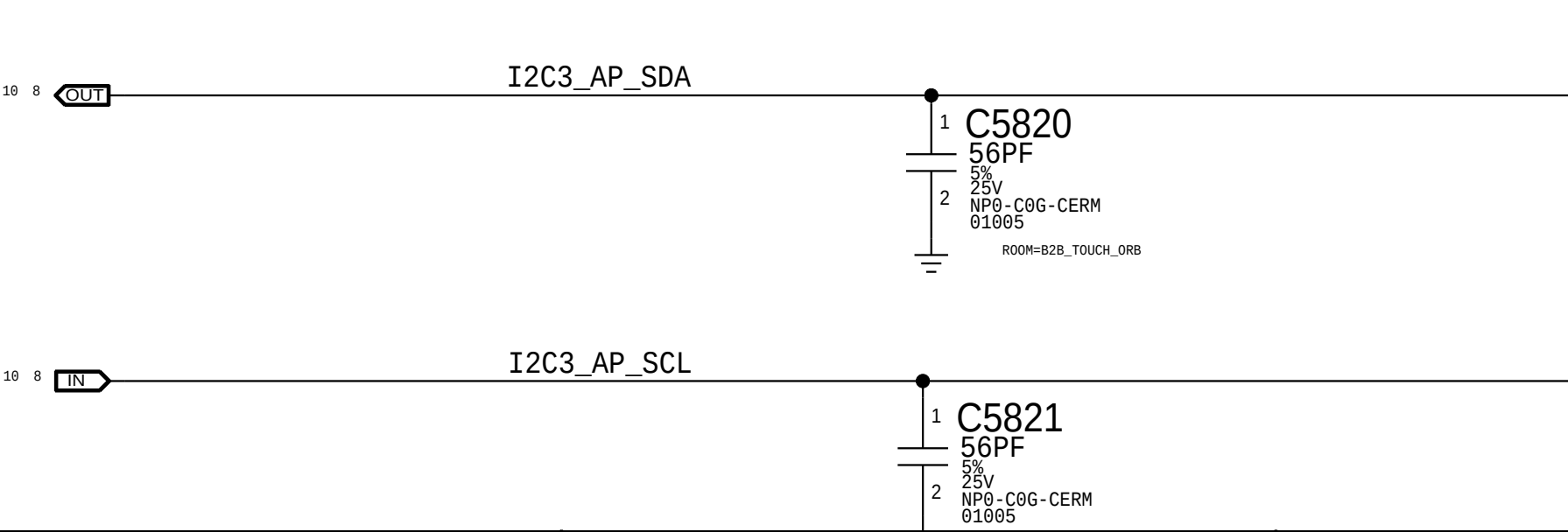
Charge Pump 3 Caps



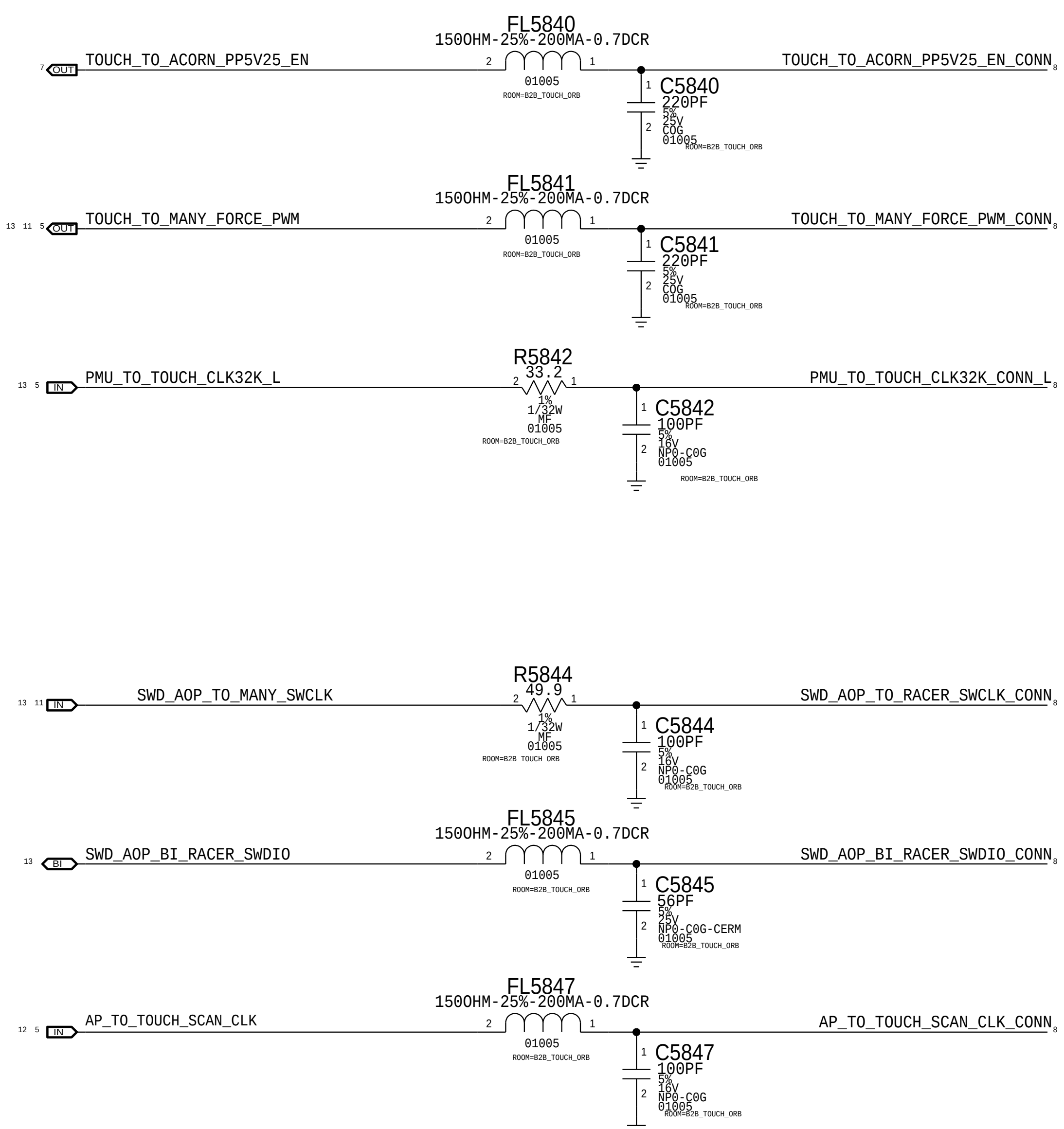
Racer I/O



AP I2C Filters



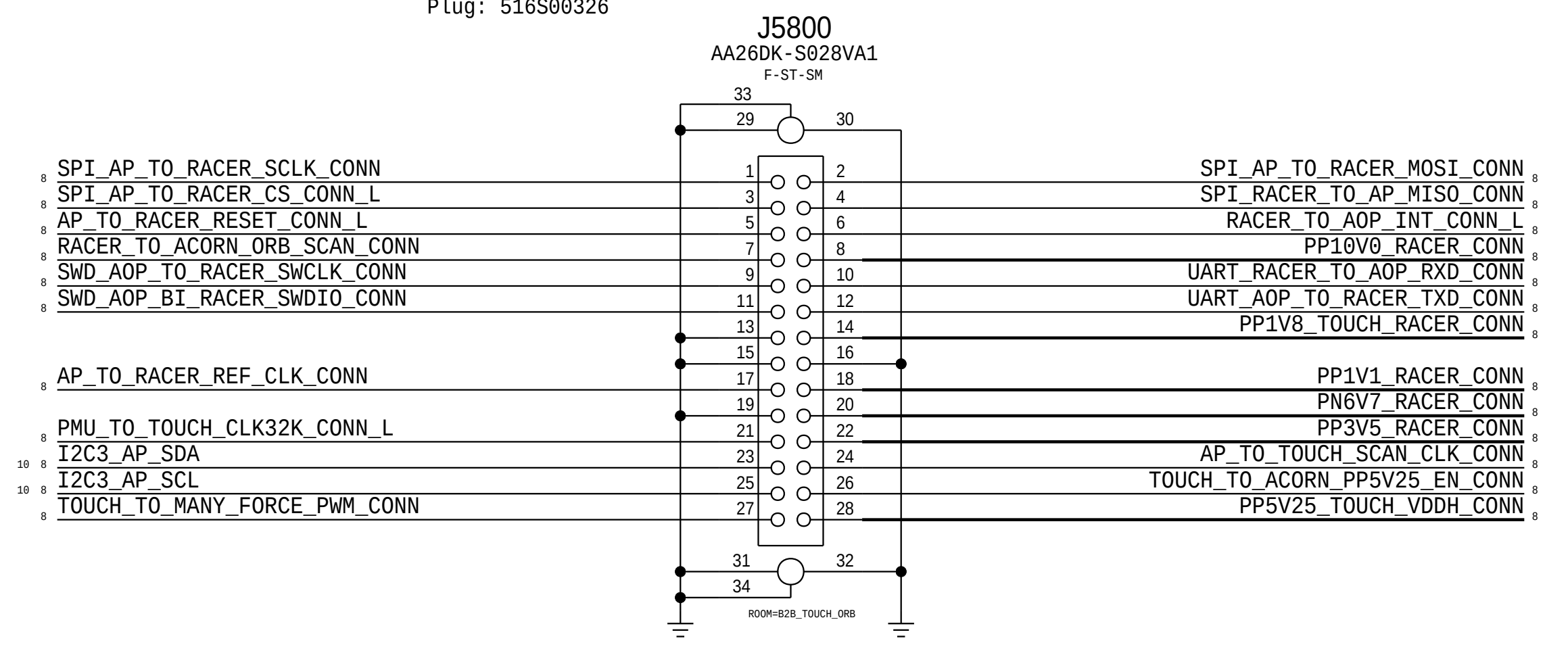
Touch and Misc I/O



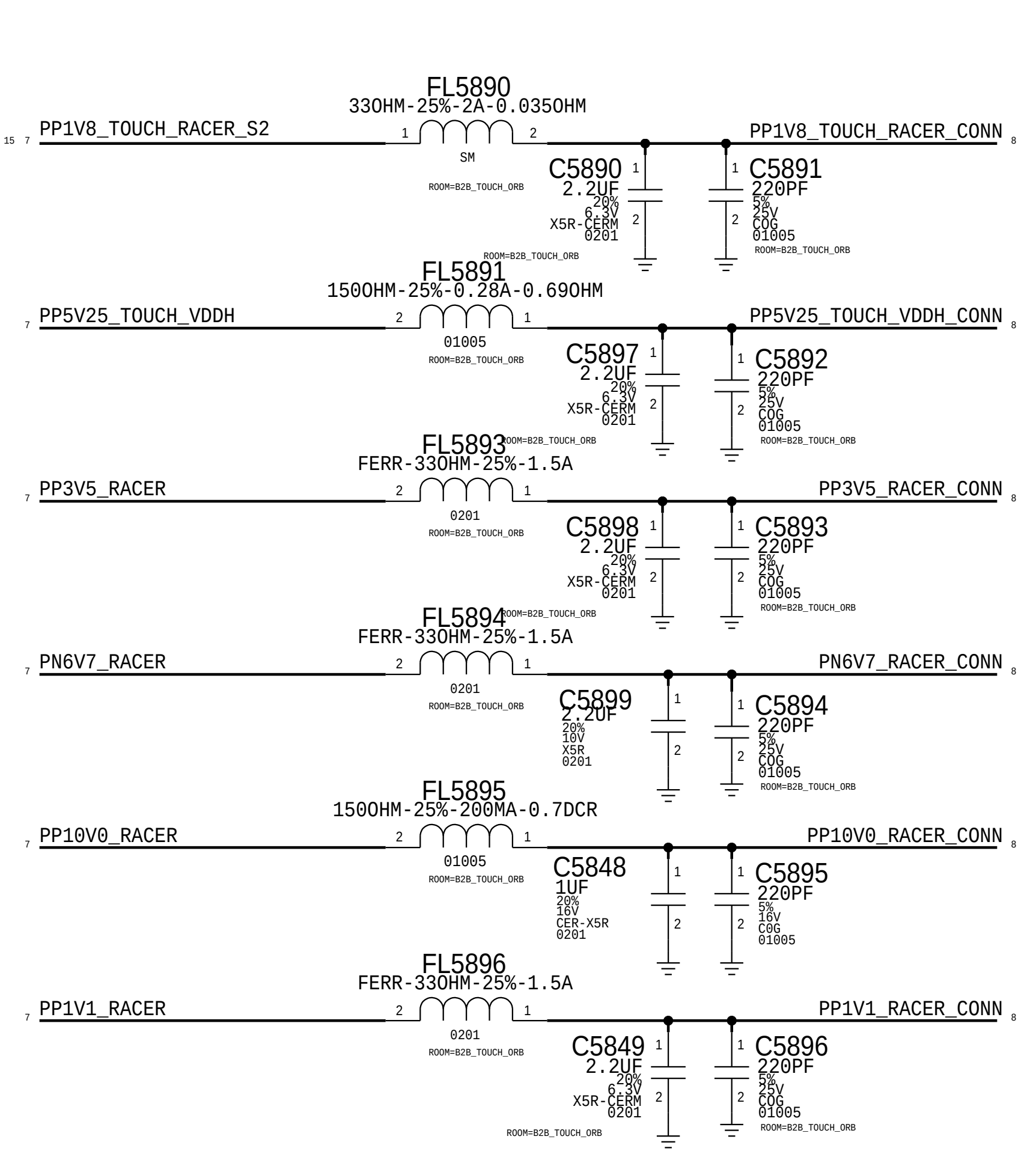
Luna + Touch Connector

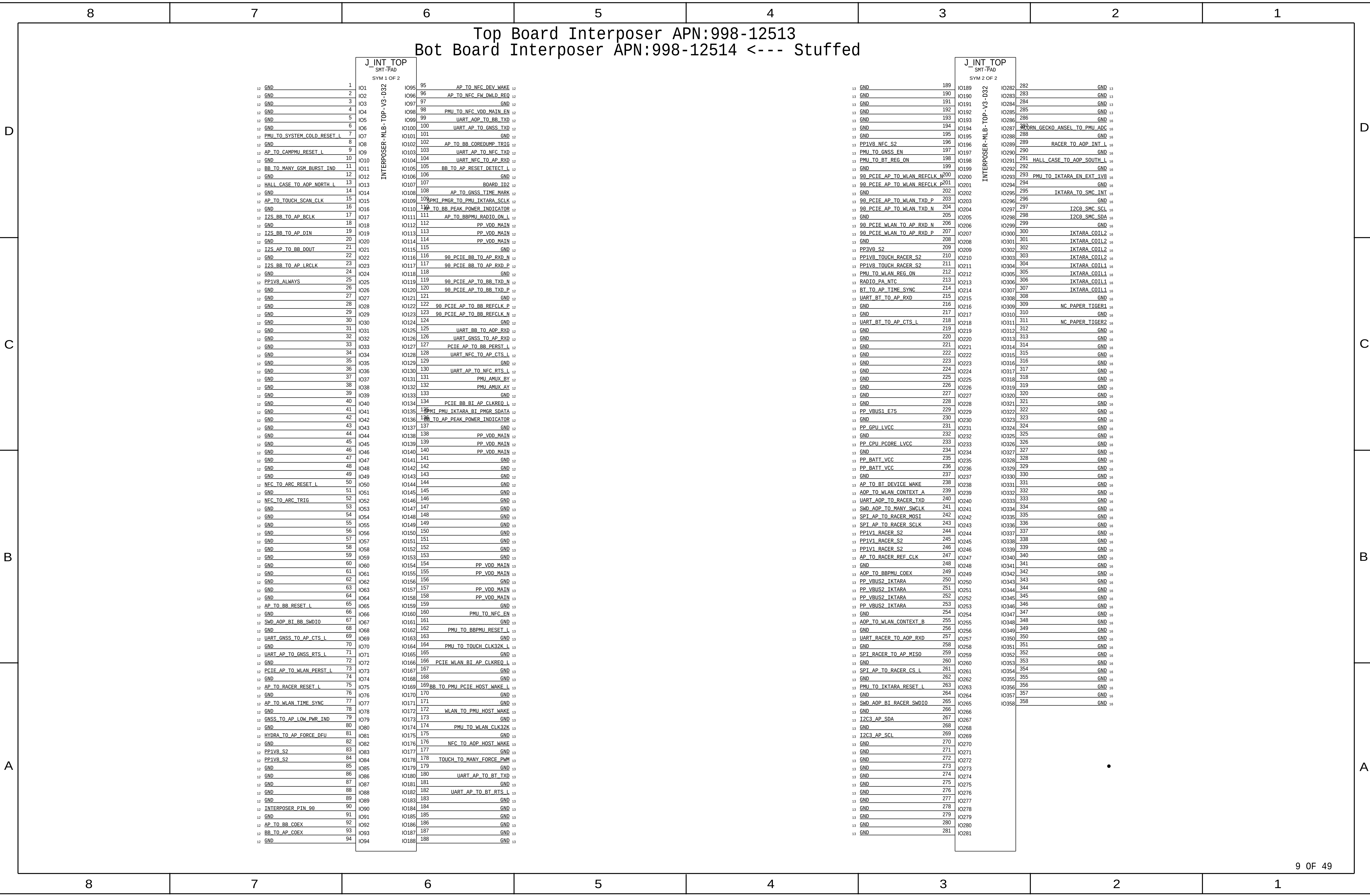
Rcpt: 516S00325
Plug: 516S00326

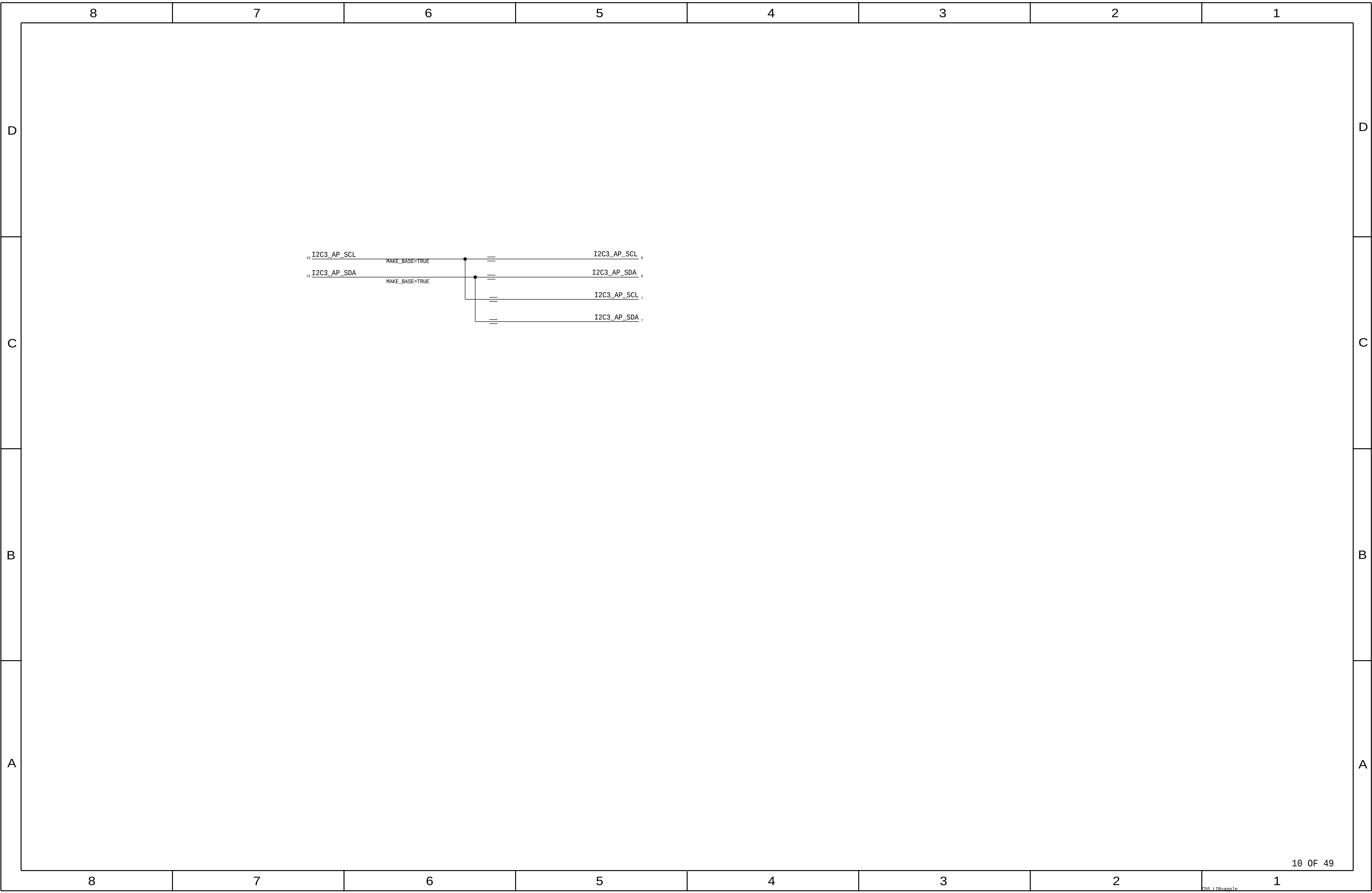
<-- This one on MLB



Touch And Racer Power

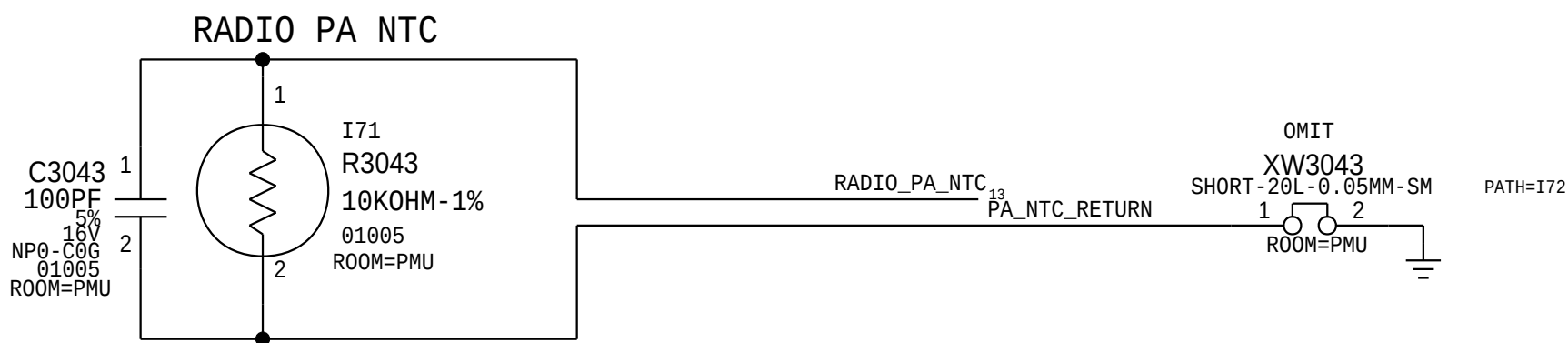






10 OF 49

CDS, LTR=apple



D

C

B

A

D

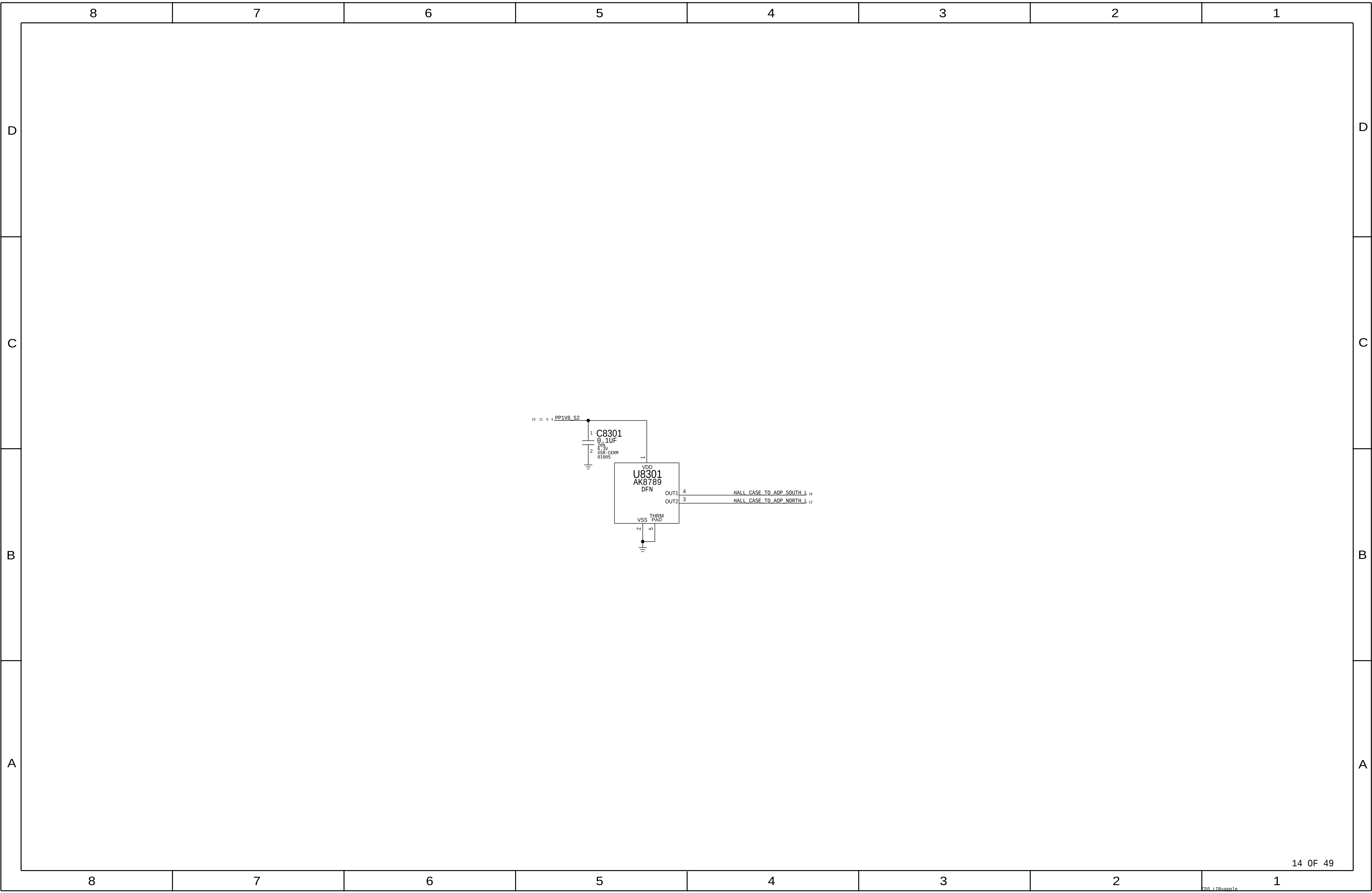
C

B

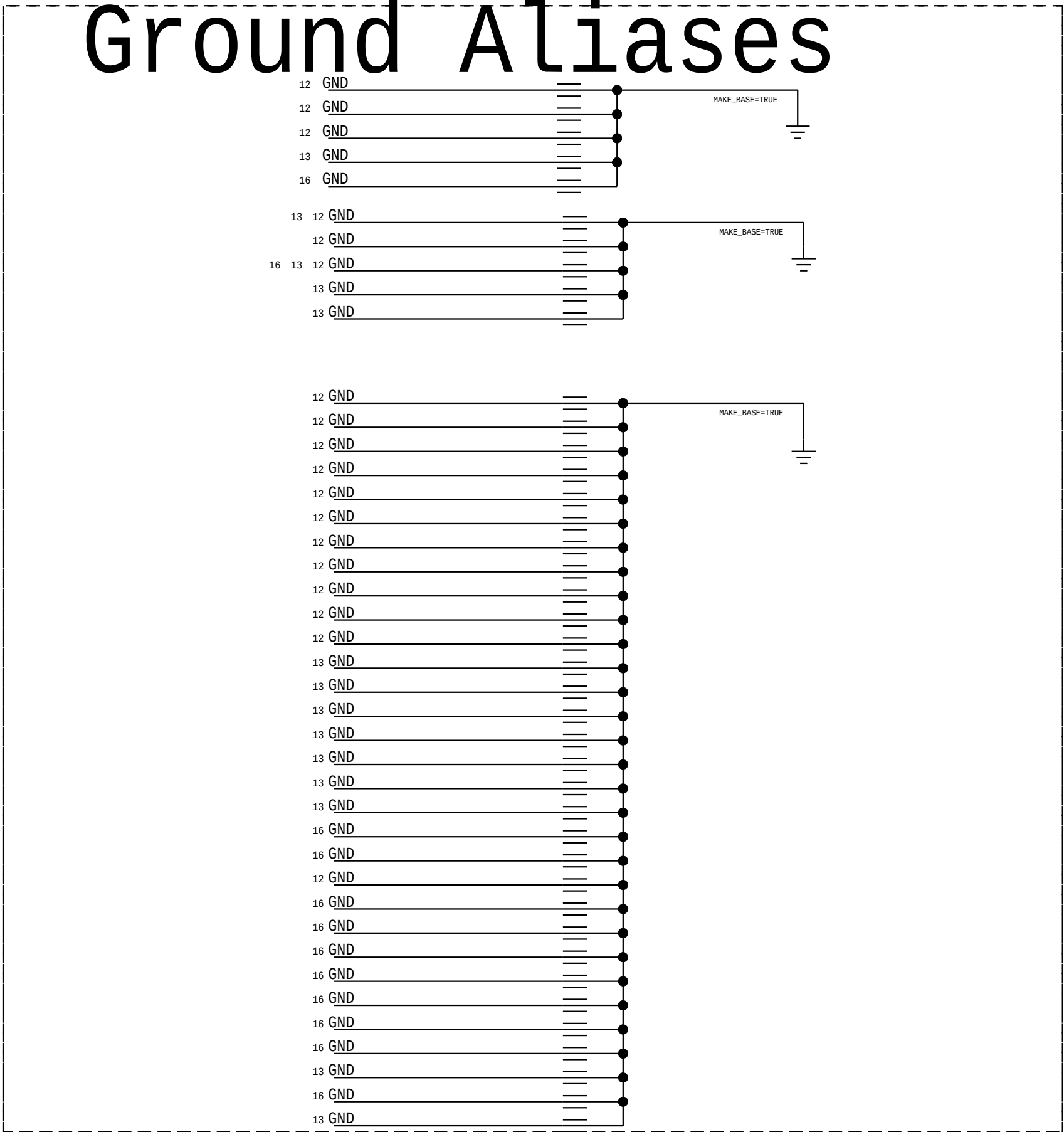
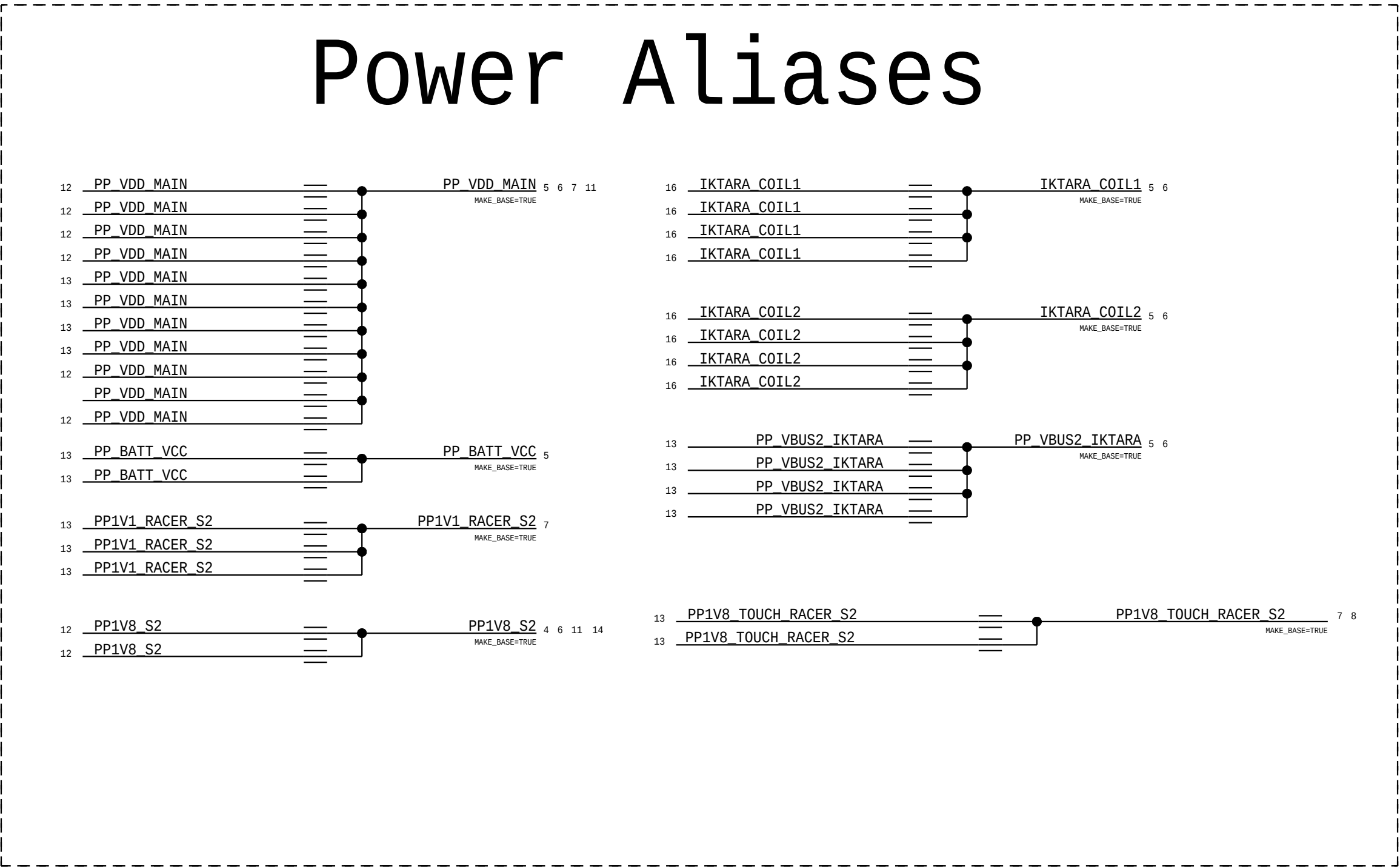
A

D

CB



Interposer Top Level Aliases



D

D

C

C

B

B

A

A

THIS SIDE HAS ATTRIBUTE
MAKE_BASE=TRUE

THIS SIDE HAS ATTRIBUTE
MAKE_BASE=TRUE

[illegible]

9 ————— GND ——— MAKE_BACE=TRUE ————— GND 12 13 15 16

9 ————— GND ——— MAKE_BACE=TRUE ————— GND 12 13 15 16

D32 HIER_RADIO_ICE - EVT

LAST_MODIFICATION= Tue Feb 6 10:49:38 2018

PAGE	CSA	CONTENTS	SYNC	DATE
17	1	RADIO: TABLE OF CONTENTS		
18	2	BOM TABLES		
19	3	ANTENNA DIAGRAM		
20	4	ANTENNA: B2BS		
21	5	ANTENNA: N-PLEX SHARED		
22	6	BBPMU: CONTROL		
23	7	BBPMU: RAILS		
24	8	BB: INTERFACE		
25	9	BB: DDR PWR & JTAG		
26	10	BB: DIGITAL PWR		
27	11	XCVR: TX & GNSS		
28	12	XCVR: INTERFACE & PWR		
29	13	XCVR: PRX DRX		
30	14	HW CONFIG OPTIONS		
31	15	ET		
32	16	LB SPAD		
33	17	HB SPAD		
34	18	UHB LMB SPAD		
35	19	BLANK PAGE		
36	20	LB DIVERSITY RECEIVE LNA		
37	21	HB DIVERSITY RECEIVE LNA		
38	22	MIMO RECEIVE LNAS		
39	23	LOWER ANTENNA & COUPLER		
40	24	UPPER ANTENNA FEEDS		
41	25	SIM: PSIM		
42	26	SIM: ESIM		
43	27	TEST POINTS		
44	28	SYMBOL: WIFI		
45	1	WIFI: TABLE OF CONTENTS		
46	2	DIETCOKE		
47	3	FEM MODULES		

HIERARCHY

SOURCE PROJECT	SUB-DESIGN NAME	VERSION	HARD/ SOFT	SYNC_DATE/TIME
D32	HIER_WIFI	0.45.0	S	2018_01_30_23:53:23

SCHEMATIC APN: 951-05145

D

C

B

A

D

C

B

A

BOM TABLES

ROW SKU

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
118S8724	1	RES,MF,0.0 OHM,1/20W,0201,HIGH FREQ	R2016_K	ROW
118S8724	1	RES,MF,0.0 OHM,1/20W,0201,HIGH FREQ	R2003_K	ROW
152S2036	1	IND,FILM,2.5NH,+/-0.1NH,600MA,UH-Q,0201	R2015_K	ROW
118S8636	1	RES,MF,4.7K OHM,1/32W,01005	R312_K	ROW

JP SKU

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
152S2054	1	IND,FILF,0.1NH,+/-3%,300MA,UH-Q,0201	R2016_K	JP
118S8724	1	RES,MF,0.0 OHM,1/20W,0201,HIGH FREQ	R2002_K	JP
152S2042	1	IND,FILM,1.0NH,+/-0.1NH,600MA,UH-Q,0201	R1403_K	JP
117S8167	1	RES,MF,3.3K OHM,1/32W,01005	R312_K	JP

NA SKU

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
118S8724	1	RES,MF,0.0 OHM,1/20W,0201,HIGH FREQ	R2016_K	NA
118S8724	1	RES,MF,0.0 OHM,1/20W,0201,HIGH FREQ	R2003_K	NA
152S2036	1	IND,FILM,2.5NH,+/-0.1NH,600MA,UH-Q,0201	R2015_K	NA
118S00124	1	RES,MF,47K OHM,1/32W,01005	R312_K	NA

4FF + VINYL

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
512S00030	1	SIM,TALL,PTH,4X3,D33	J_SIM	VINYL
990-12938	1	VINYL,N10,BIRCH 4.0	U_SIM	VINYL
117S8158	1	RES,MF,100K OHM,1/32W,01005	R550_K	VINYL
118S8636	1	RES,MF,4.7K OHM,1/32W,01005	R30_SIM	VINYL

SINGLE 4FF

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
512S00030	1	SIM,TALL,PTH,4X3,D33	J_SIM	SINGLE
117S8158	1	RES,MF,100K OHM,1/32W,01005	R551_K	SINGLE

ETIC CAP ALTERNATES

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00167	138S00041	BOM_TABLE_ALTS	C1204_K	CAP,2.2UF
138S00167	138S00041	BOM_TABLE_ALTS	C1205_K	CAP,2.2UF
138S00167	138S00041	BOM_TABLE_ALTS	C1206_K	CAP,2.2UF

CAP ALTERNATES

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S0031	138S00049	BOM_TABLE_ALTS	ALL	CAP,2.2UF
138S00006	138S00084	BOM_TABLE_ALTS	ALL	CAP,20UF
138S0719	138S1103	BOM_TABLE_ALTS	ALL	CAP,4.7UF
138S00128	138S00133	BOM_TABLE_ALTS	ALL	CAP,0.47UF

EEPROM ALTERNATES

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
335S00013	335S00094	BOM_TABLE_ALTS	EEPROM_K	EEPROM,8KBIT,I2C

RULER_RULE_SET=MSAP_2017								
MANUFACTURING CONFIGURATION								
SPACING MULTIPLES	DIELECTRIC MULTIPLES	PAD TO X RATIO	VIA TO X RATIO	VOID SPACE RATIO	MECH HOLE ASPECT RATIO	LAZER HOLE ASPECT RATIO	CURRENT DENSITY (AMPS/SQR MM)	BOARD SIZE (LONGEST SIDE LENGTH MM)
1	1.65, 2, 2D, 3.2D, 4D	?	?		?	?	2	? 50
LAYERS				MINIMUM CU WIDTH RATIO		MINIMUM CU SPACING RATIO		MINIMUM TO DEFAULT RATIO
TOP,BOTTOM,ISL4,ISL5				1.0		1.0		1.0
ISL2,ISL3,ISL6,ISL7				1.0		1.0		1.0

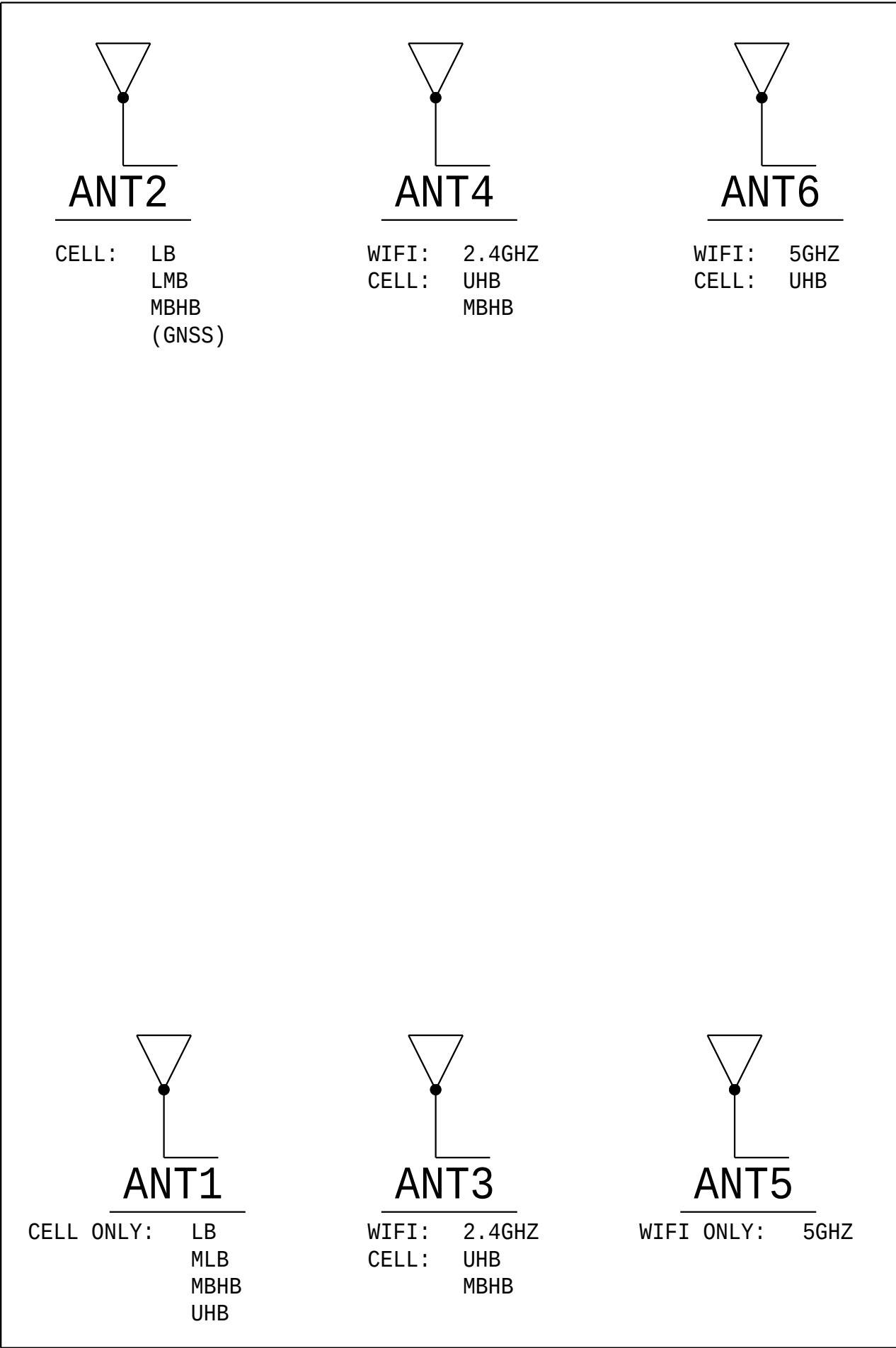
LOAD_CURRENT=50
LOAD_CURRENT=150
LOAD_CURRENT=500
LOAD_CURRENT=1000
LOAD_CURRENT=3000

TABLE_IMPEDANCE_HYBRID=50_HYBRID	
LAYERS	IMPEDANCE RULE
TOP	50_OHM_SE
ISL3	50_OHM_SE
ISL6	50_OHM_SE_6R4R8
ISL7	50_OHM_SE
ISL5	50_OHM_SE

TABLE_IMPEDANCE_HYBRID=50_HYBRID_WIDE	
LAYERS	IMPEDANCE RULE
TOP	50_OHM_SE_1R3
ISL3	50_OHM_SE
ISL6	50_OHM_SE_6R4R8
ISL7	50_OHM_SE
ISL5	50_OHM_SE

8	7	6	5	4	3	2	1
D							D
C							C
B							B
A							A
8	7	6	5	4	3	2	1

RADIO ANTENNAS



UAT

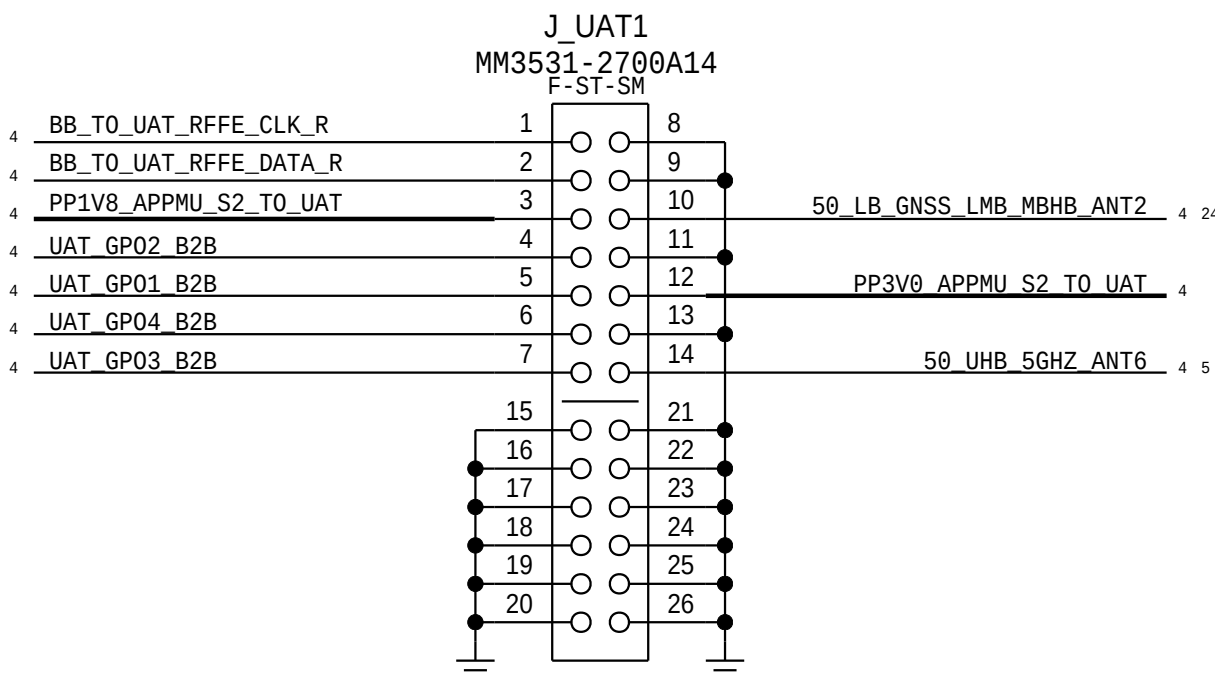
LAT

ANTENNA B2BS

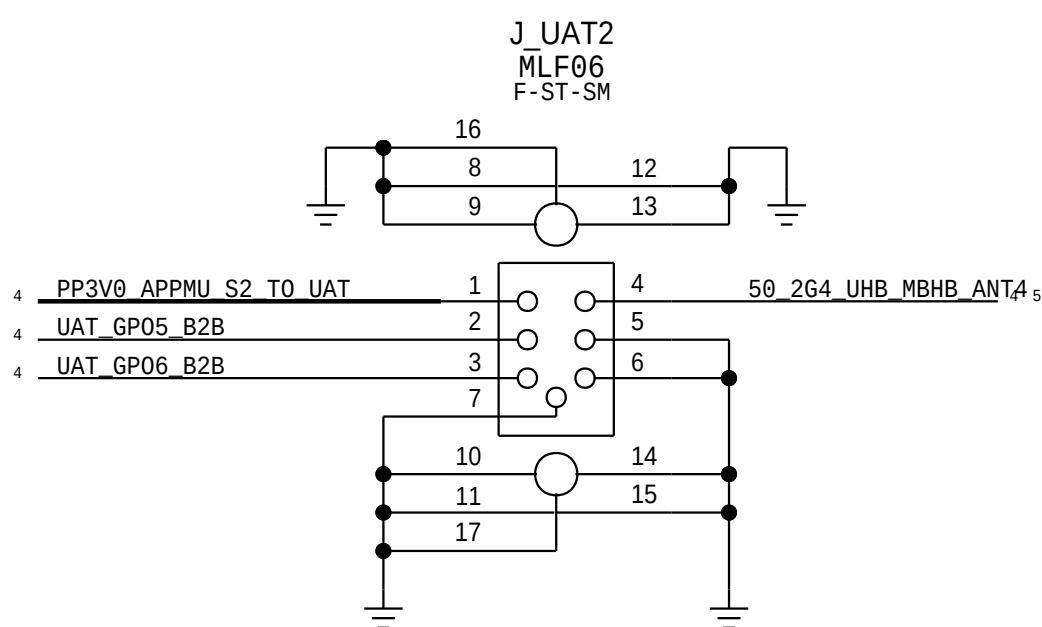
UAT

D3X UAT B2BS

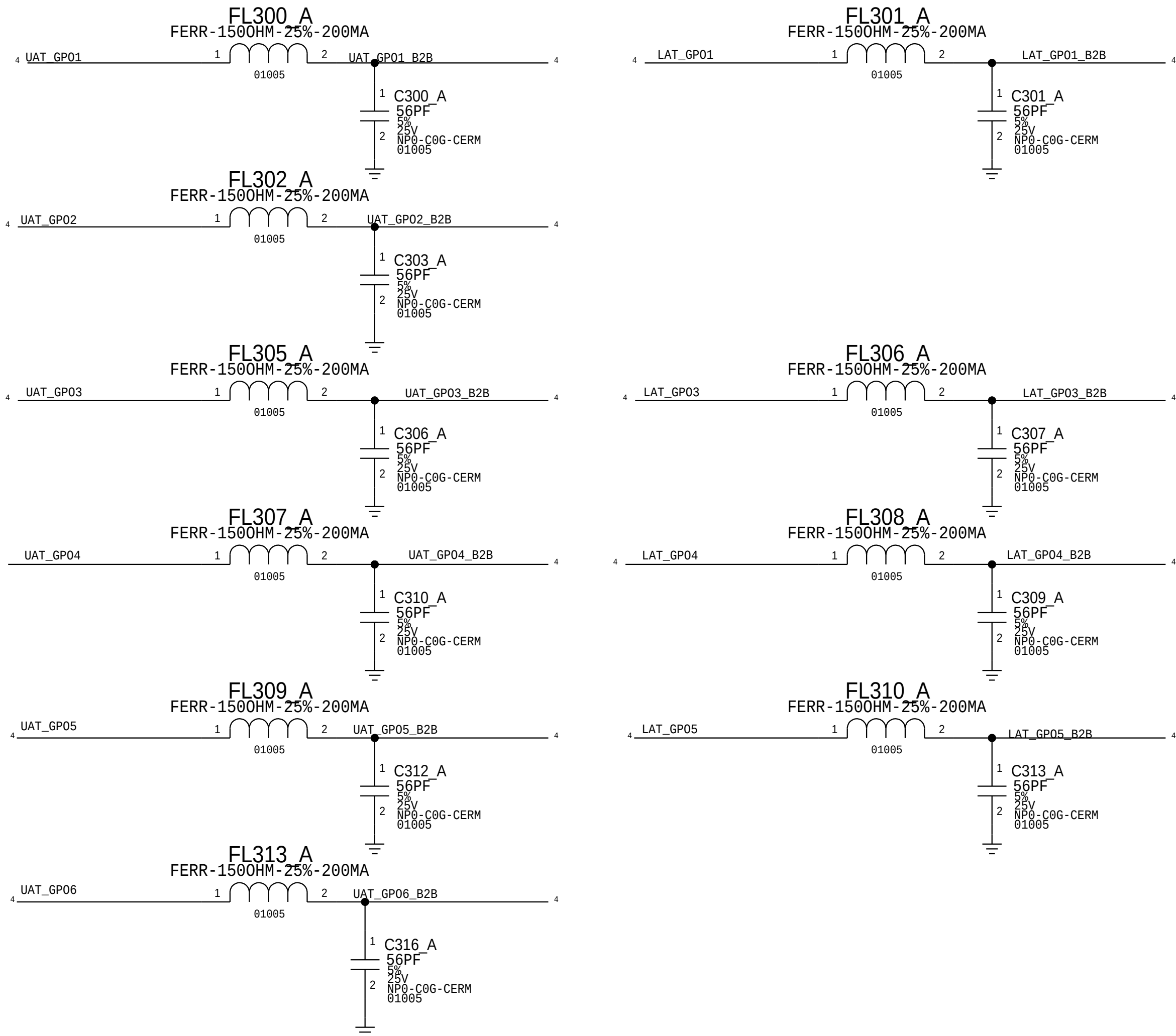
RCPT ON MLB BOT (518S00167)
PLUG ON FLEX (518S00166)



RCPT ON MLB BOT (516S00434)
PLUG ON FLEX (516S00435)



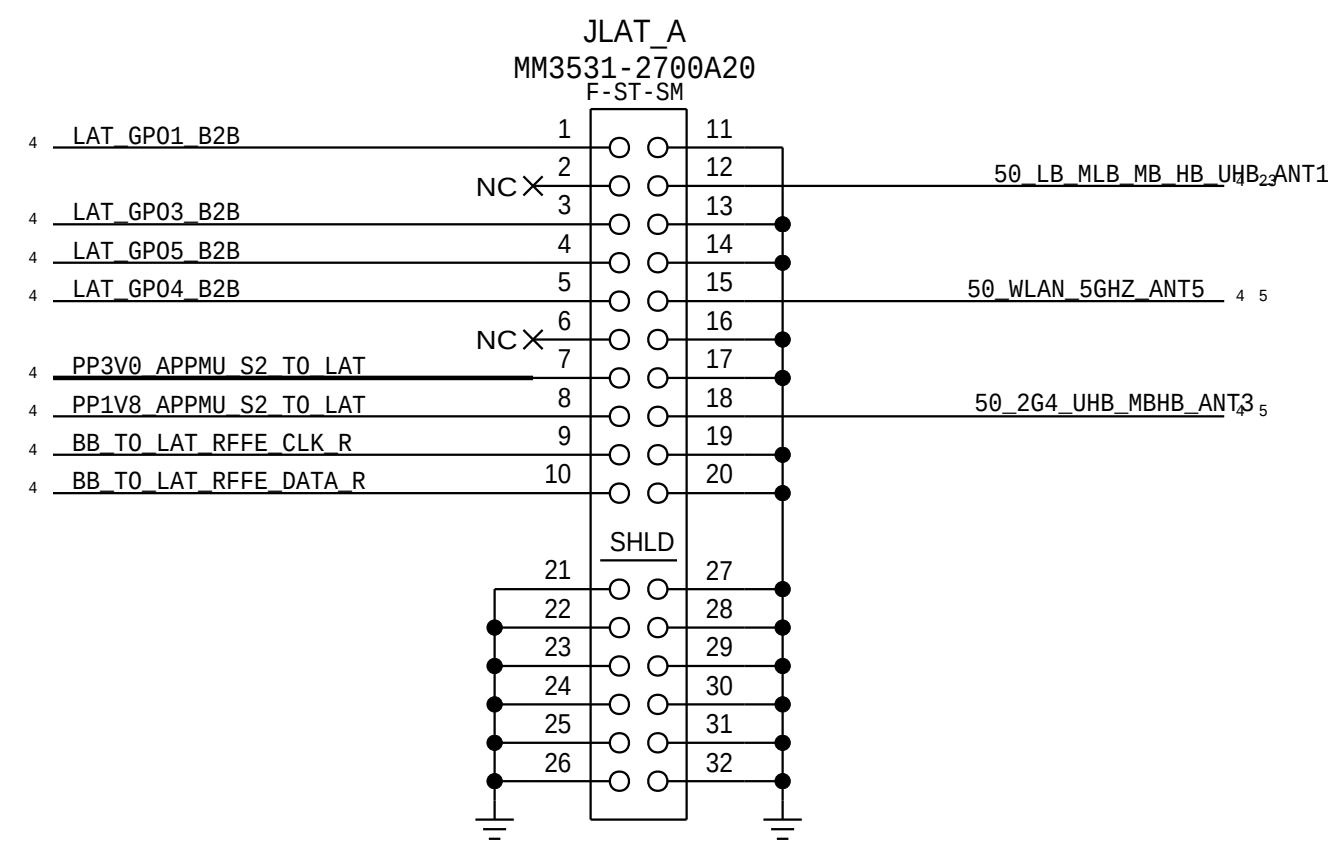
GPO FILTERS



LAT

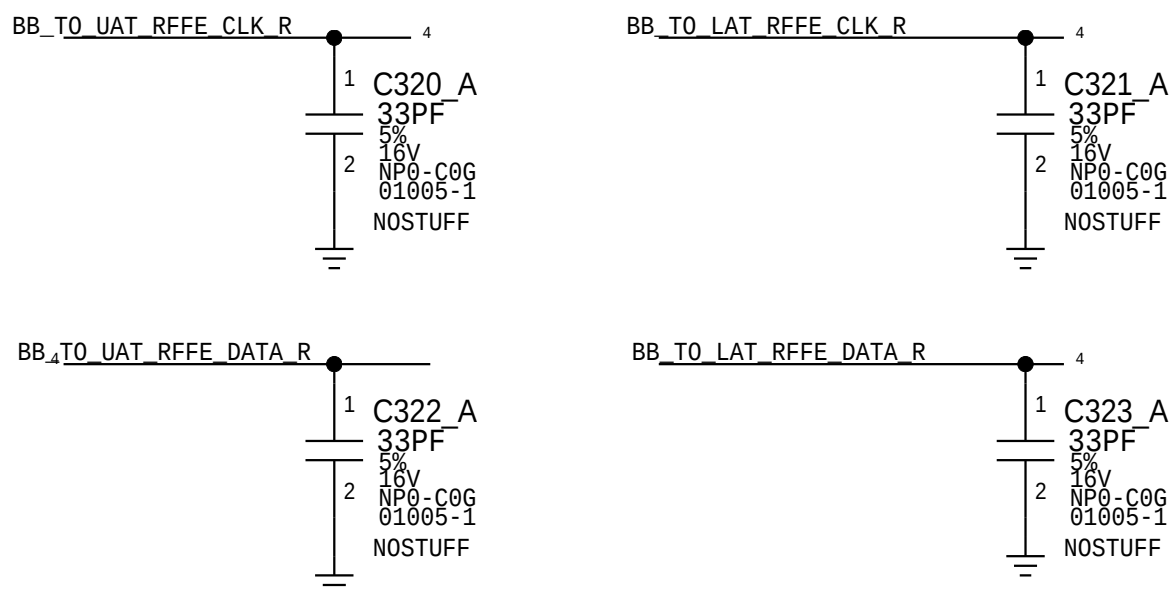
D32 LAT B2B

RCPT ON MLB BOT (518S00149)
PLUG ON FLEX (518S00150)

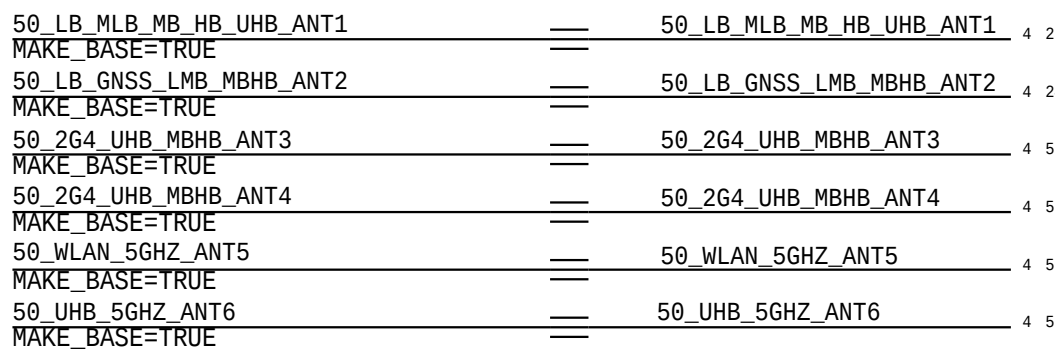


D32 ONLY

RFFE FILTERS



ANTENNA CONNECTIONS



87654321

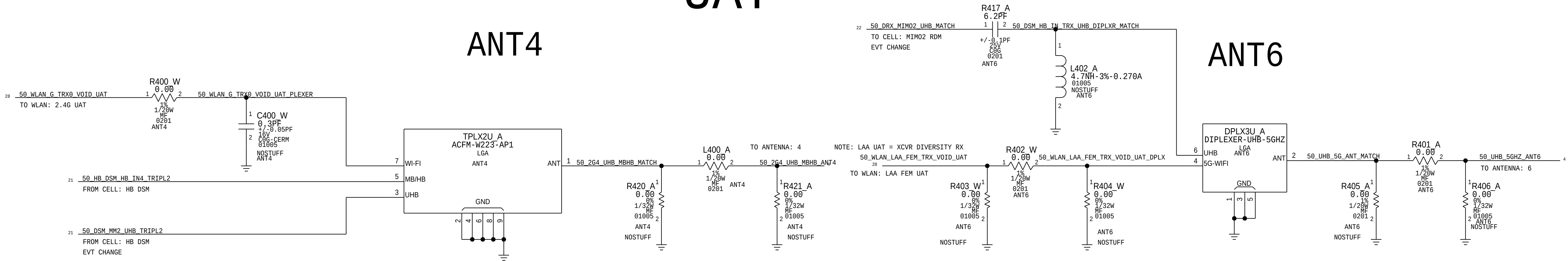
SHARED N-PLEXERS

D

UAT

ANT4

ANT6



MHB-UHB/2.4G WLAN TRIPLEXER2

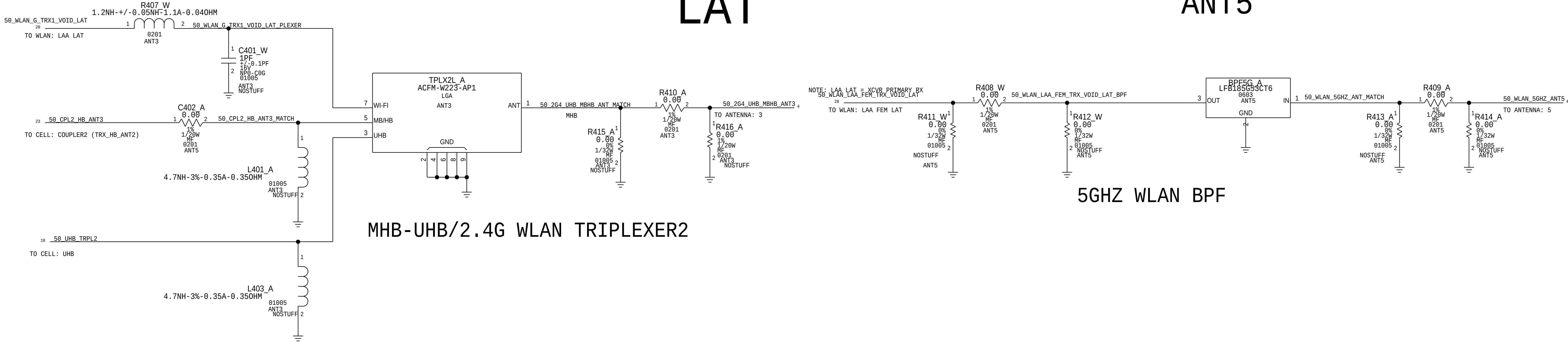
UHB-5G WLAN DIPLEXER3

C

LAT

ANT3

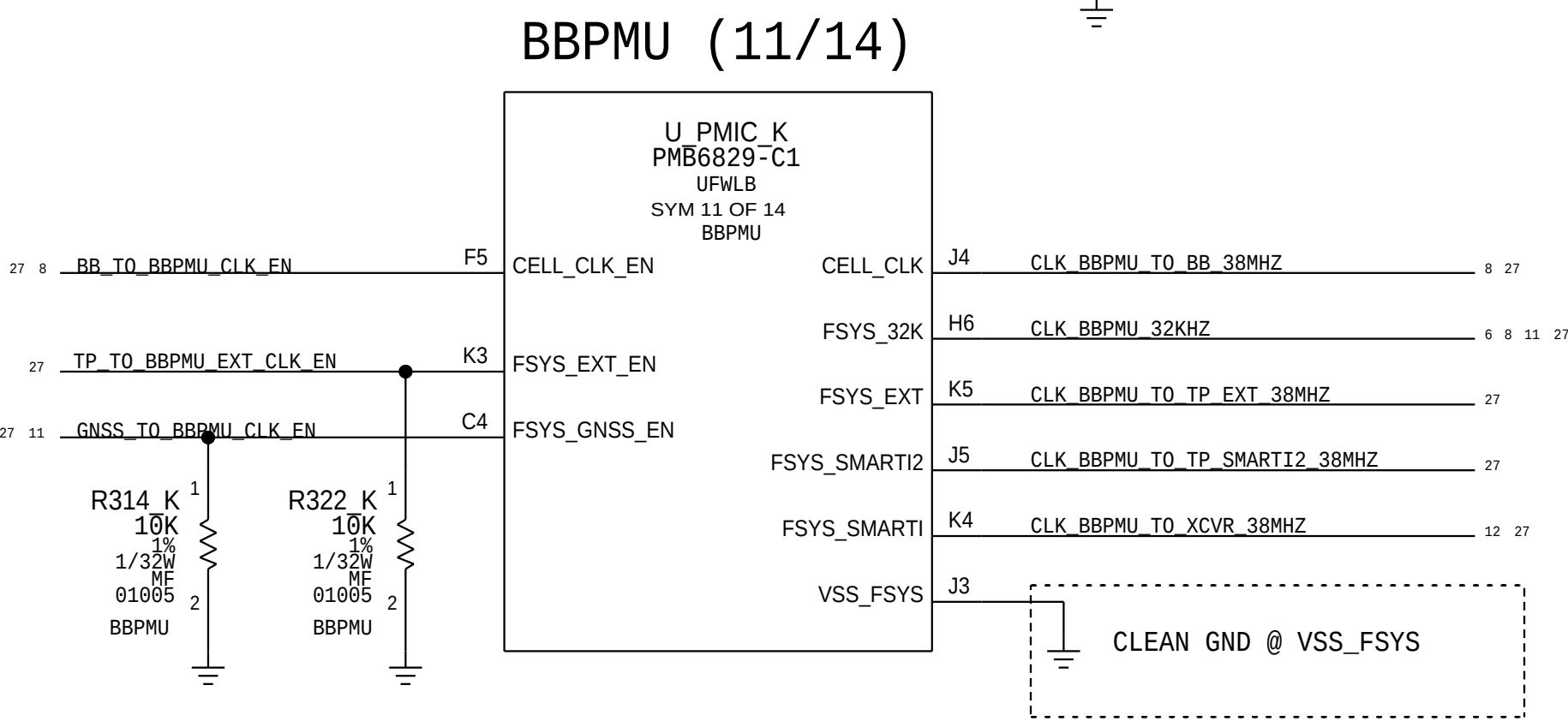
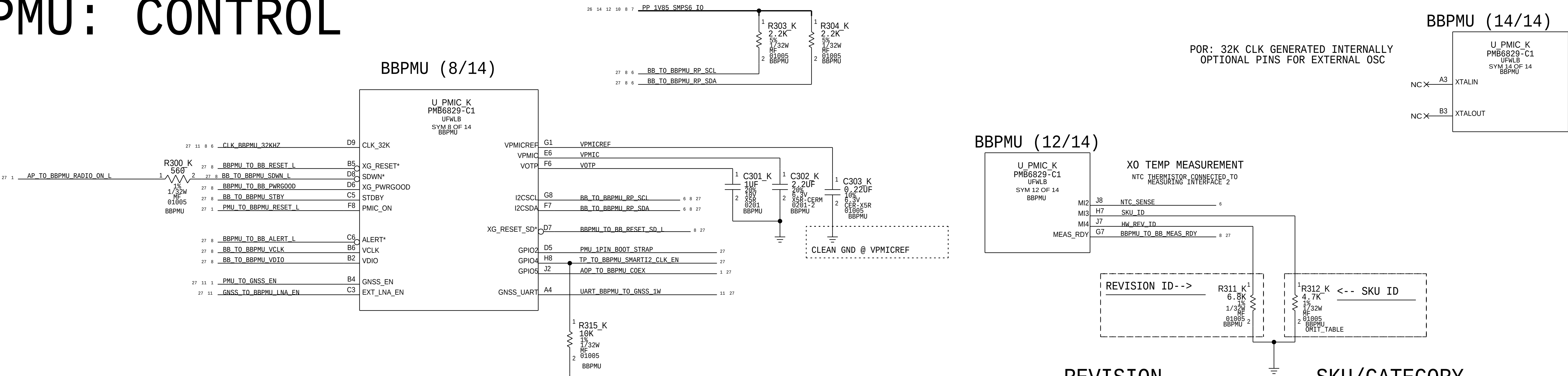
ANT5



MHB-UHB/2.4G WLAN TRIPLEXER2

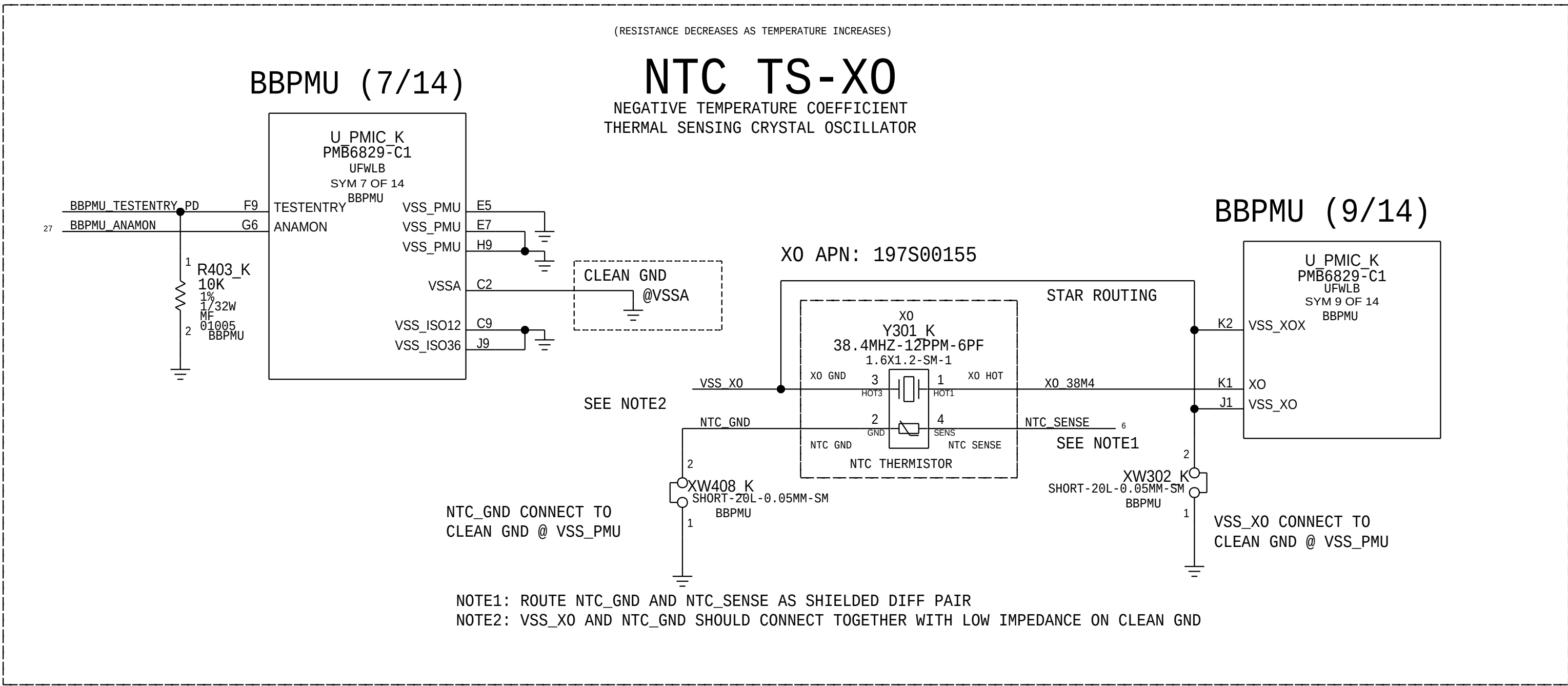
5GHZ WLAN BPF

BBPMU: CONTROL



REVISION			SKU/CATEGORY		
R311_K	MLB	RF DEV	R312_K	HW REVISION	X-CODE
0.0	T/POC/PRE-PROT00	1.0	0.0	RFDEV JP	
1.2K	P0	1.1	1.2K	ICE 18.0 JP	X1344
2.2K	P1	2.0	2.2K	ICE 18.0 ROW	(ICE 18.0)
3.3K	P2	2.1	3.3K	ICE 18.1 JP	X1049
4.7K	P2 WITH S7.1	3.0	4.7K	ICE 18.1 ROW	(ICE 18.1)
6.8K	EVT		6.8K	ICE 18.2 JP	X1210
8.2K	EVT 1.5	4.0	8.2K	ICE 18.2 ROW	(ICE 18.2)
10K	CARRIER		10K	RFDEV ROW	
12K	DVT	5.0	12K	ICE 18.5 JP	X1170
15K	PVT	5.1	15K	ICE 18.6 JP	X1176
18K			18K		
22K		6.0	22K	ICE 18.5 US	X1170
27K		6.0	27K	ICE 18.6 US	X1176
33K			33K	ICE 18.0 NA	X1344
39K			39K	RFDEV NA	
47K			47K	ICE 18.1 NA	X1049
56K	DARWIN		56K	ICE 18.2 NA	X1210
68K	JP_SKWS		68K		
82K			82K	KAR00 INTERNAL	
100K	P1 DOE2, TXC		100K	KAR00 INTERNAL	
120K	P1 DOE1, S7 B1		120K	KAR00 INTERNAL	
150K			150K		

HWID TABLE RDAR://32880011
ADC TABLE RDAR://27081897
BBPMU ADC TABLES

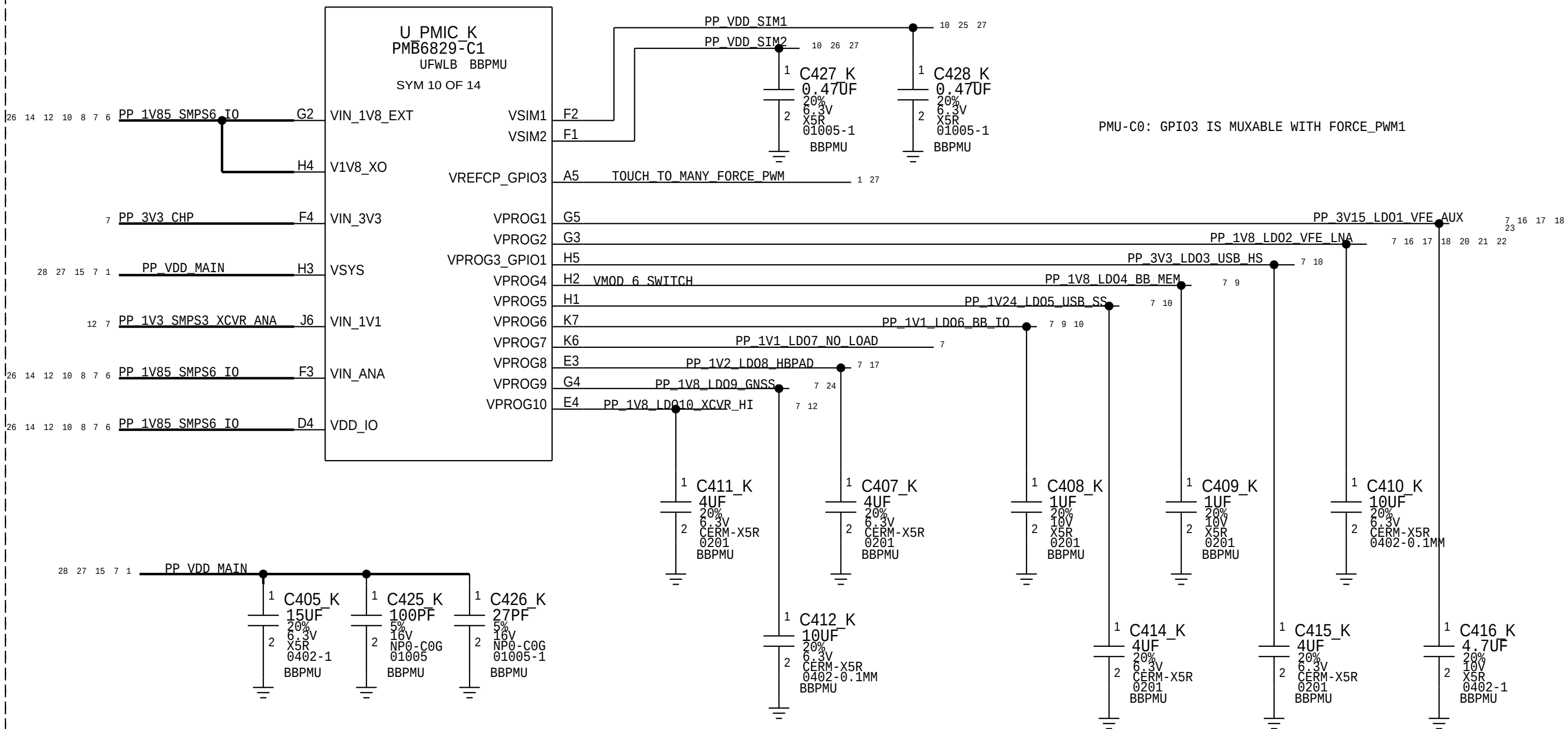


NOTE1: ROUTE NTC_GND AND NTC_SENSE AS SHIELDED DIFF PAIR
NOTE2: VSS_XO AND NTC_GND SHOULD CONNECT TOGETHER WITH LOW IMPEDANCE ON CLEAN GND

BBPMU: RAILS

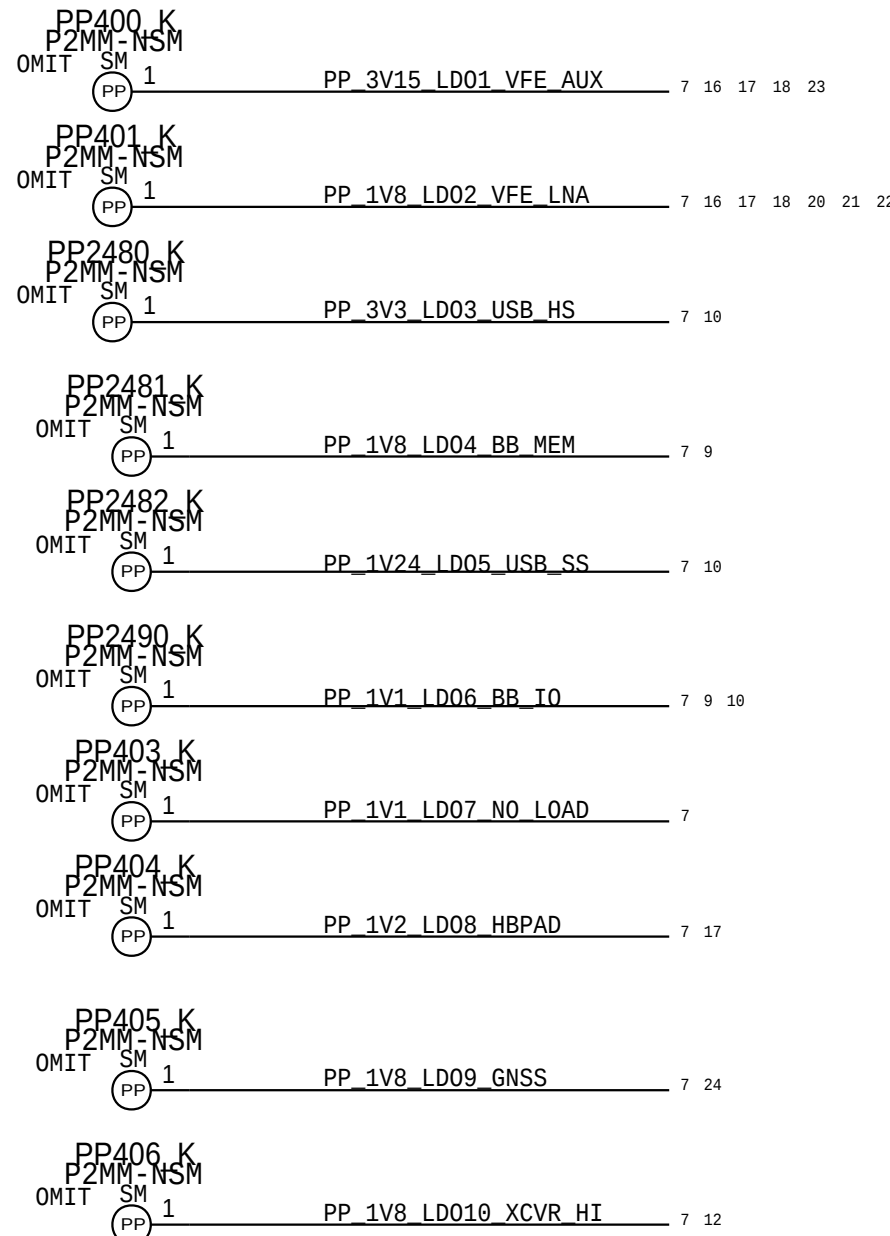
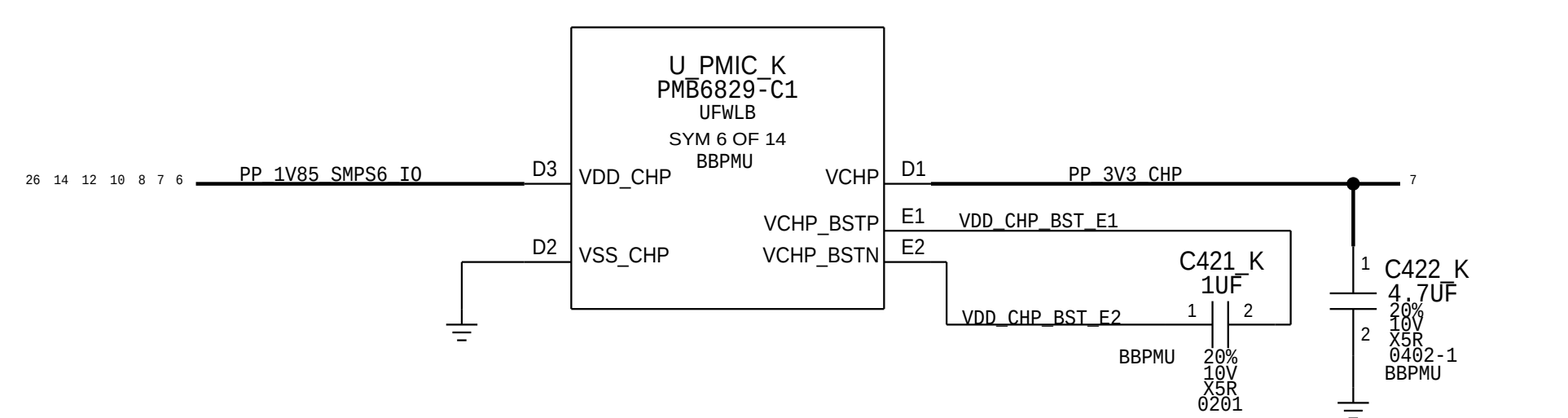
PROGRAMMABLE LDOS

BBPMU (10/14)



CHARGE PUMP

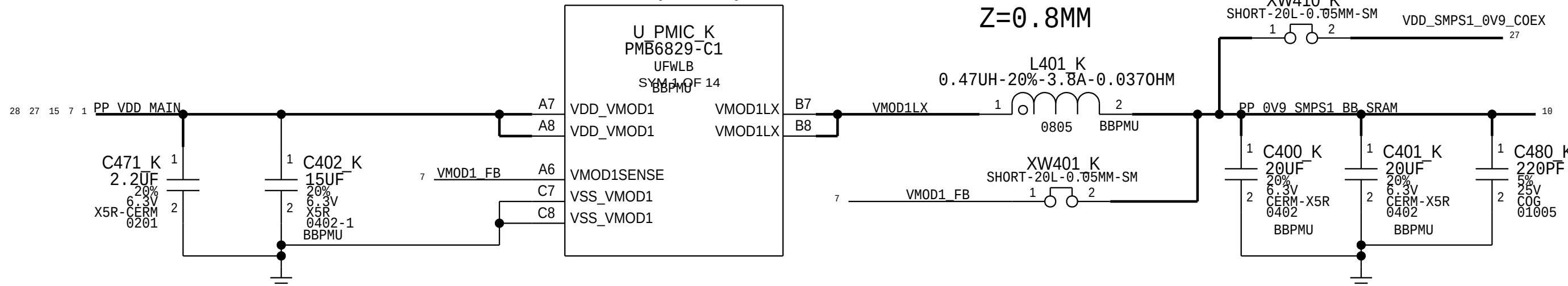
BBPMU (6/14)



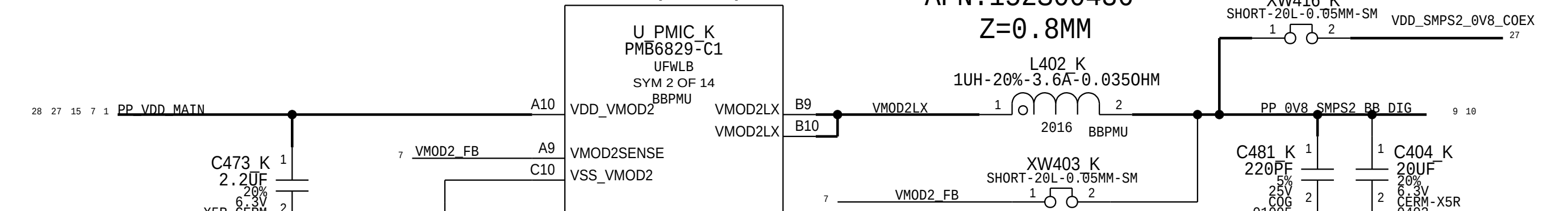
SWITCHERS

SMPS1/2/4 SHARED VDD_MAIN INTERNALLY
SMPS3/5/6 SHARED VDD_MAIN INTERNALLY

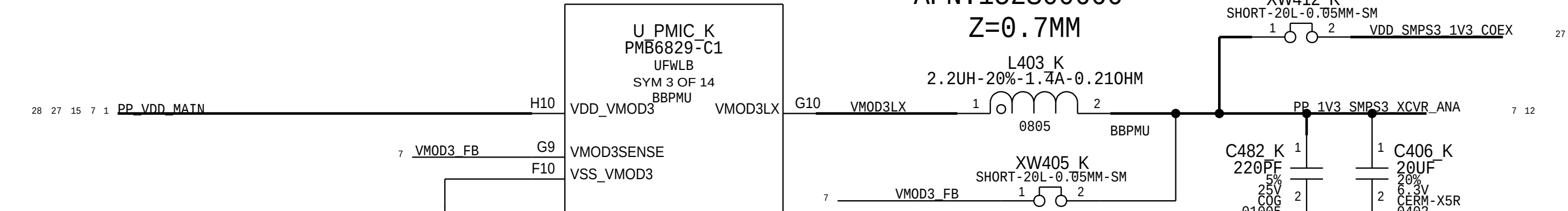
BBPMU (1/14)



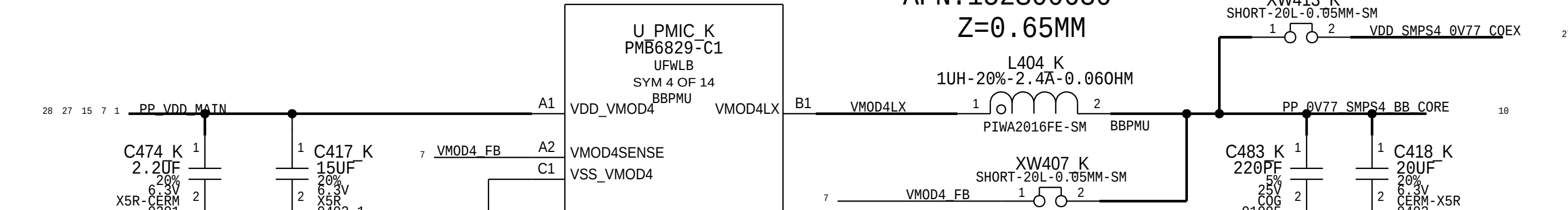
BBPMU (2/14)



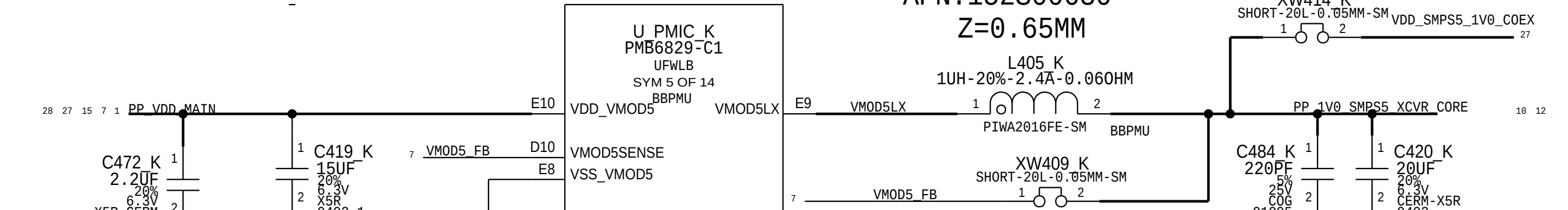
BBPMU (3/14)



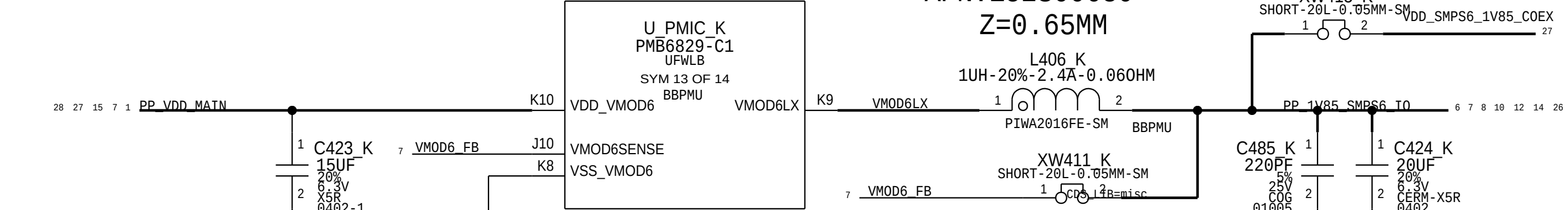
BBPMU (4/14)



BBPMU (5/14)

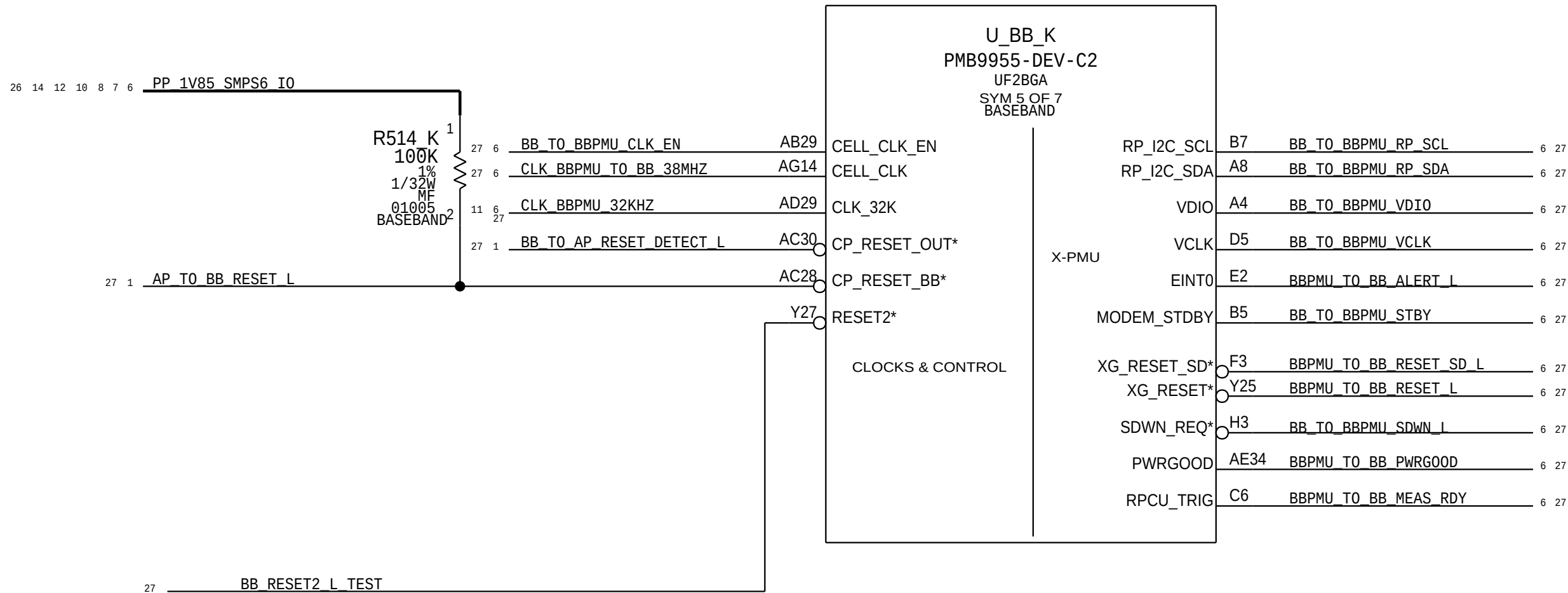


BBPMU (13/14)

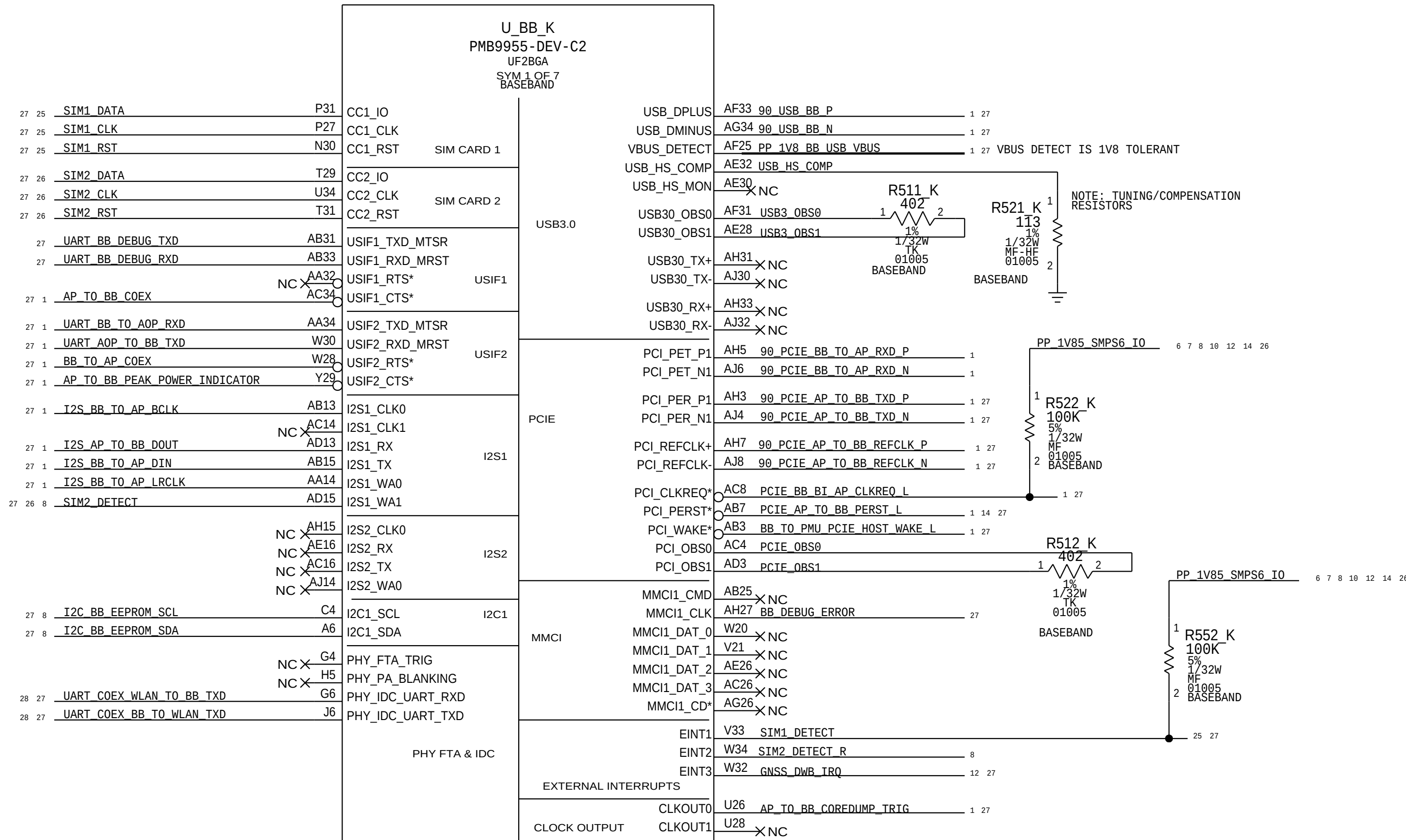


BB: INTERFACE

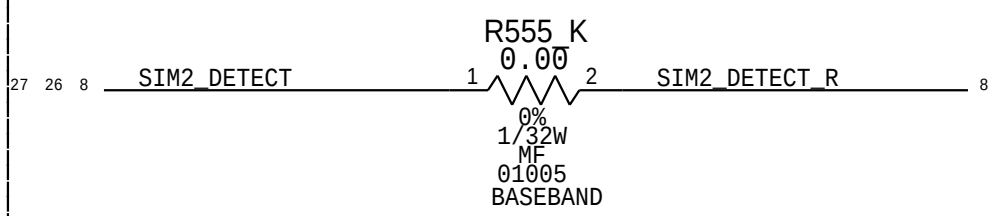
XG756 (5/7)



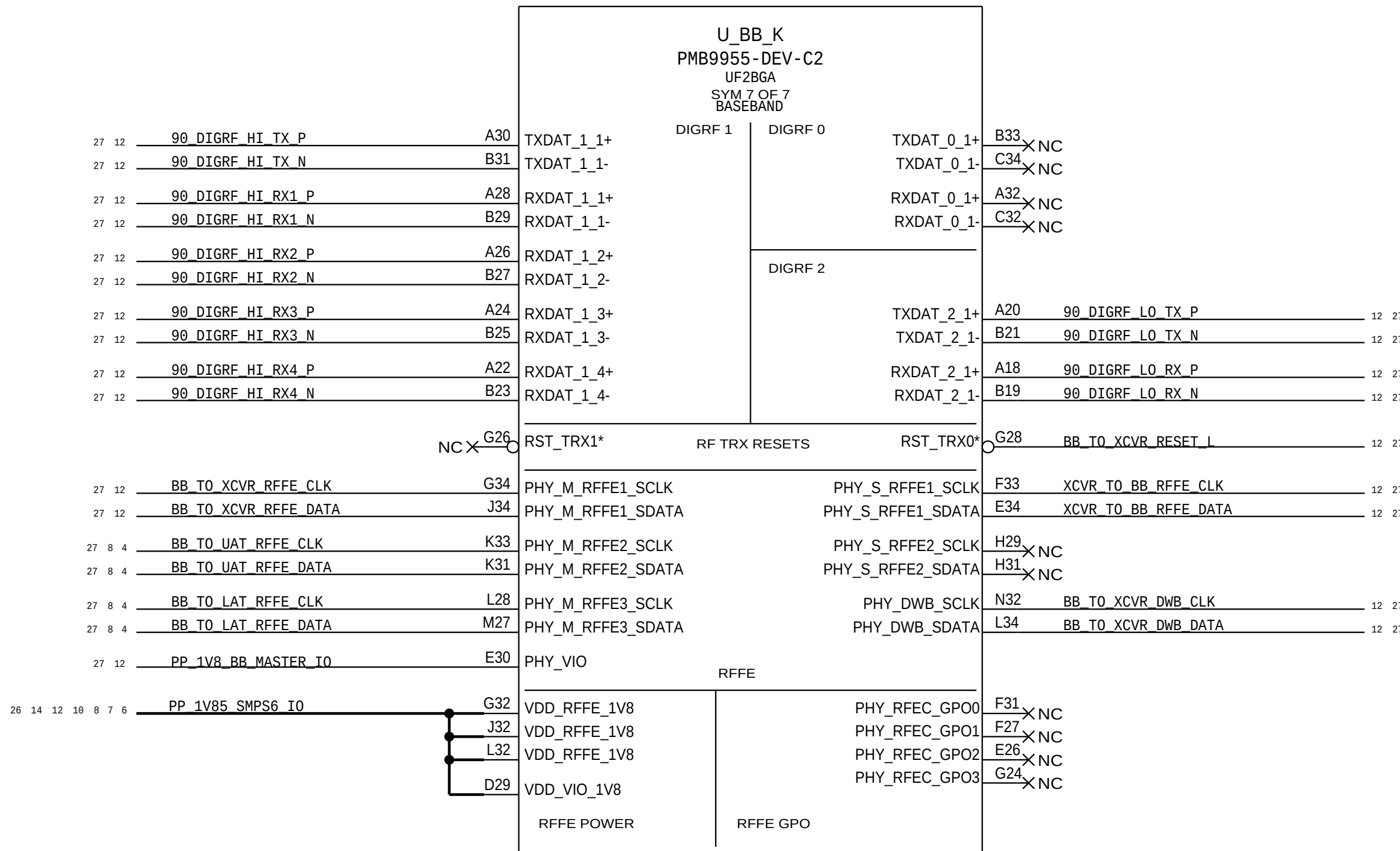
XG756 (1/7)



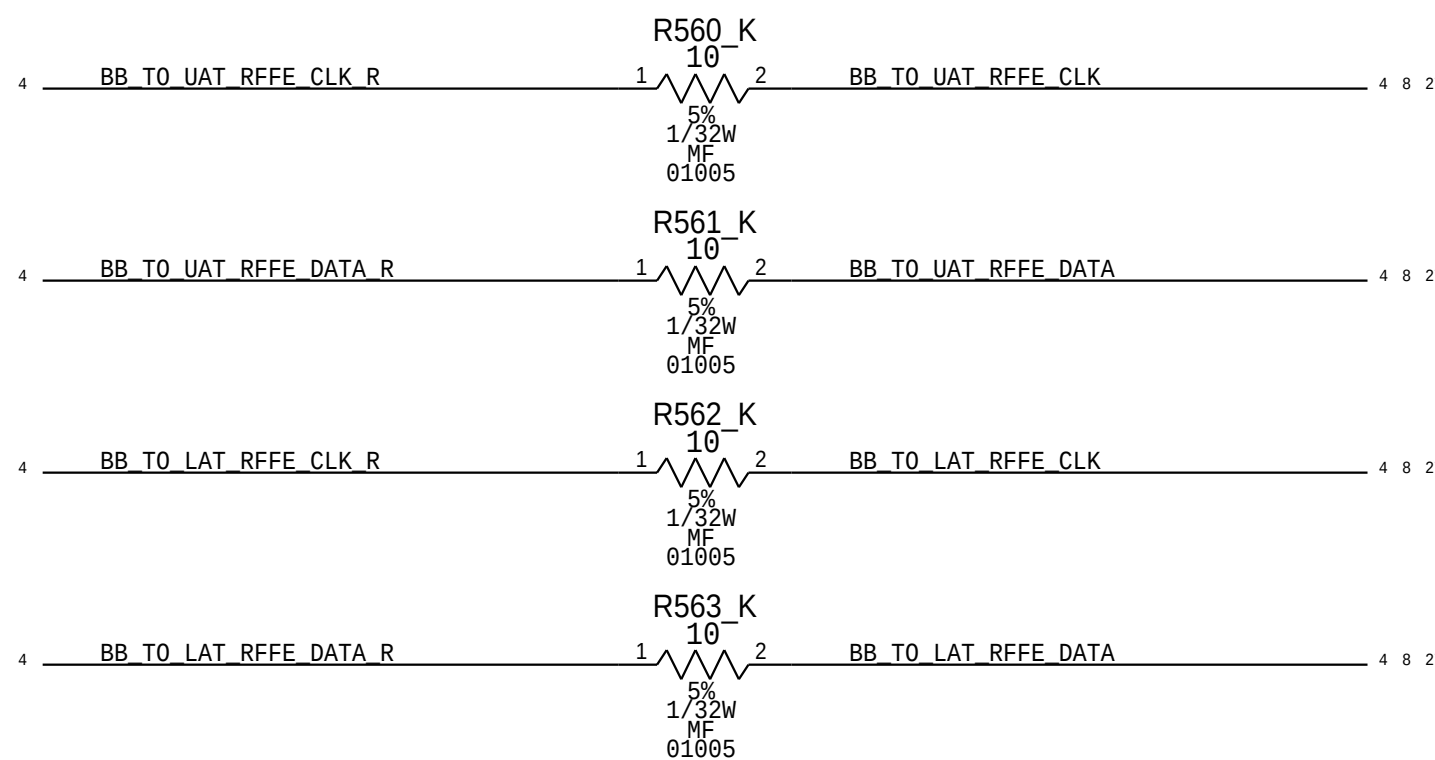
SIM2_DETECT TO EINT2



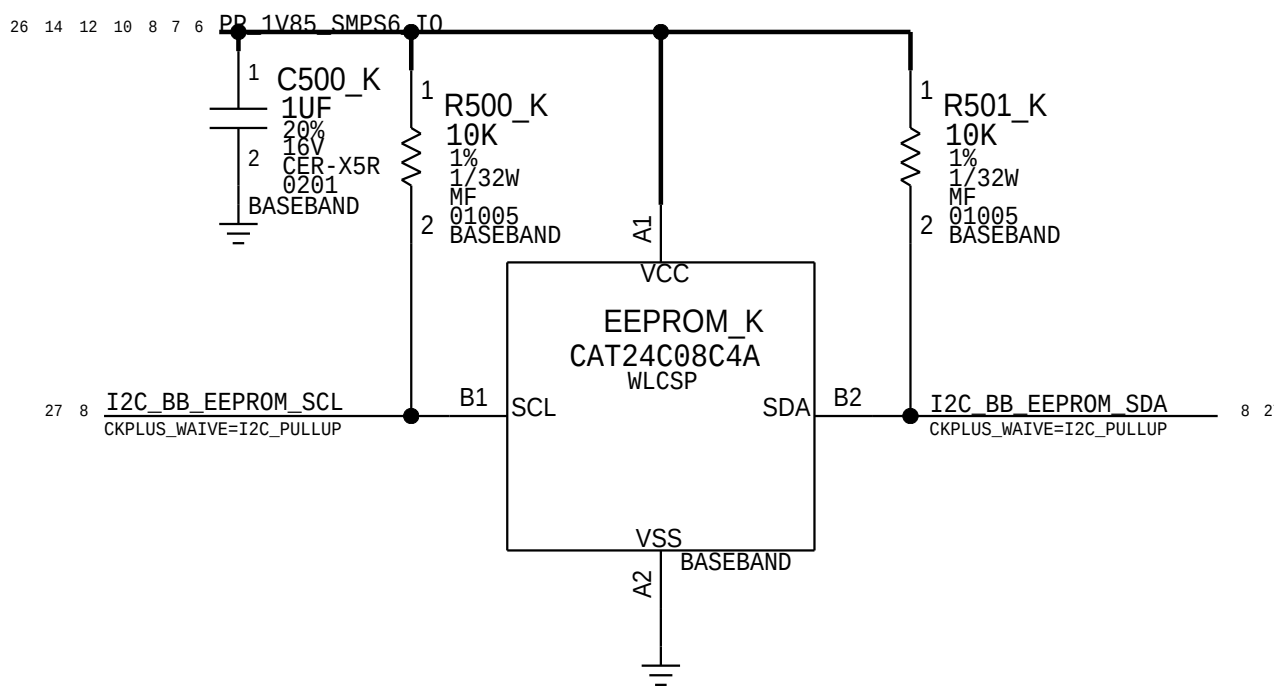
XG756 (7/7)



BB RFFE DAMPING RESISTORS



BB EEPROM



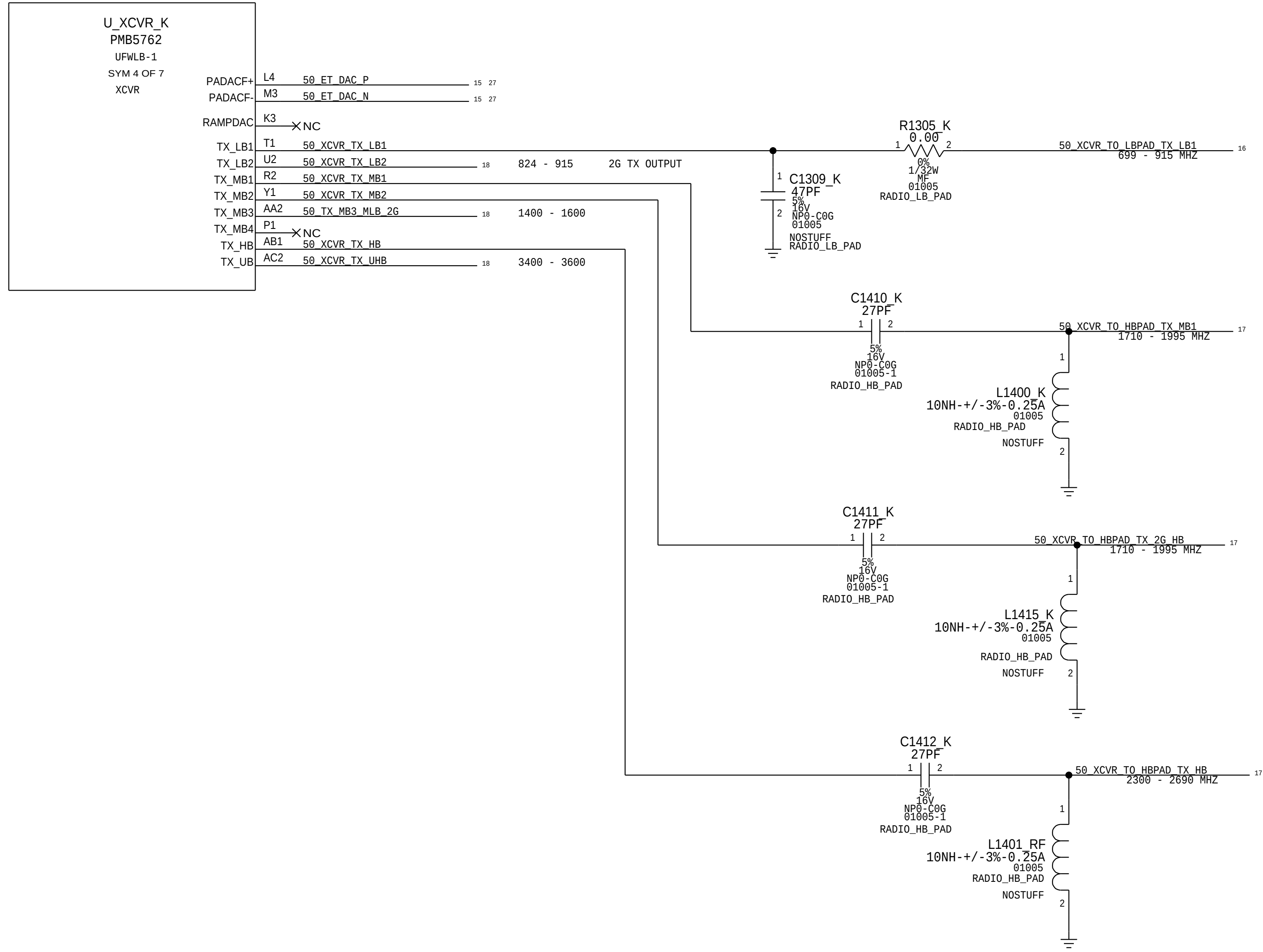
D



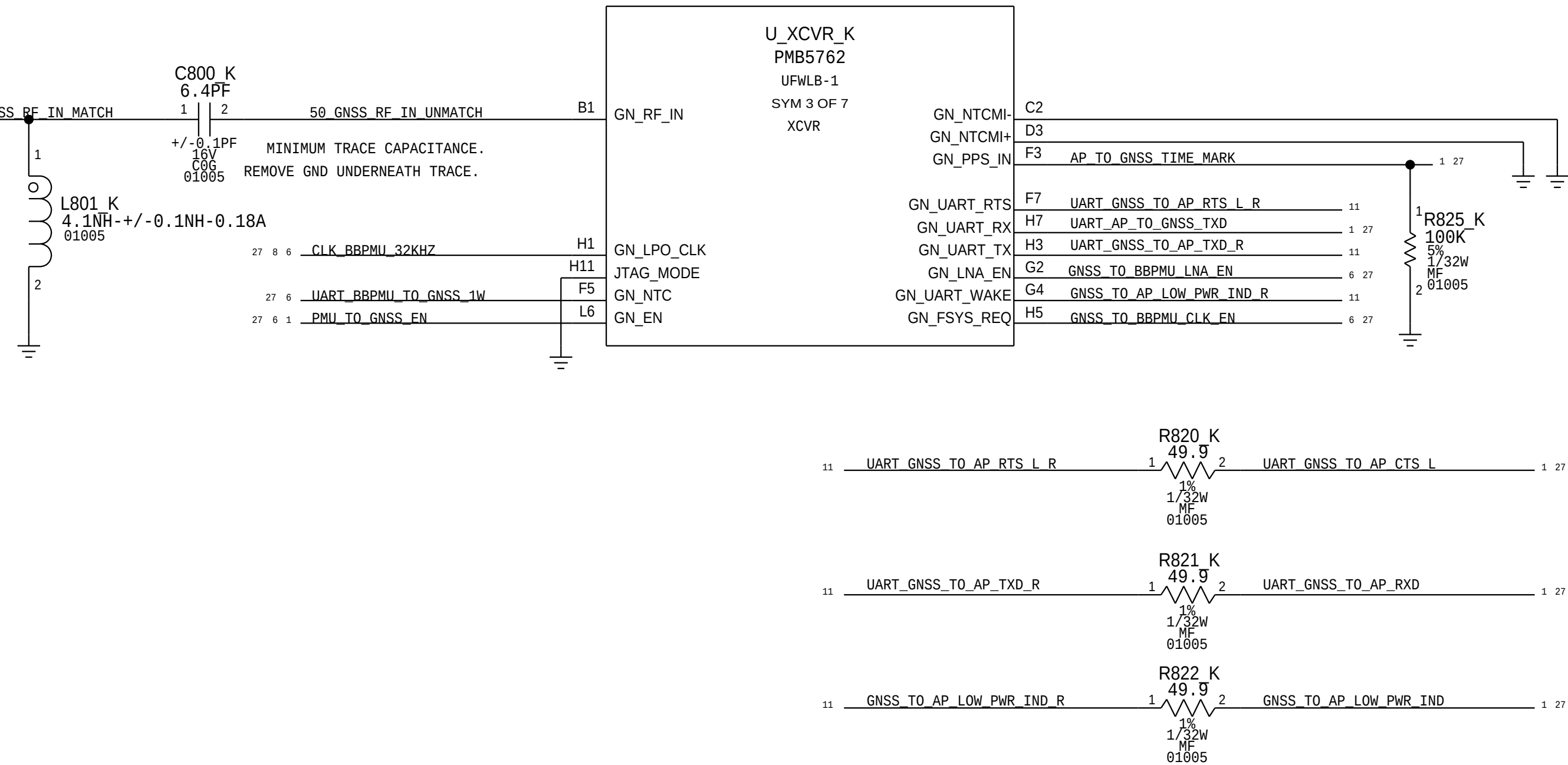
A

XCVR: TRANSMIT & GNSS

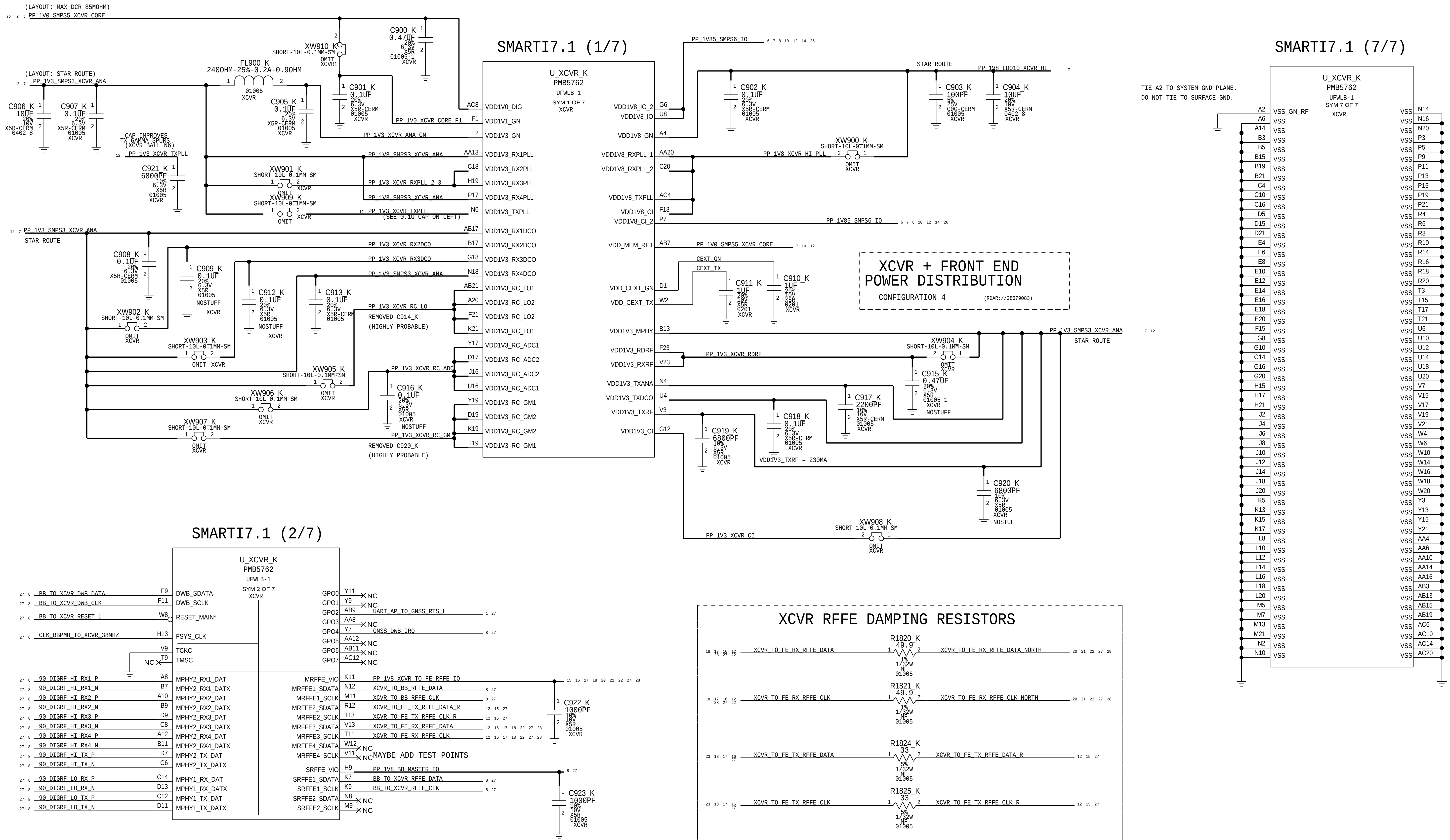
SMARTI7.1 (4/7)



SMARTI7.1 (3/7)

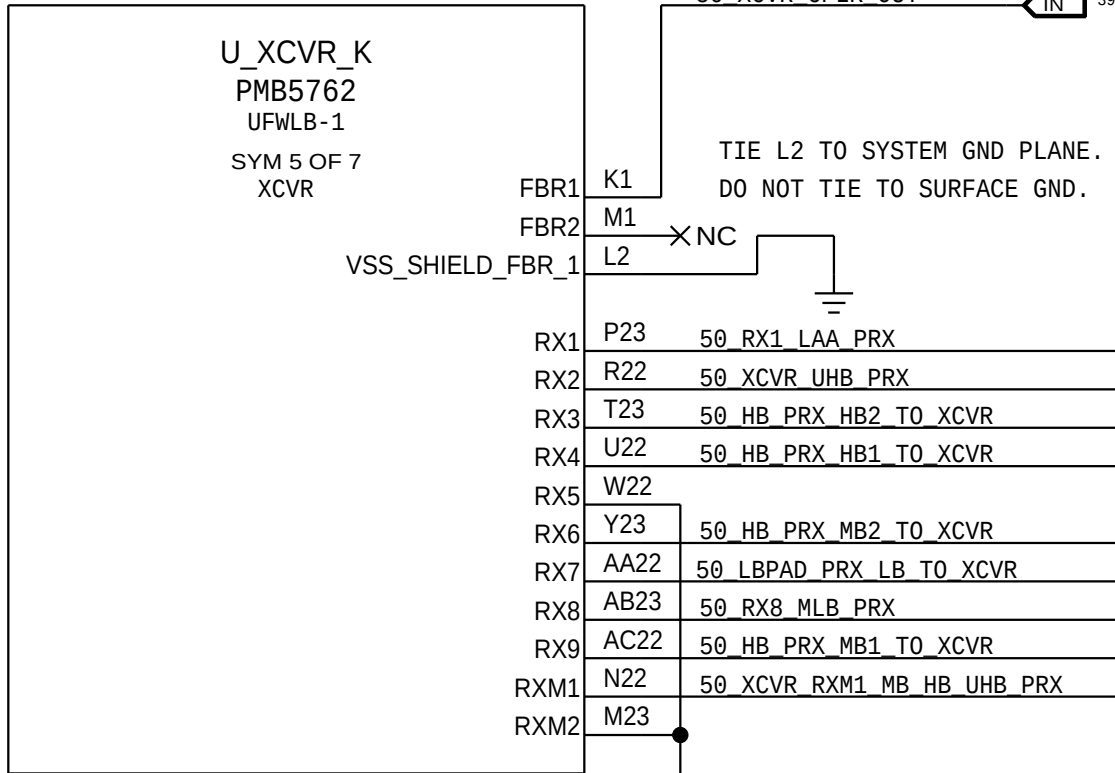


XCVR: INTERFACE & PWR

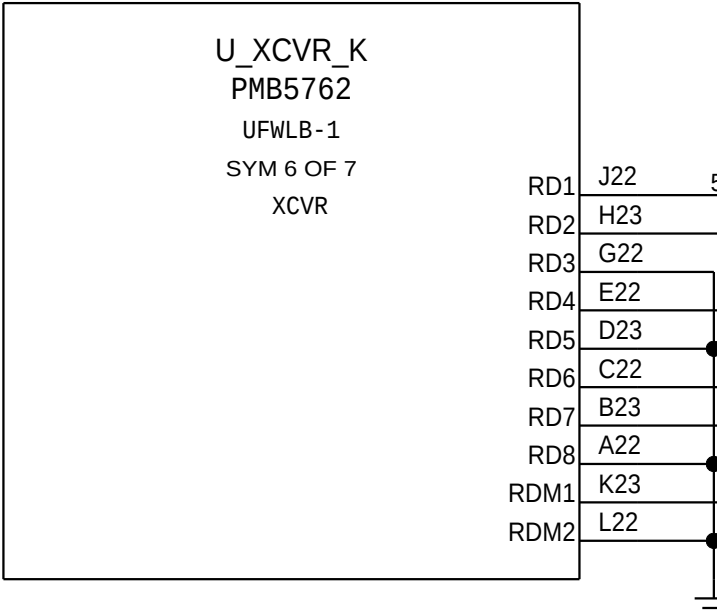


XCVR: PRIMARY/DIVERSITY RX

SMARTI7.1 PRX (5/7)



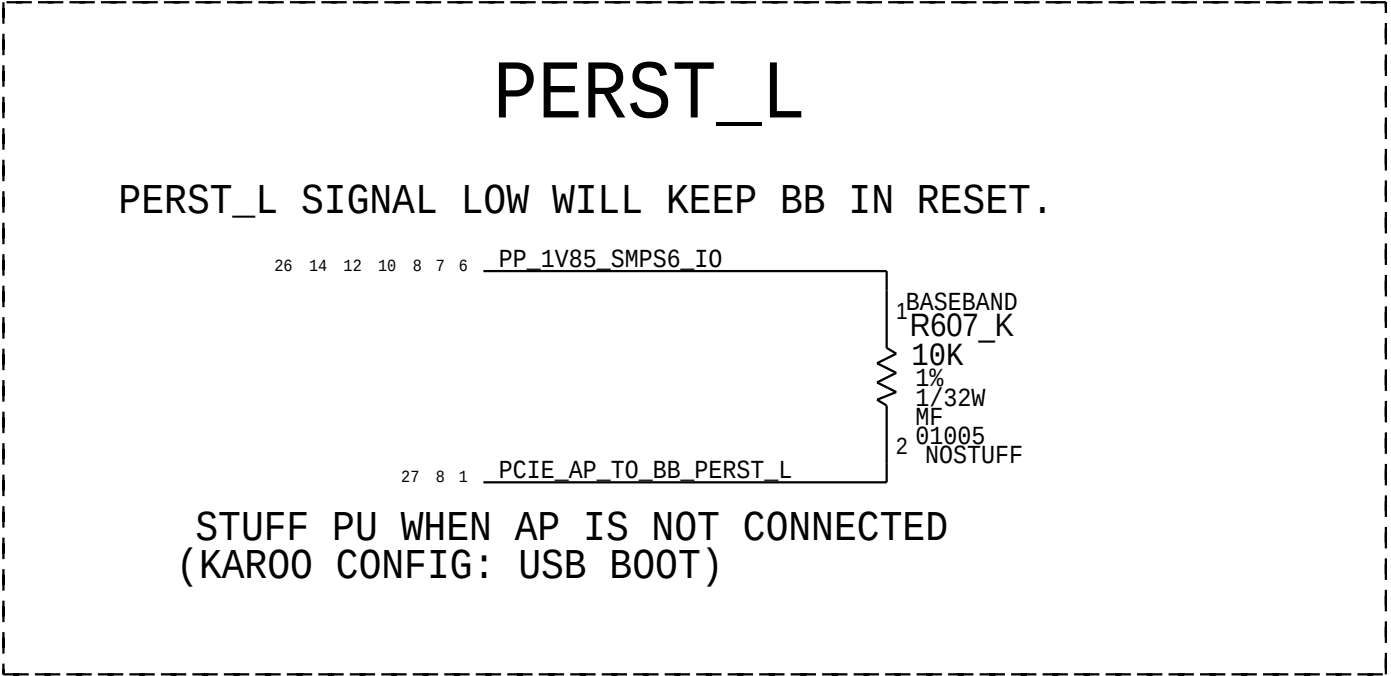
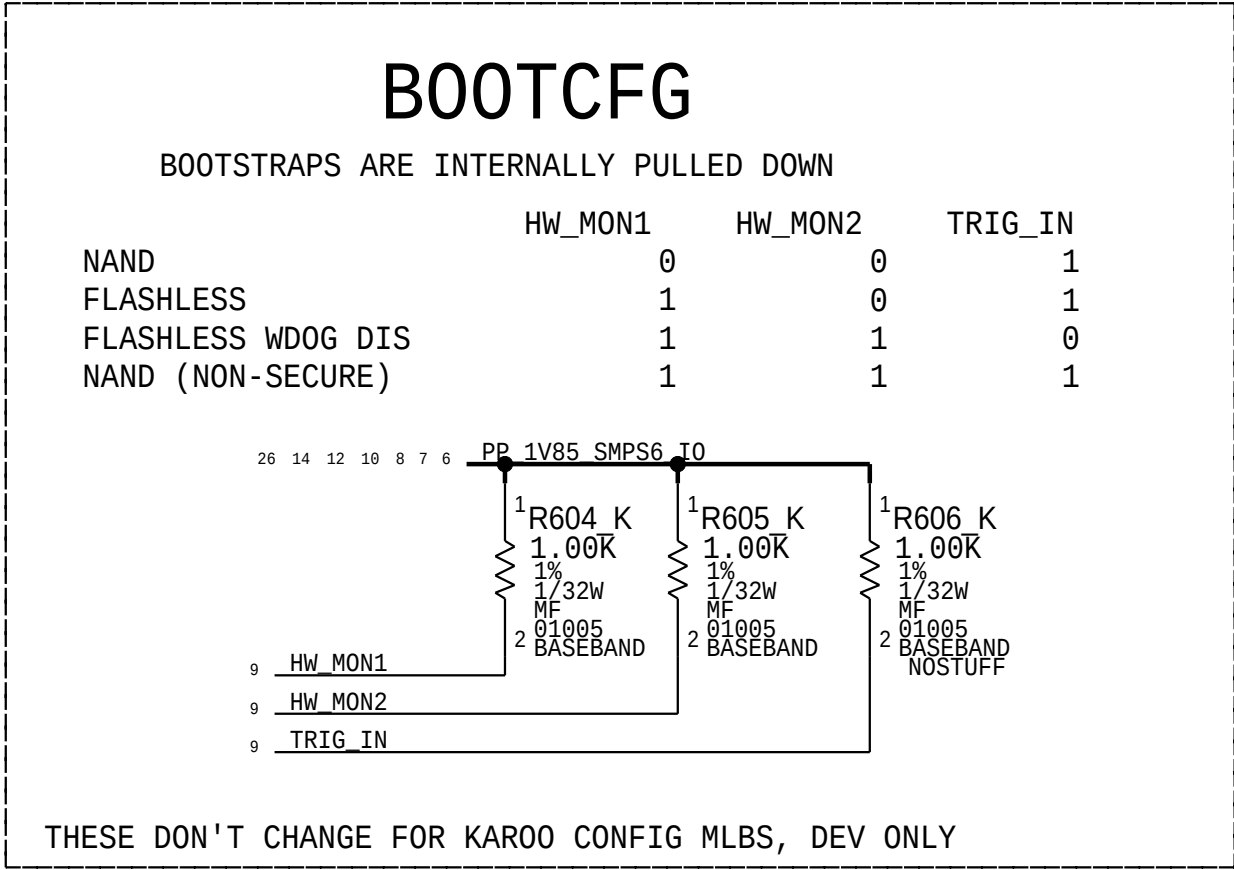
SMARTI7.1 DRX (6/7)



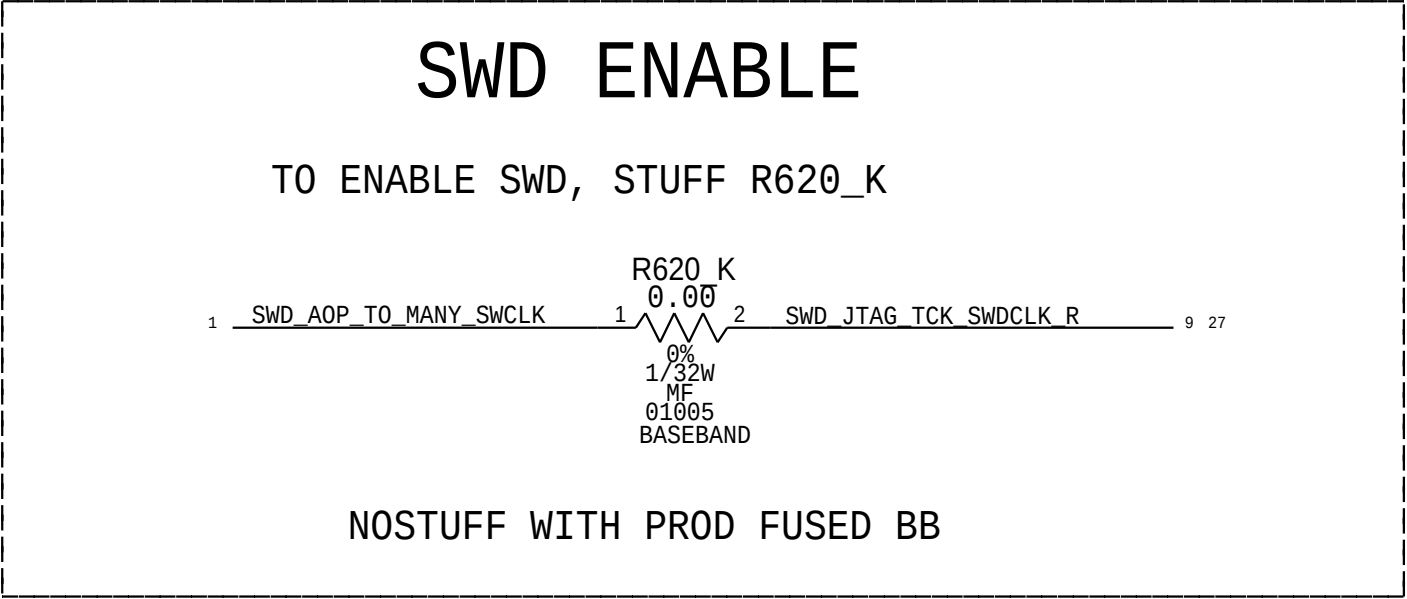
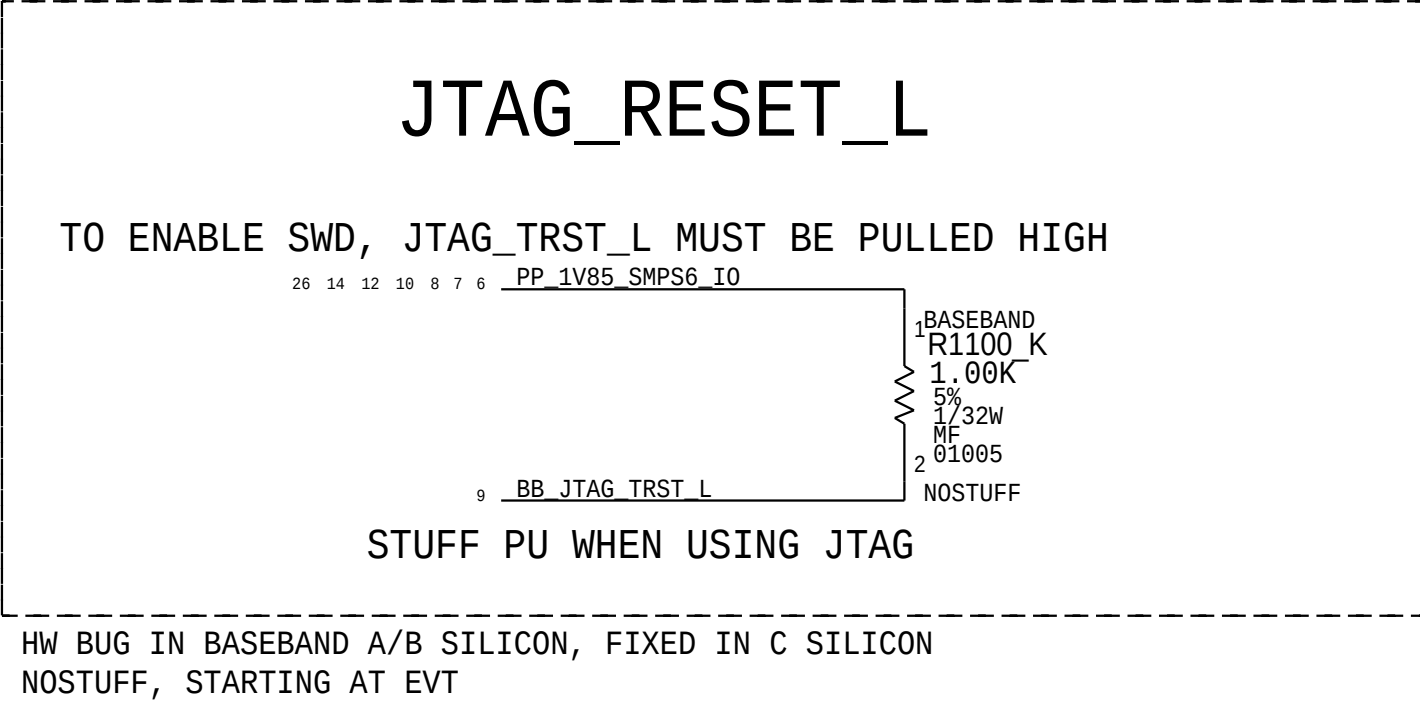
S7 PORT MAP CHANGE 10/11/2016:
USE RD6 INSTEAD OF RD3.
CONNECT ALL UNUSED RD PORTS TO GND.

HW CONFIG OPTIONS

BOOT

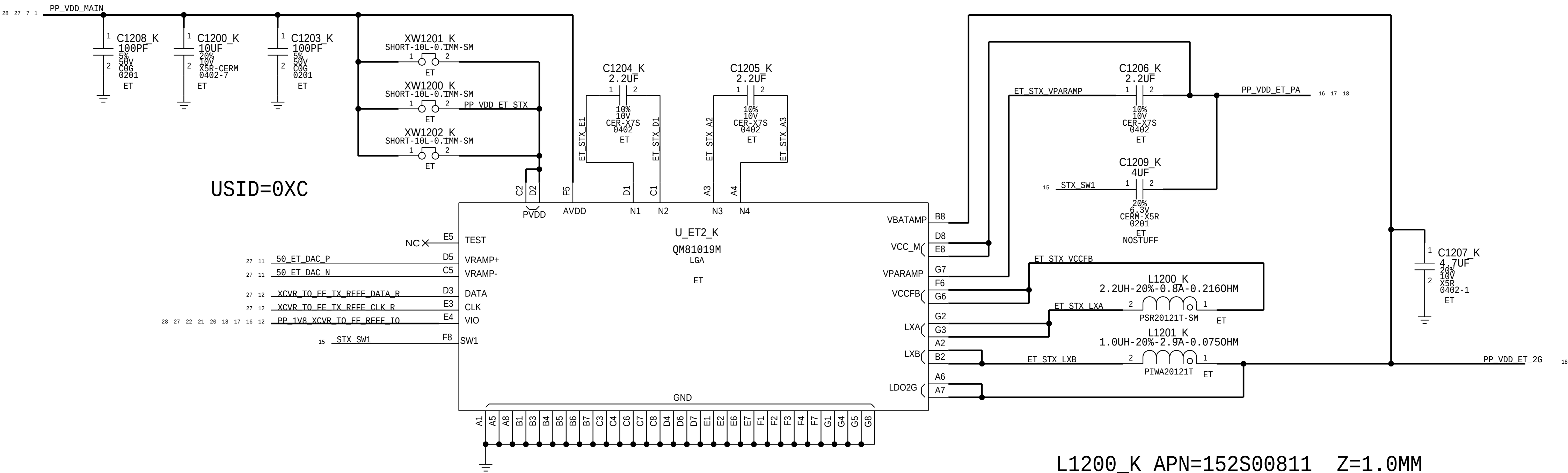


JTAG & SWD



ET MODULATOR

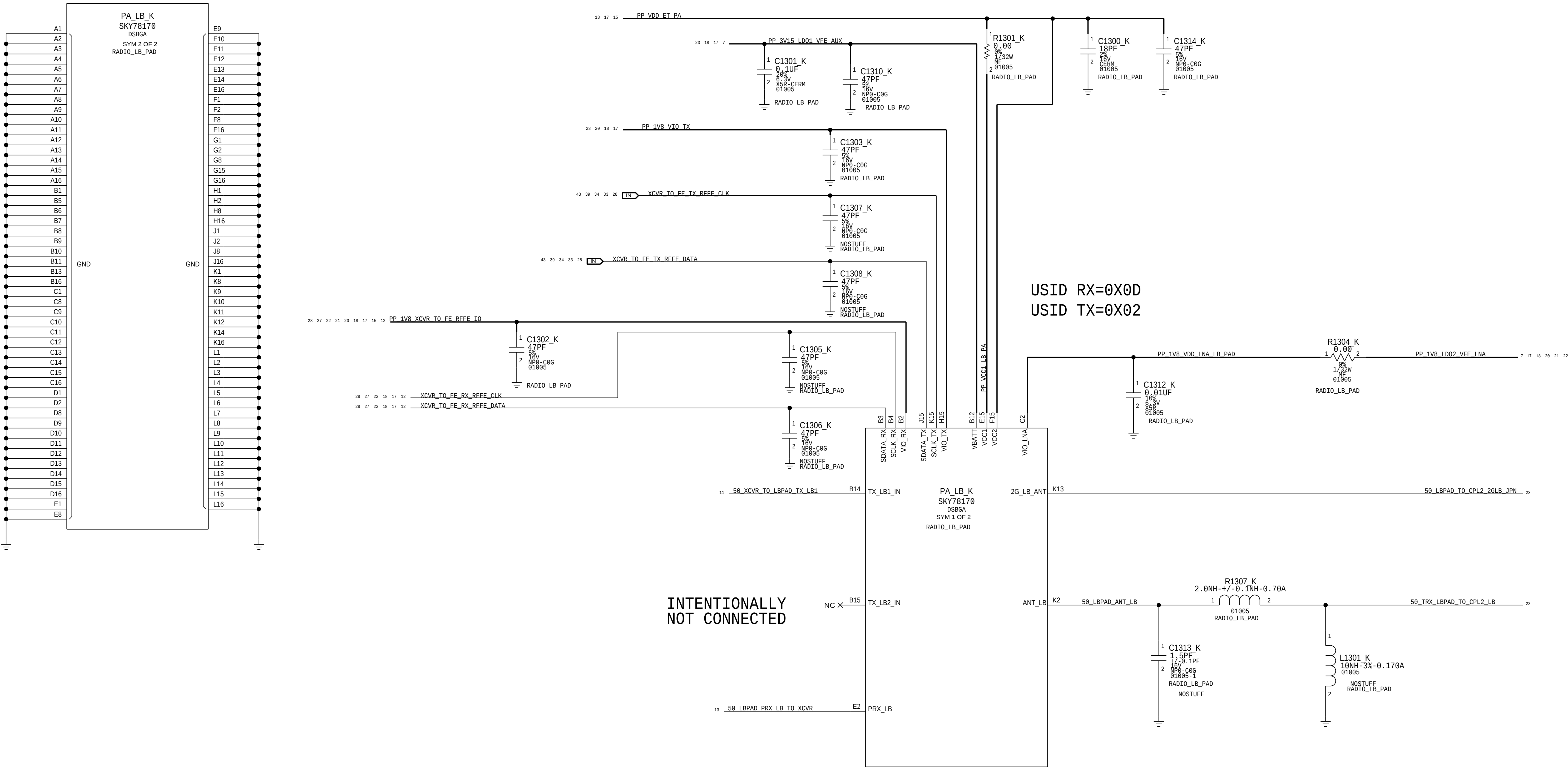
ALPES STX E7.2.1 MODULE



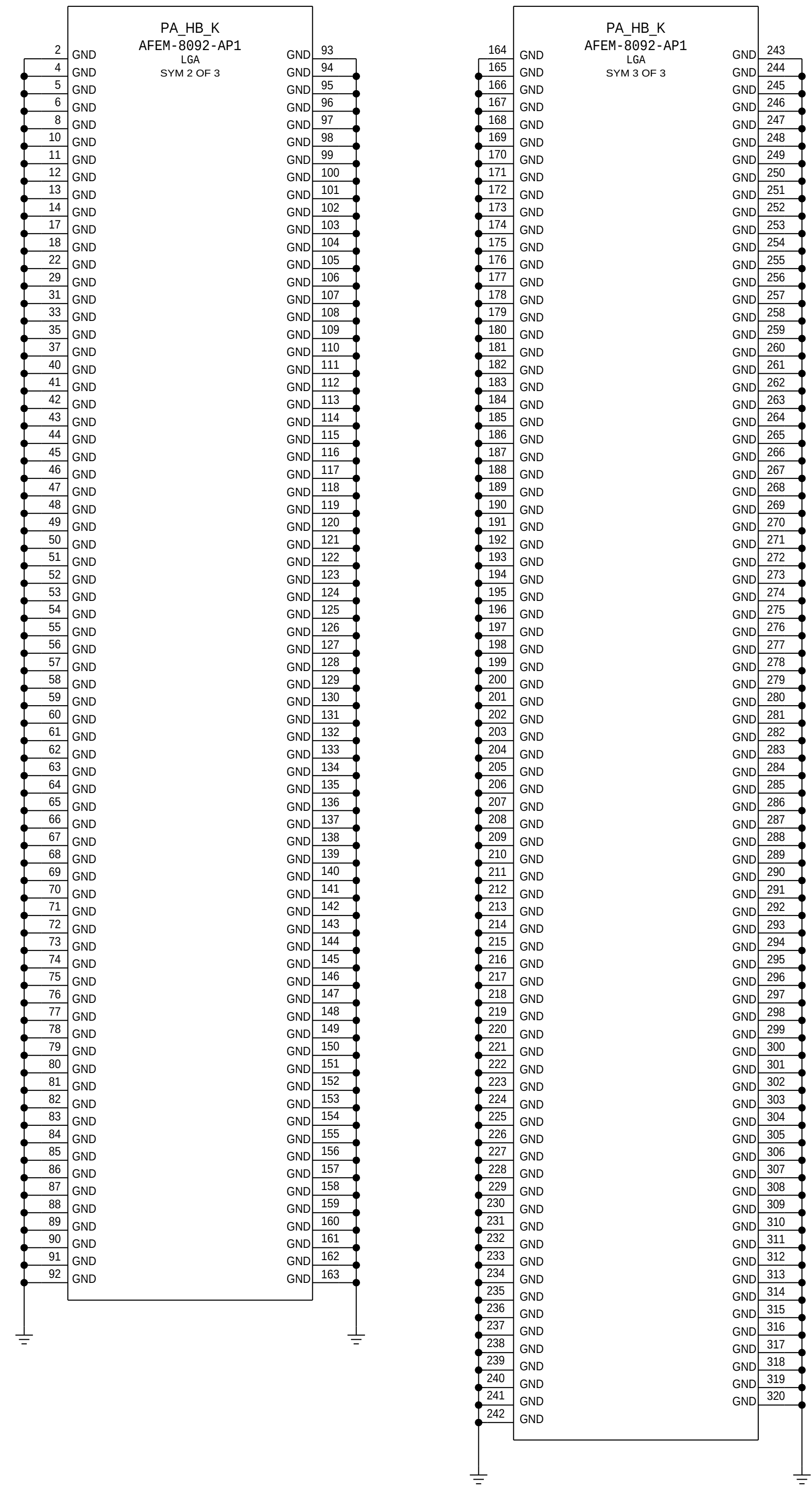
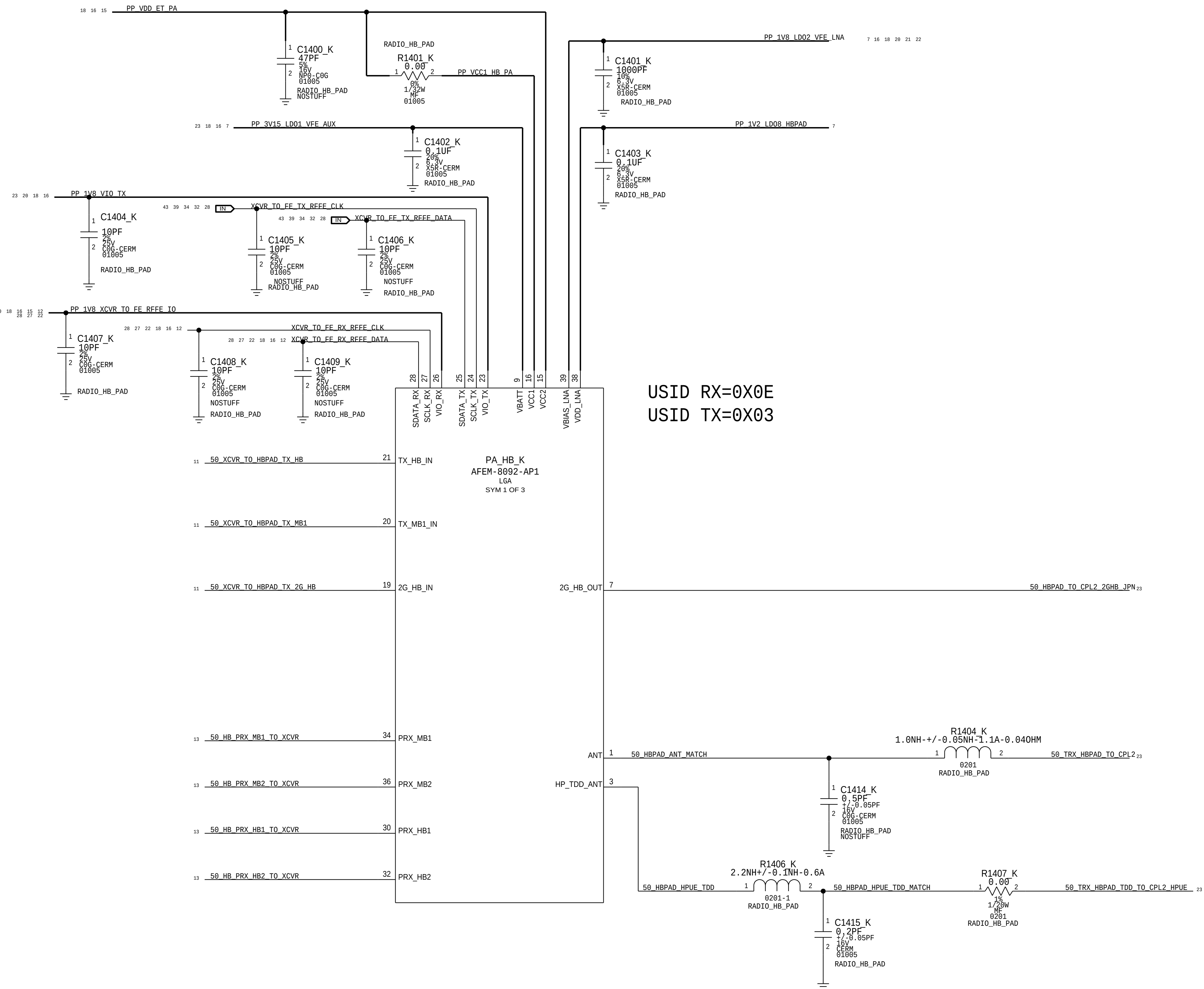
L1200_K APN=152S00811 Z=1.0MM
L1201_K APN=152S00814 Z=1.0MM

C1204_K, C1205_K, C1206_K: KYOCERA APN=138S00167
C1204_K, C1205_K, C1206_K ALTERNATE: MURATA APN=138S00041

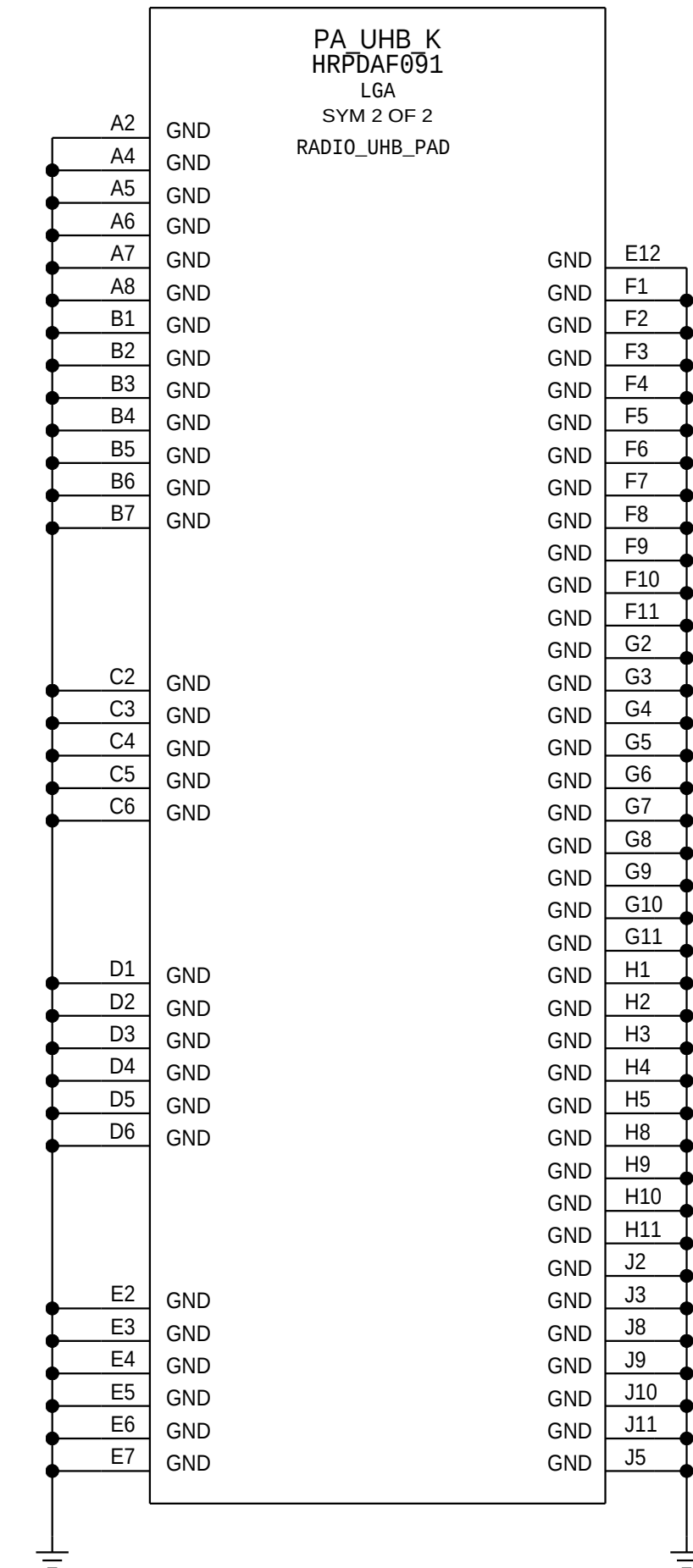
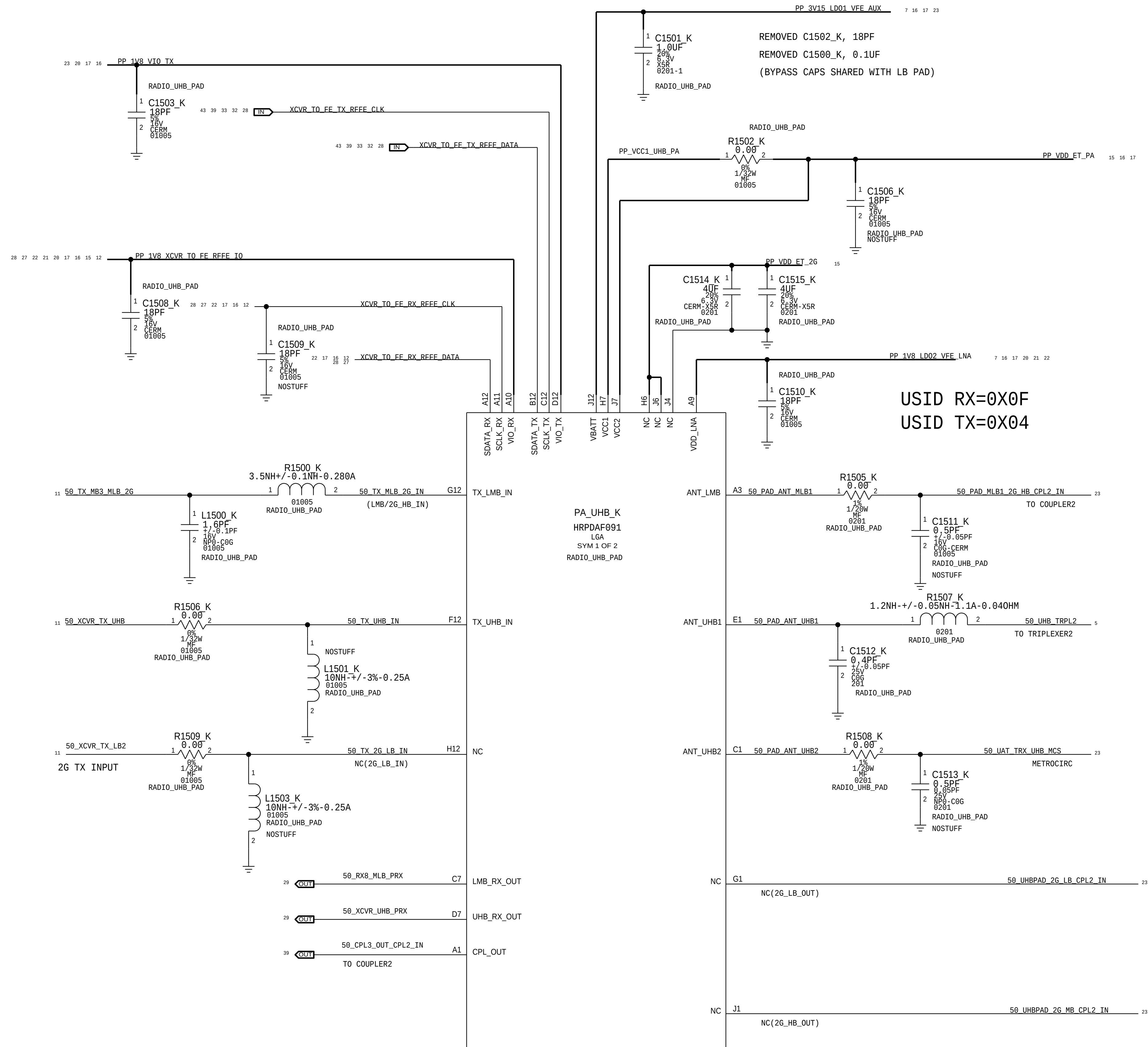
LB SPAD



HB SPAD



UHB LMB SPAD



8

7

6

5

4

3

2

1

BLANK PAGE

D

D

C

C

B

B

A

A

8

7

6

5

4

3

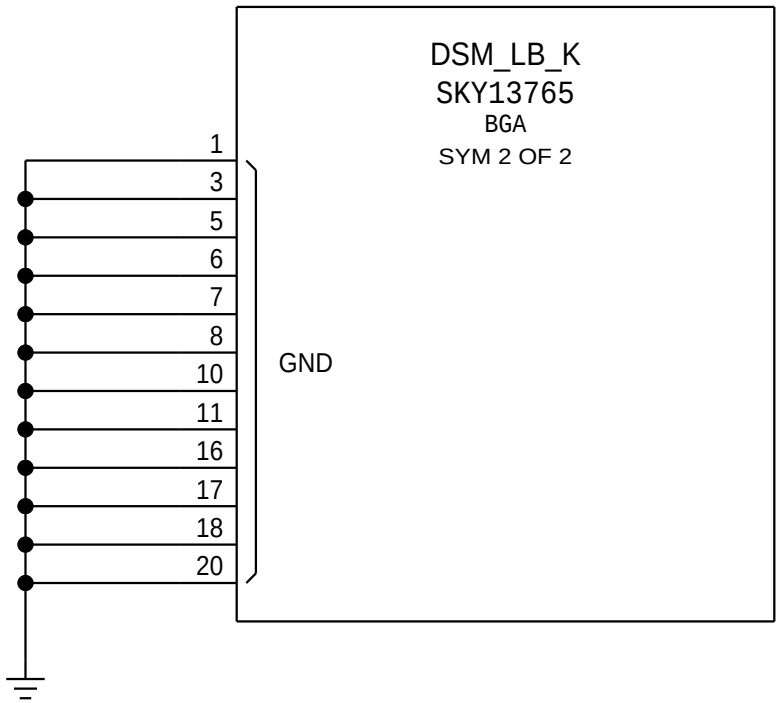
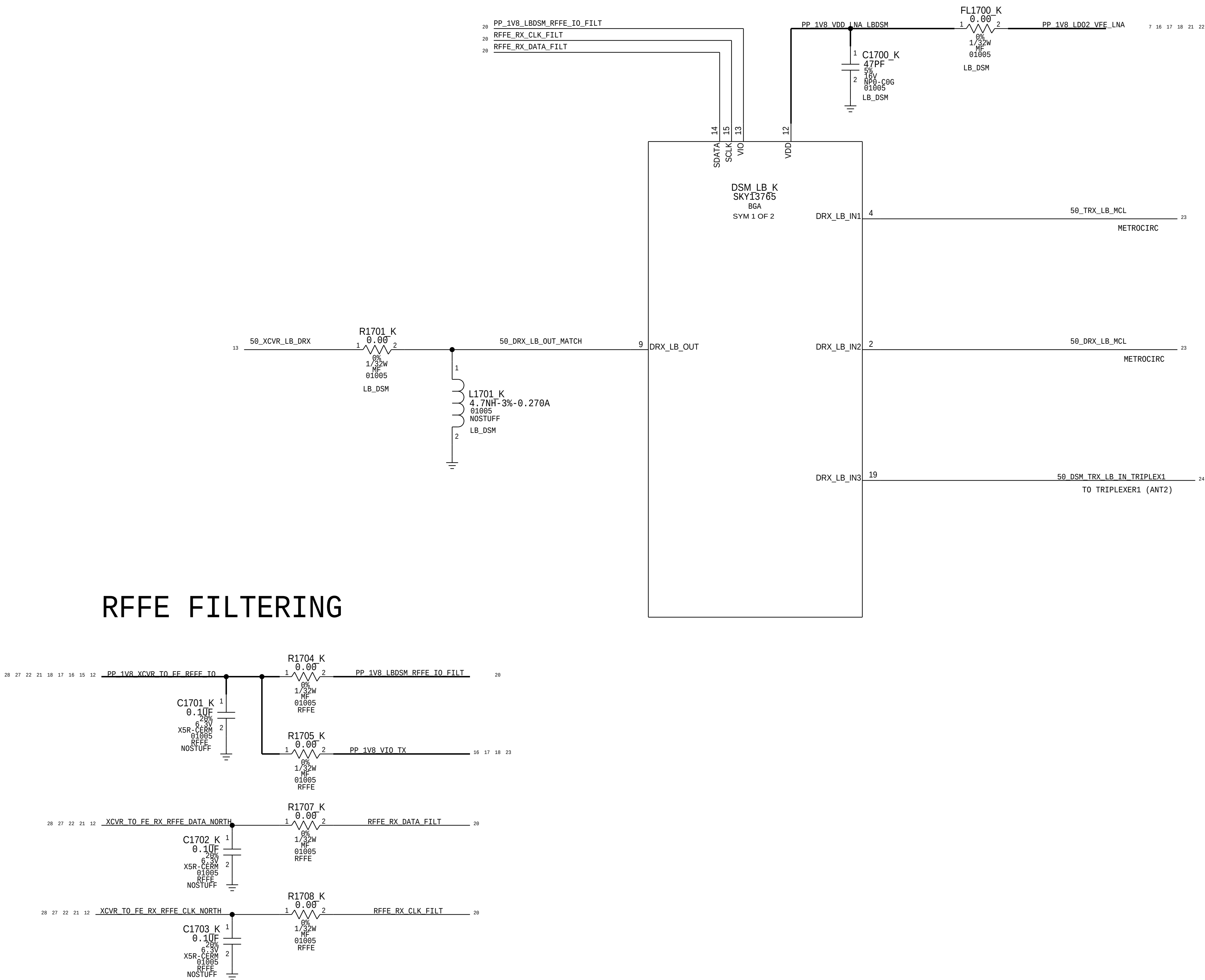
2

1

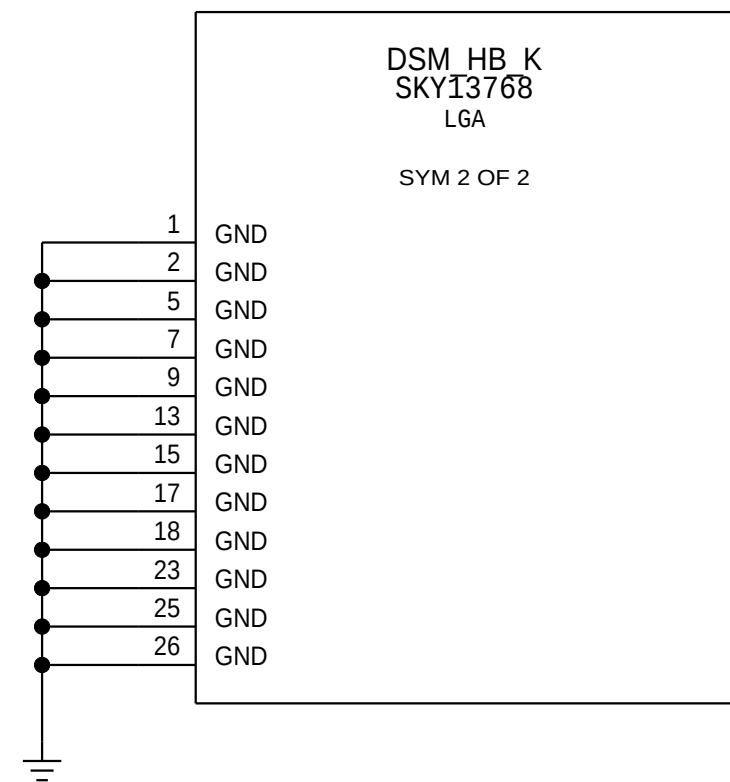
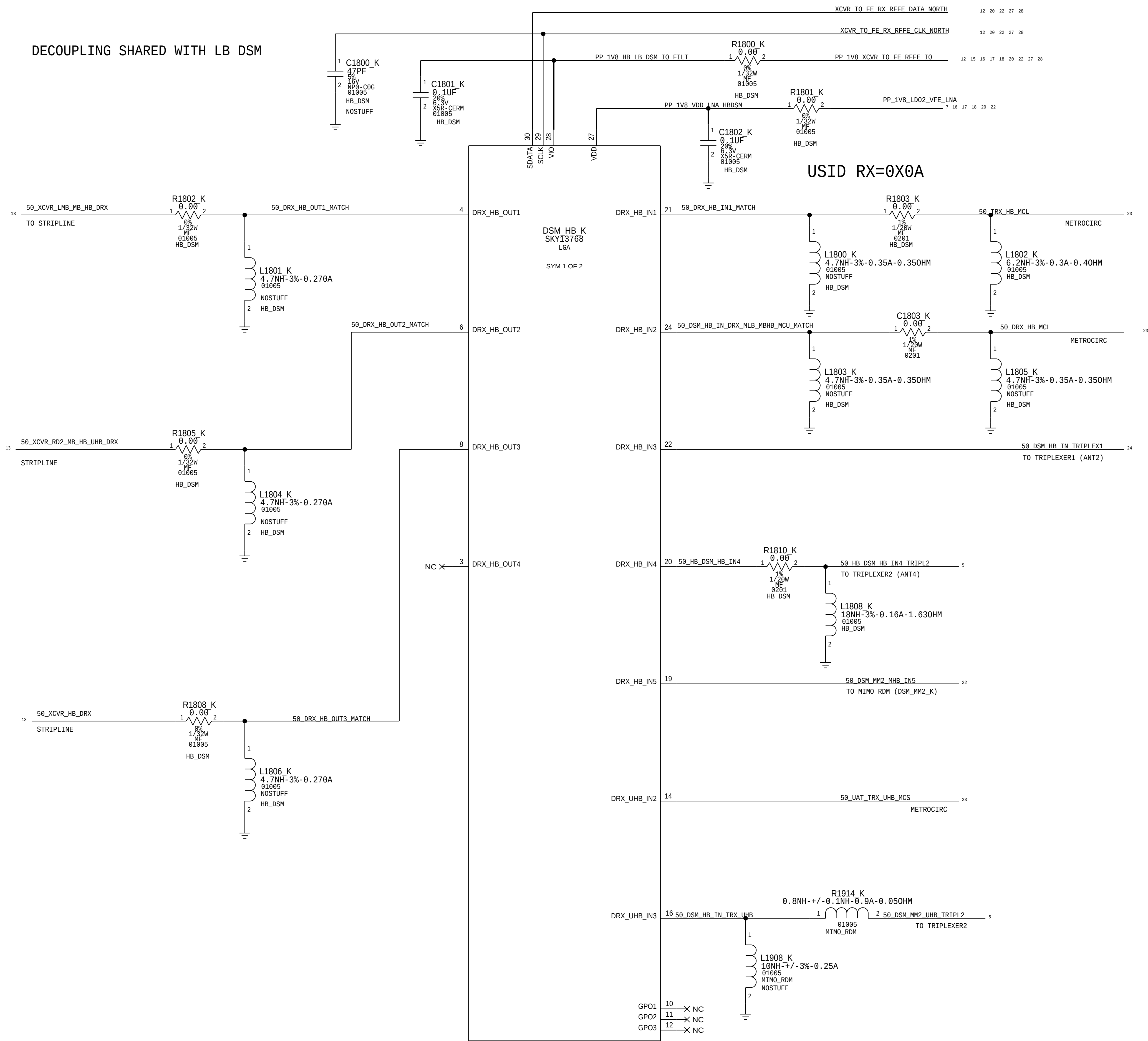
LB DIVERSITY RECEIVE LNA

USID RX=0X09

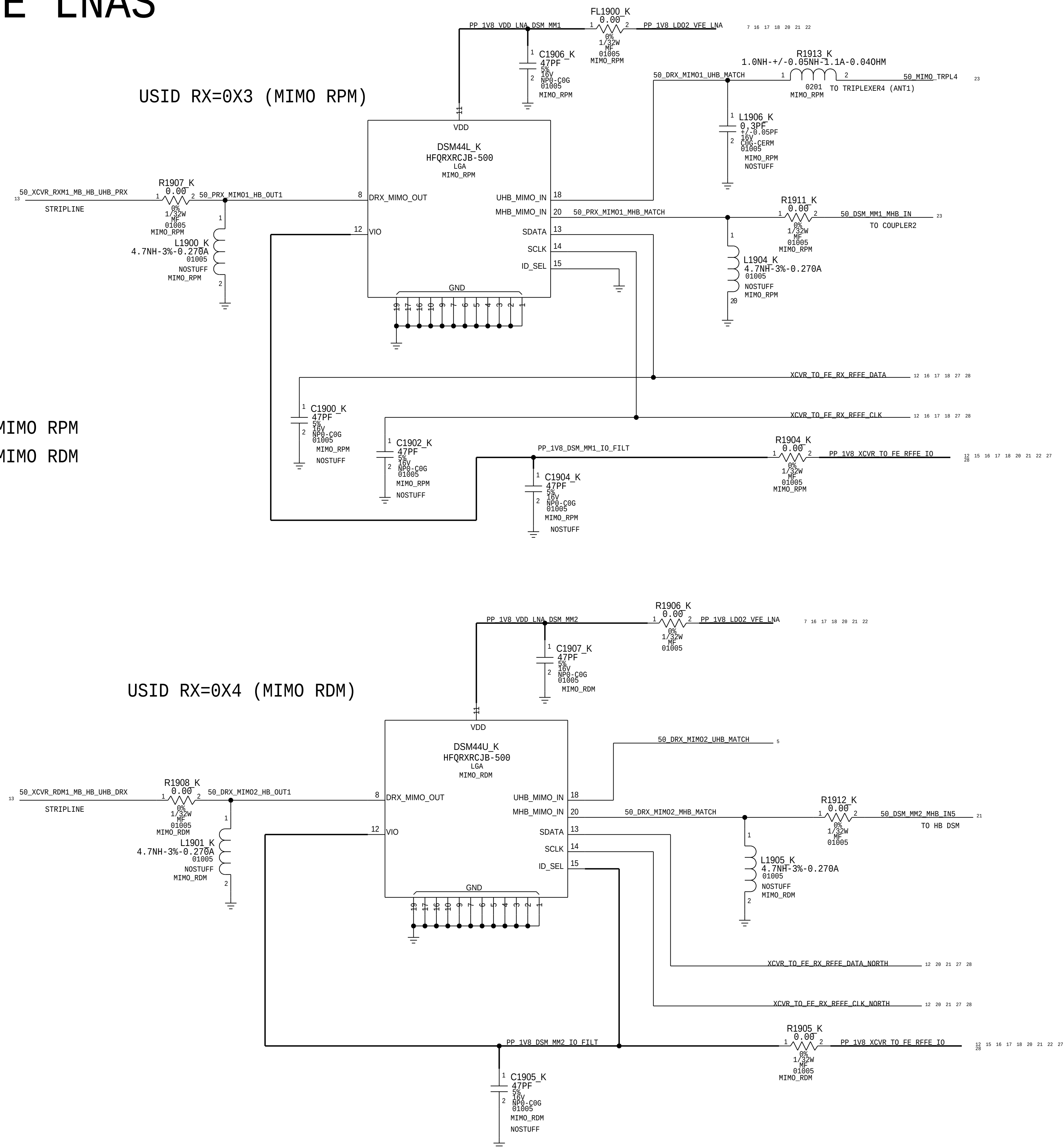
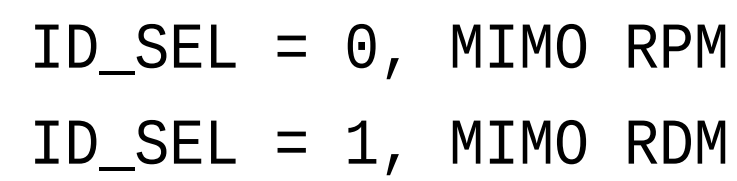
RFFE FILTERING



HB DIVERSITY RECEIVE LNA



MIMO RECEIVE LNAs



LOWER ANTENNA AND COUPLER

USID TX=0X06

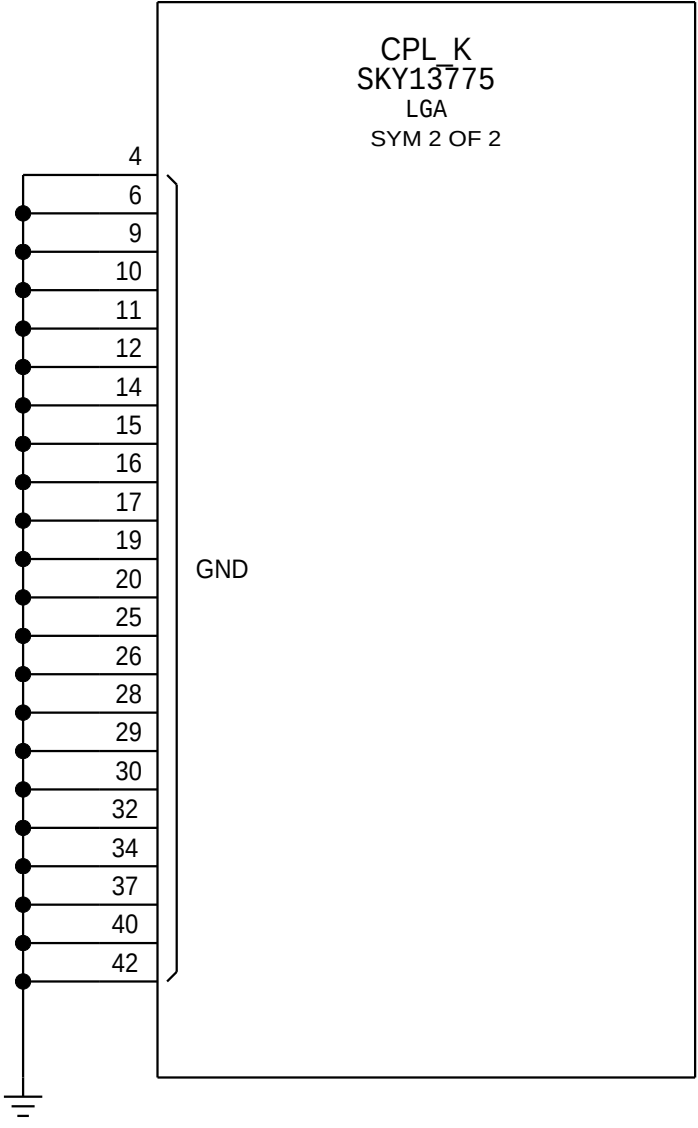
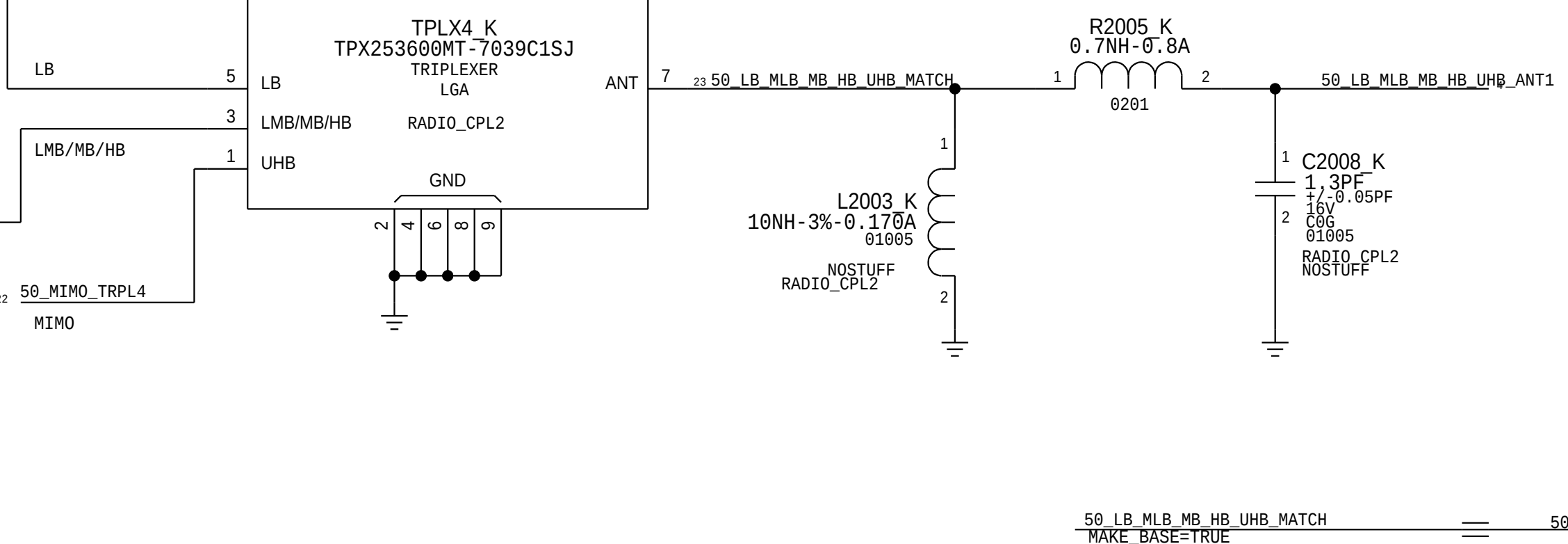
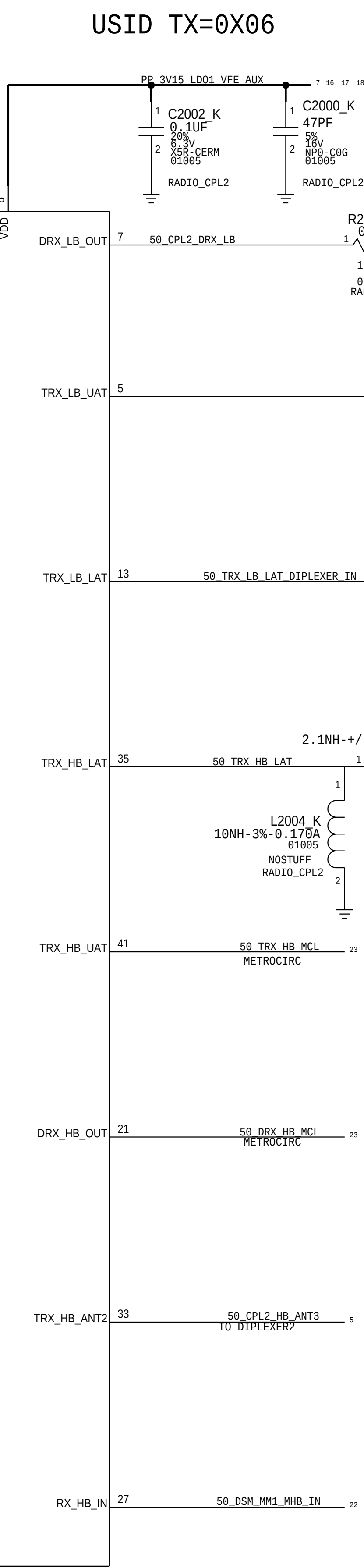
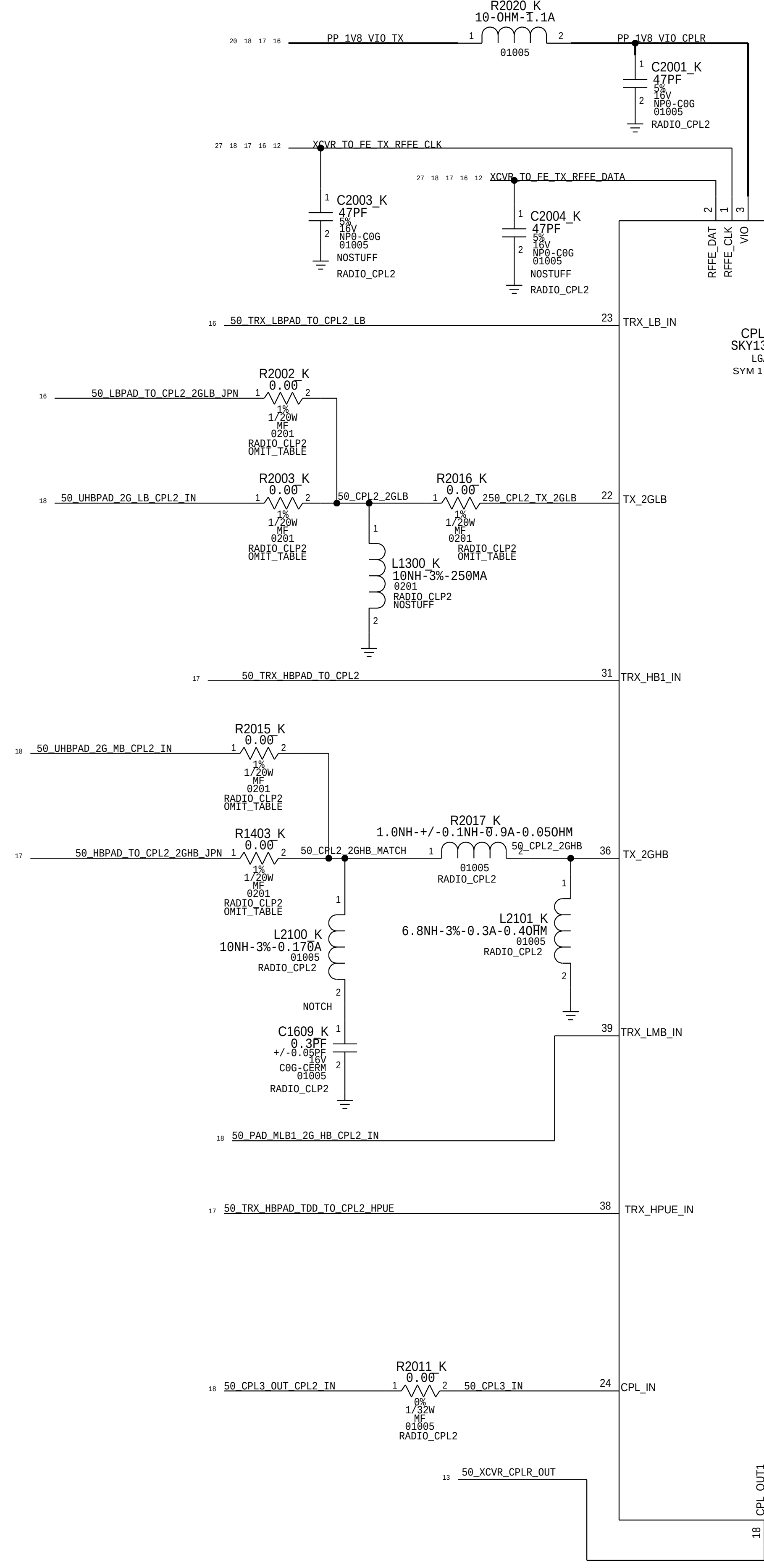
METROCIRC REPLACEMENT
(CPL2 END) (DSM END)

23	50_DRX_LB_MCL	==	50_DRX_LB_MCL	20
23	50_DRX_HB_MCL	==	50_DRX_HB_MCL	21
18	50_UAT_TRX_UHB_MCS	==	50_UAT_TRX_UHB_MCS	21
23	50_TRX_HB_MCL	==	50_TRX_HB_MCL	21
23	50_TRX_LB_MCL	==	50_TRX_LB_MCL	20

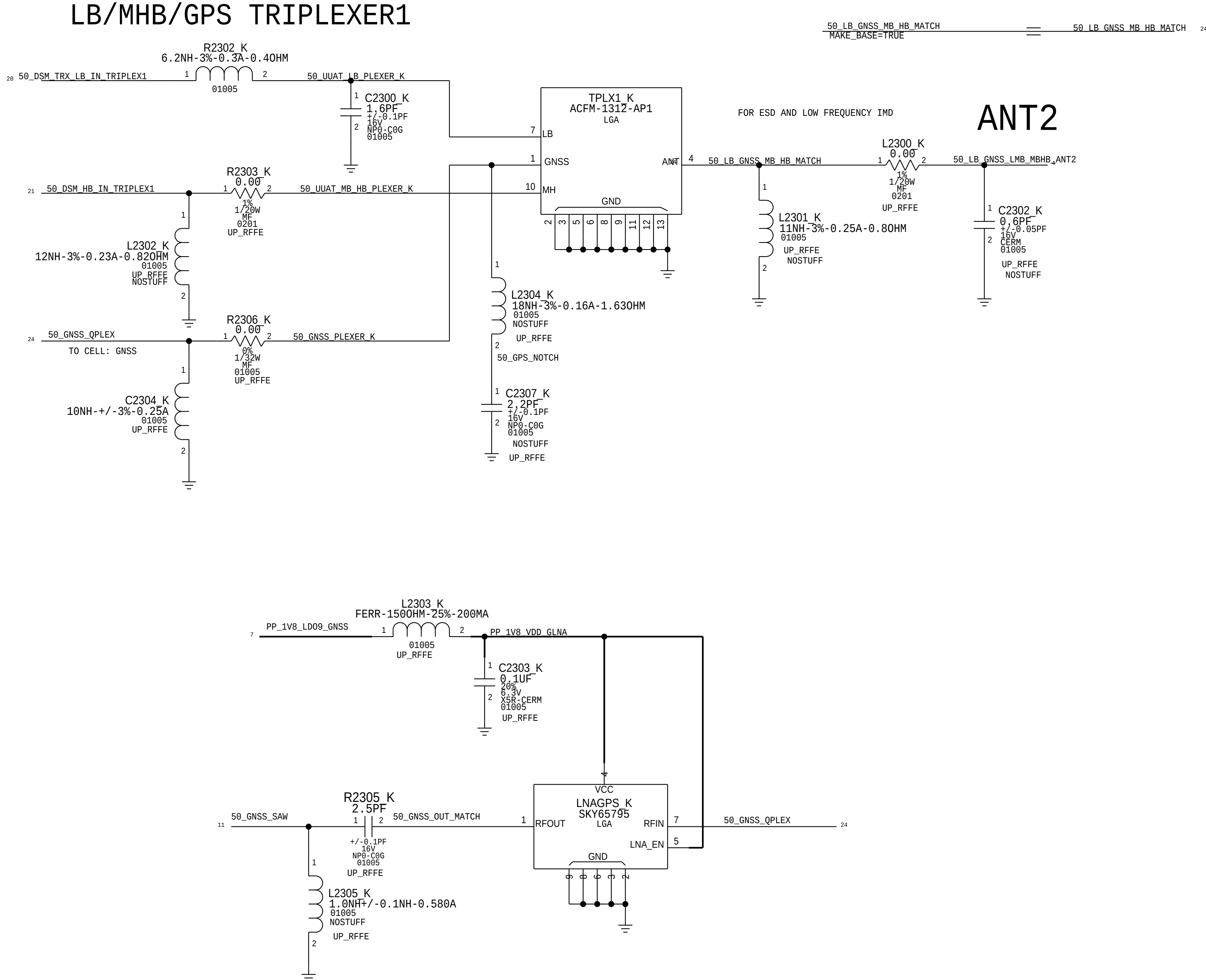
ANT1

LB-MHB-UHB TRIPLEXER4 (ANT1)

ALIAS FOR LAYOUT PURPOSES



UPPER ANTENNA FEEDS



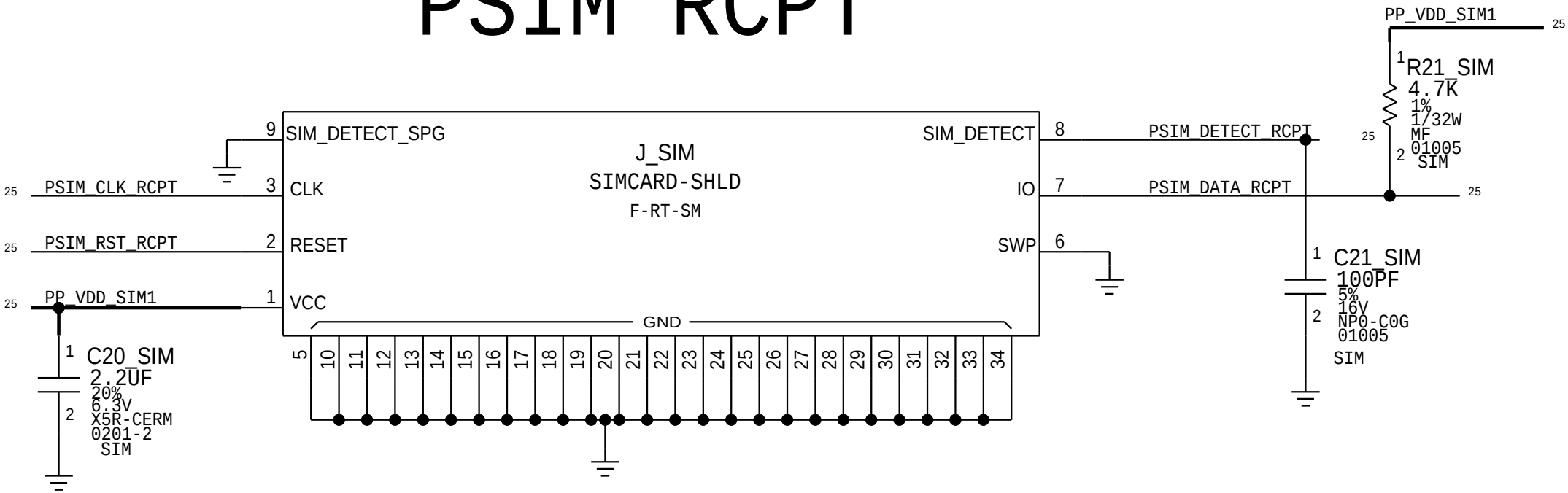
PSIM

```
27 8 SIM1_CLK == SIM1_CLK 25
    MAKE_BASE=TRUE
27 8 SIM1_DATA == SIM1_DATA 25
    MAKE_BASE=TRUE
27 8 SIM1_RST == SIM1_RST 25
    MAKE_BASE=TRUE
27 8 SIM1_DETECT == SIM1_DETECT 25
    MAKE_BASE=TRUE
27 10 7 PP_VDD_SIM1 == PP_VDD_SIM1 25
    MAKE_BASE=TRUE
```

ICE18.1 PHYSICAL SIM

ICE17 PART OFFERS MORE SHIELDING AND PLACEMENT AREA

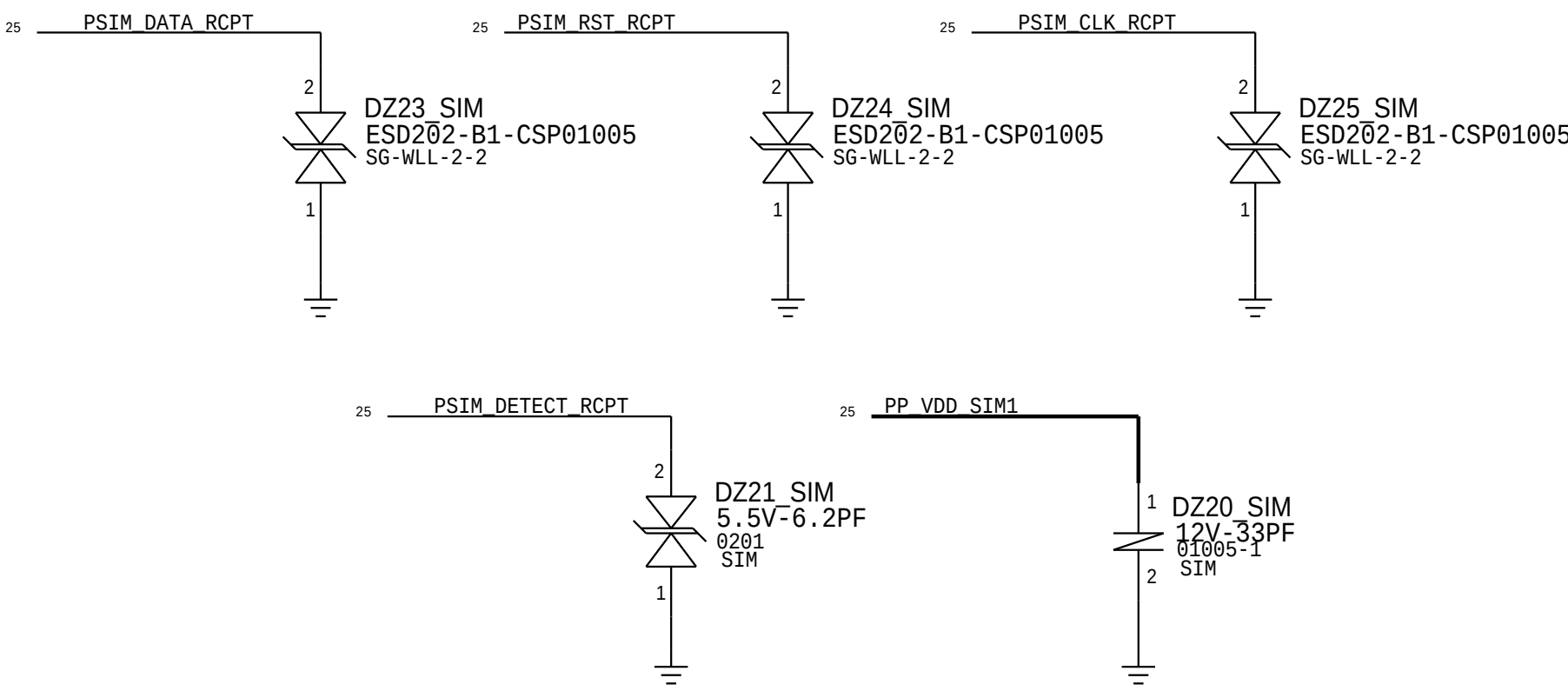
PSIM RCPT



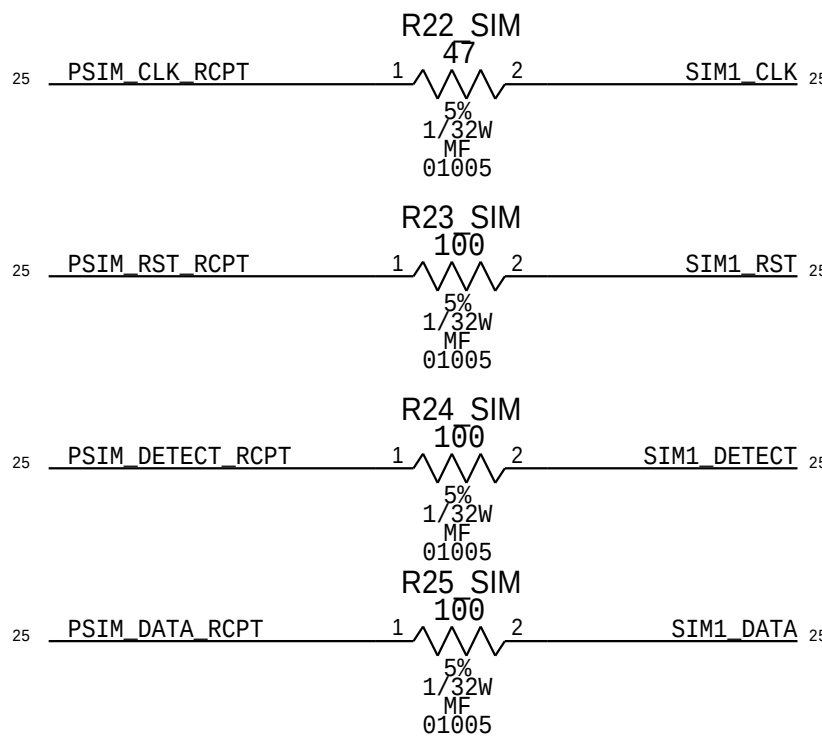
ICE18.1 ONLY

APN:512S00030

ESD DIODES

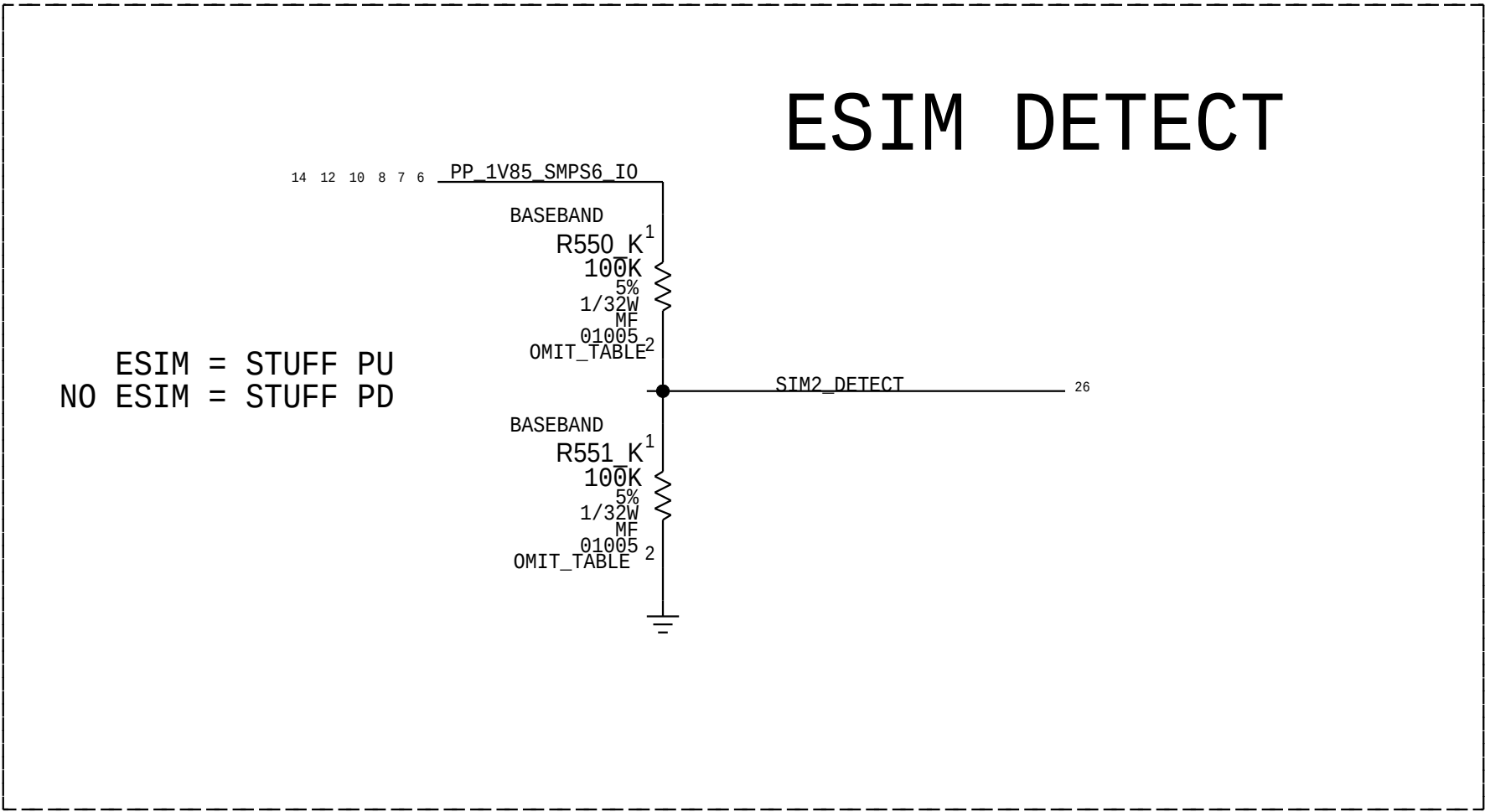
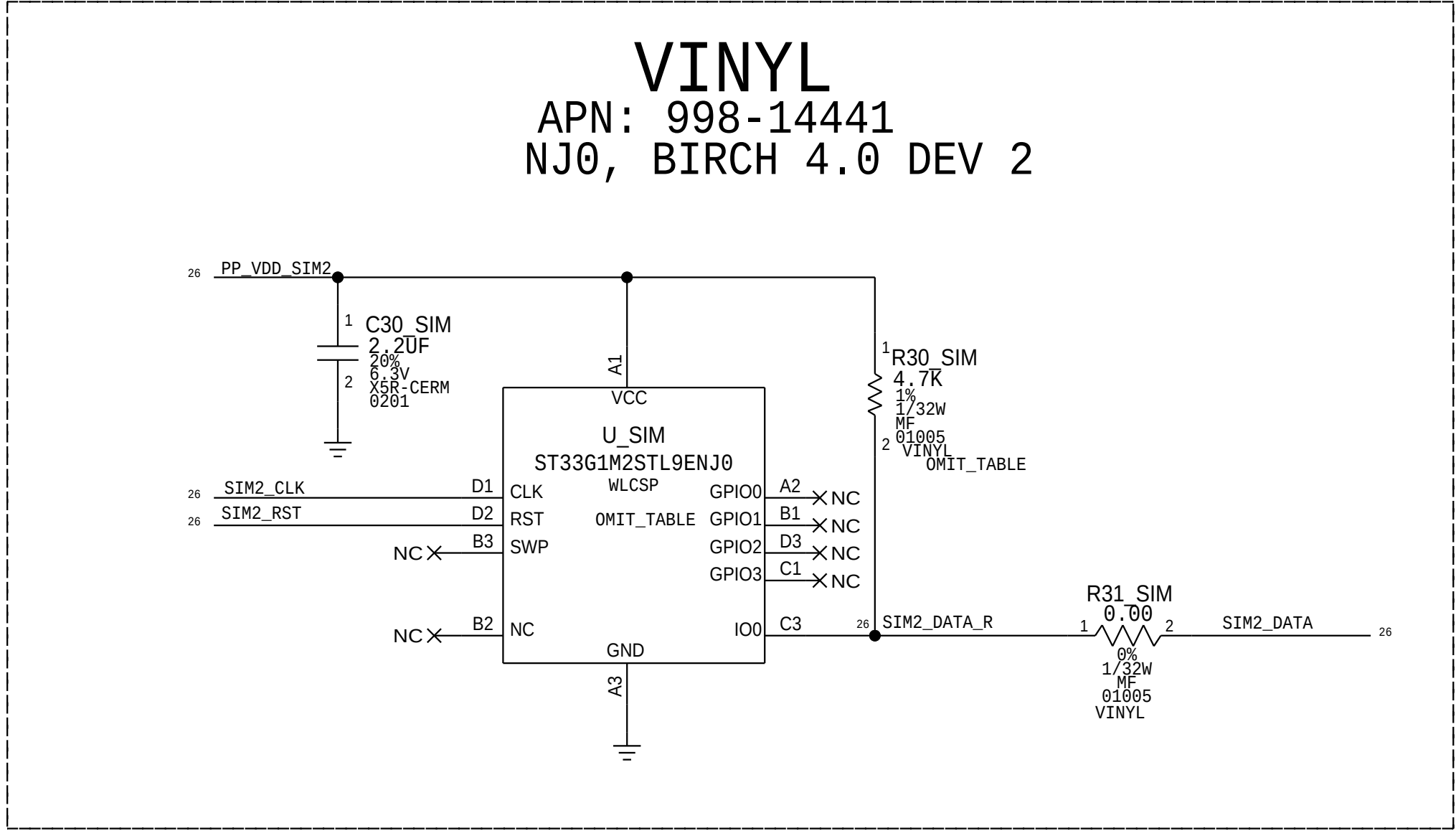


EOS RESISTORS

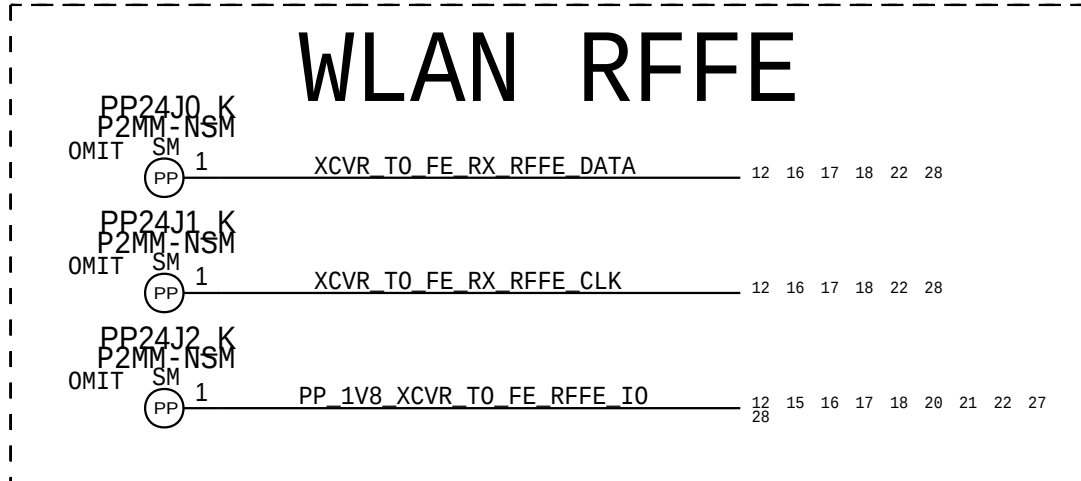
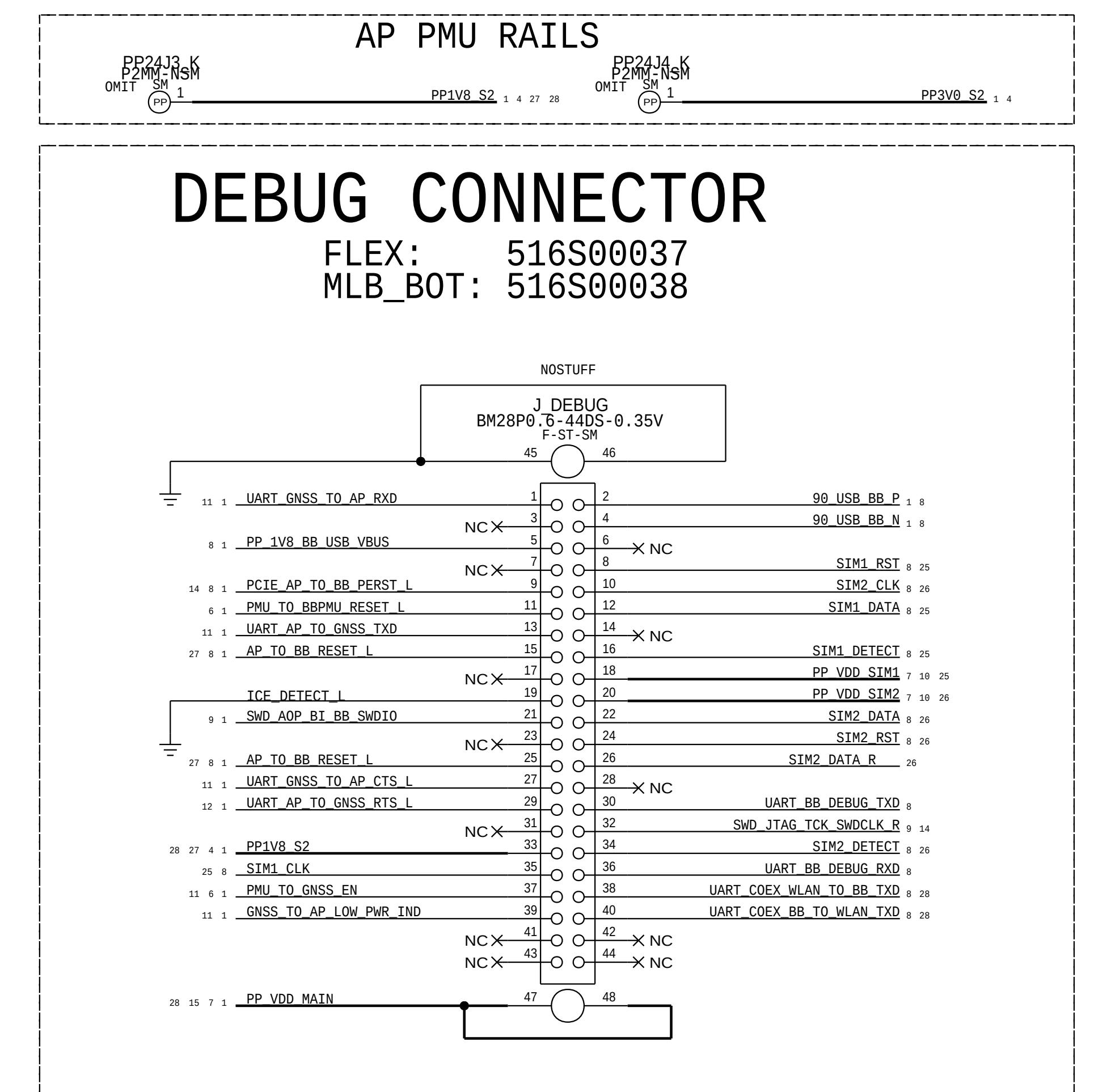
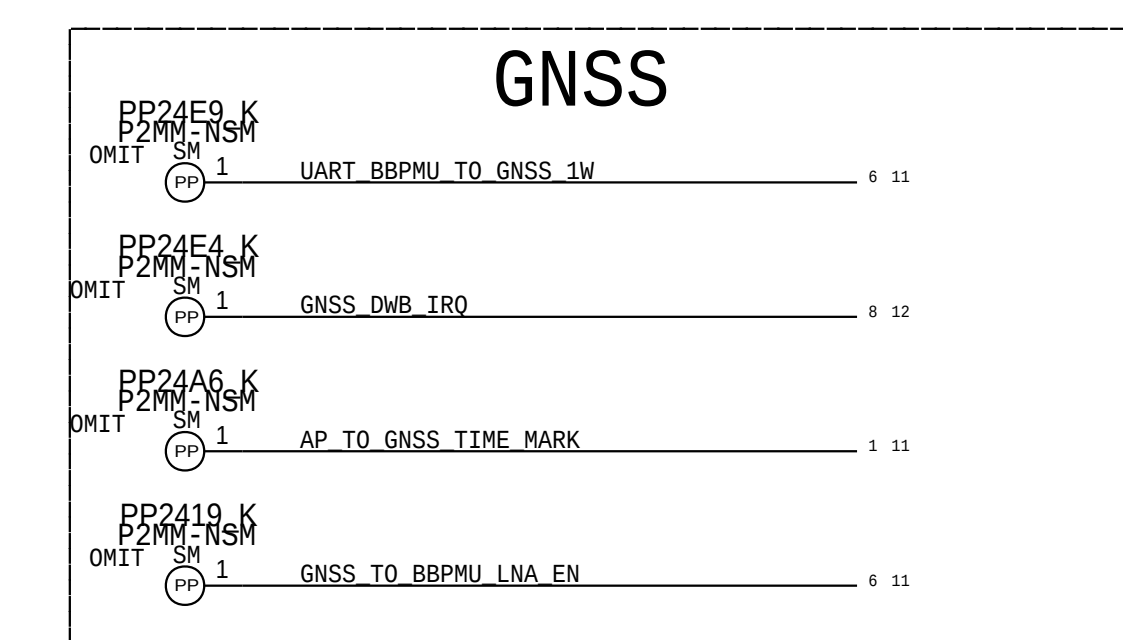
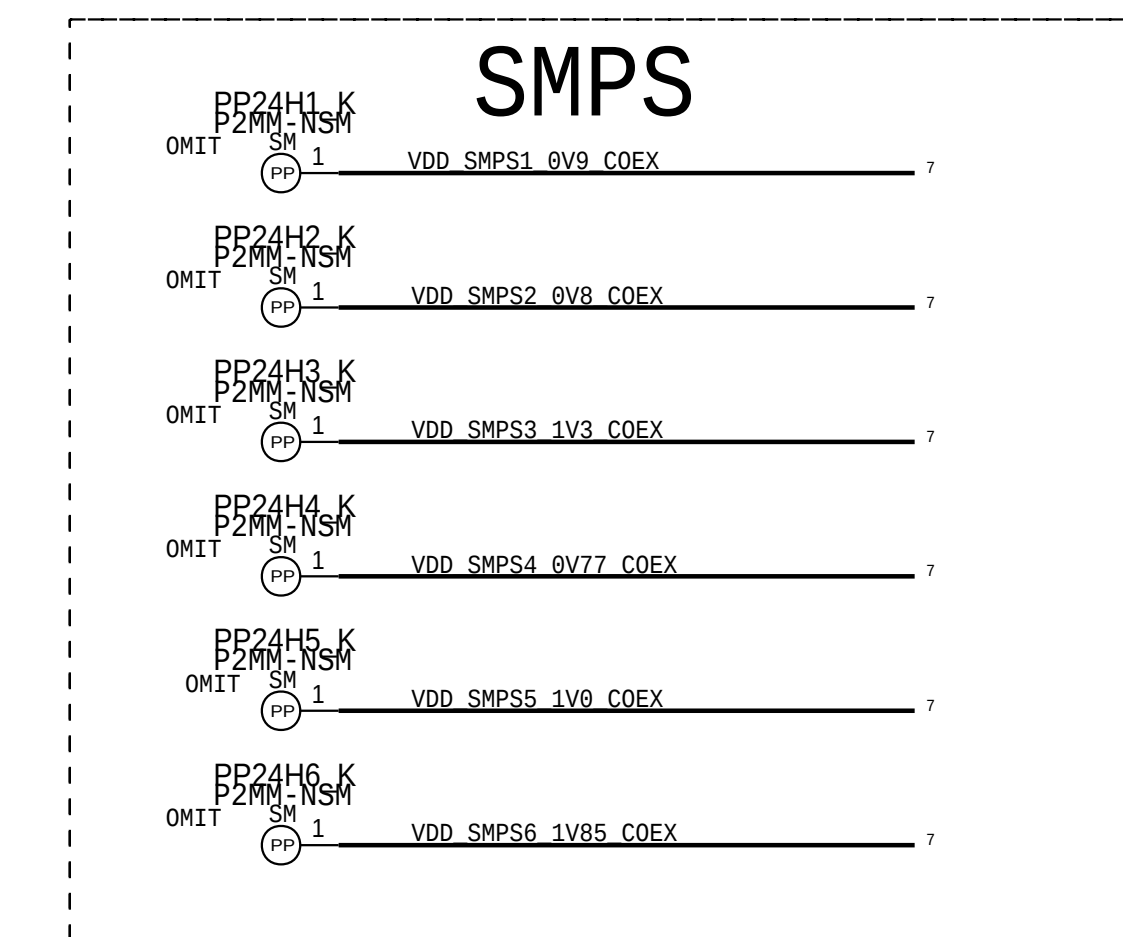
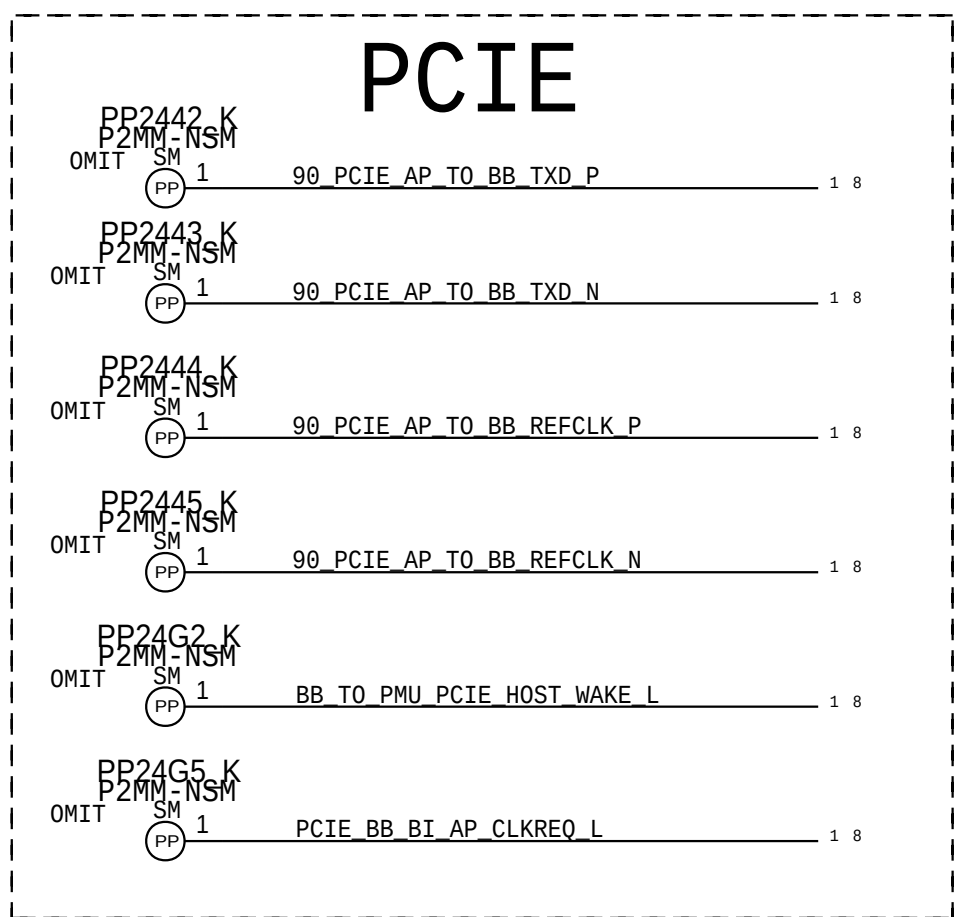
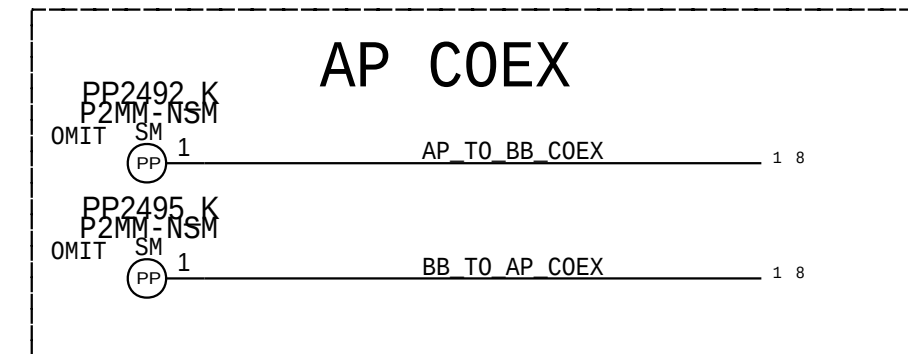
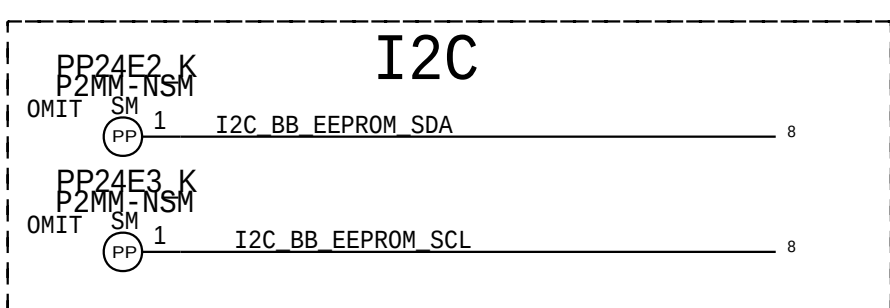
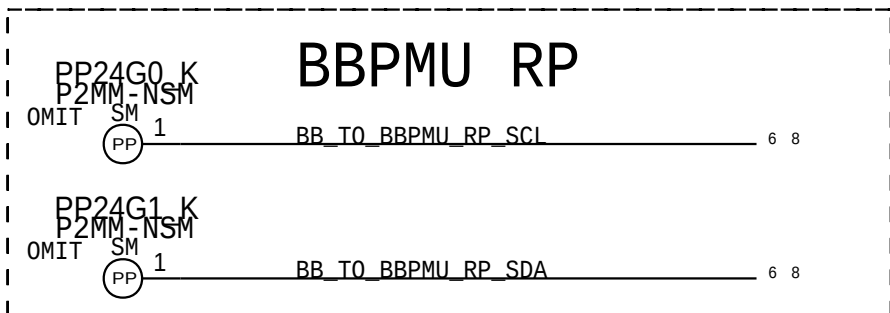
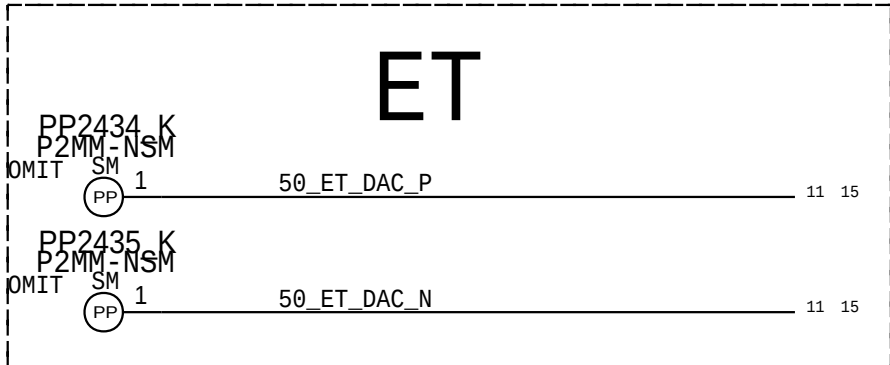
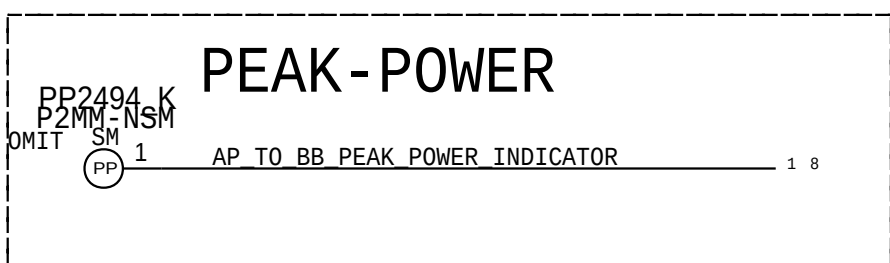
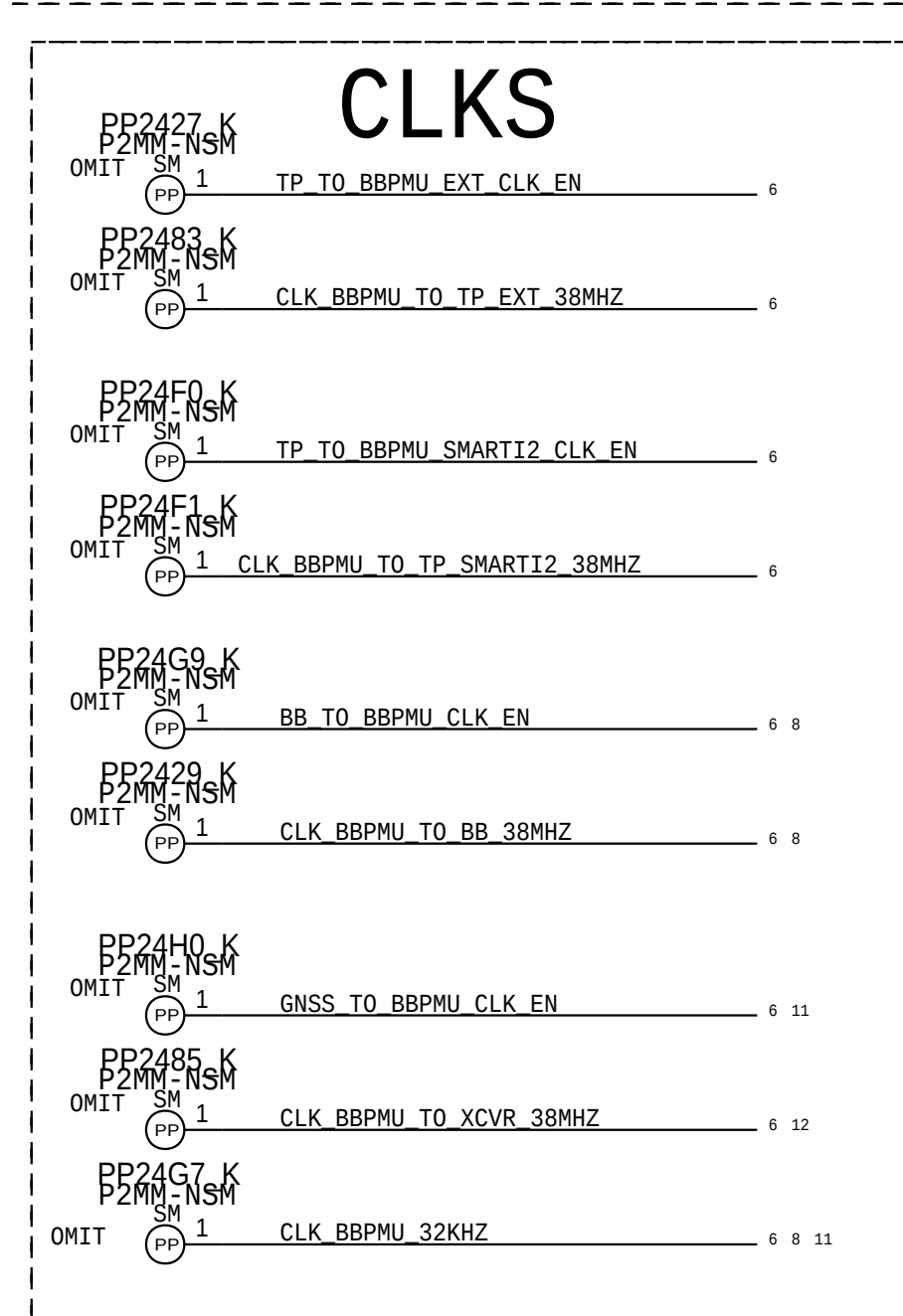
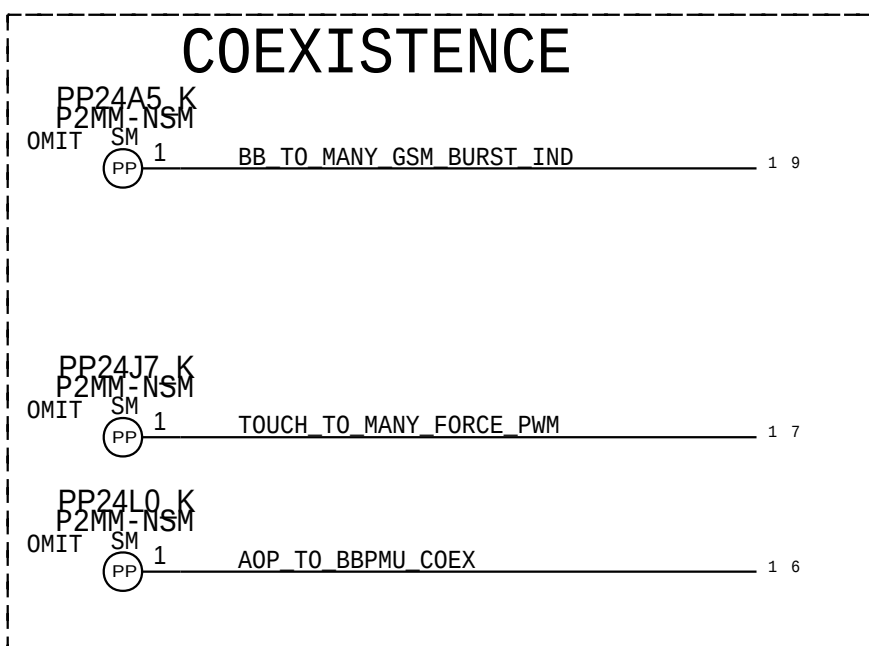
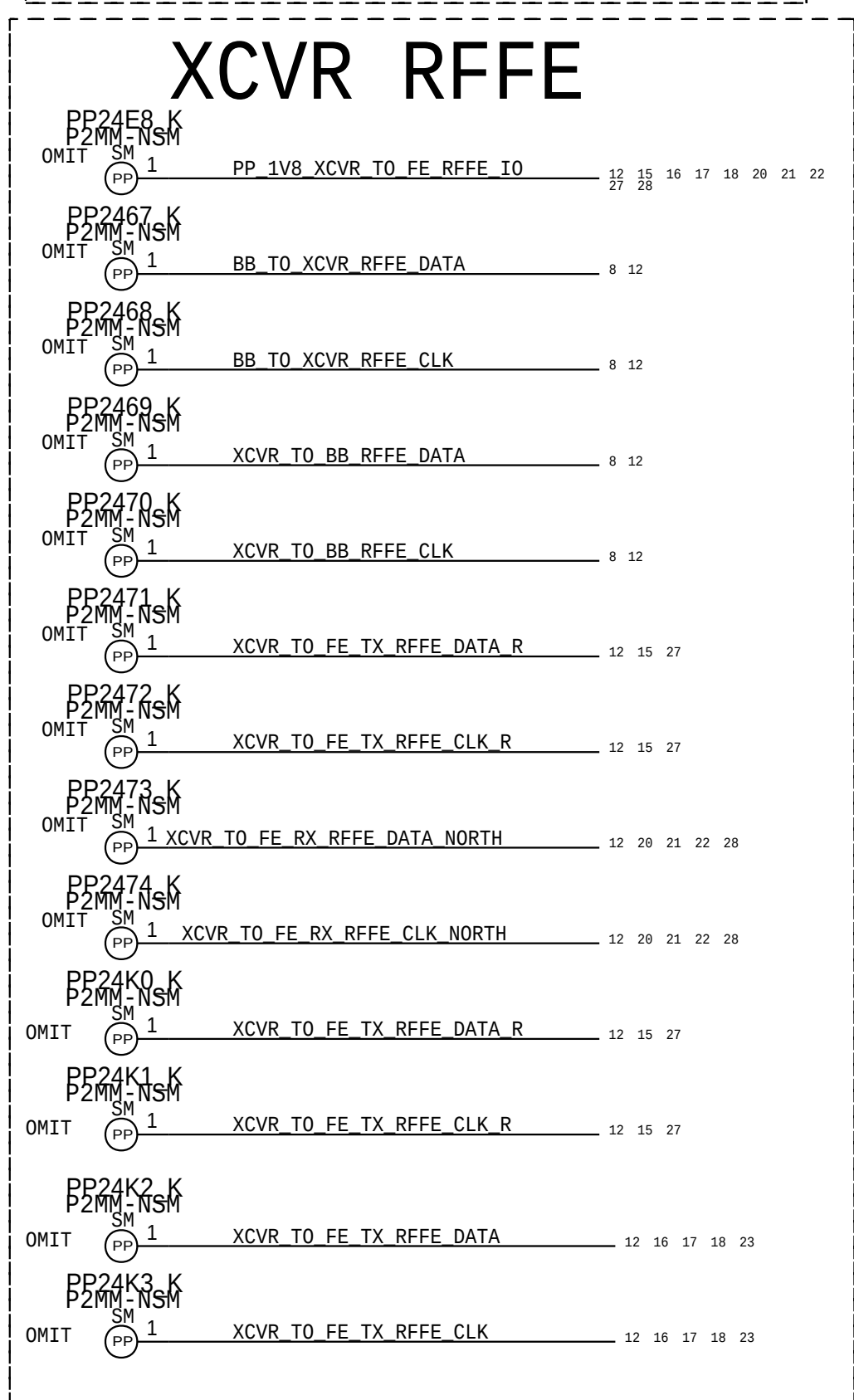
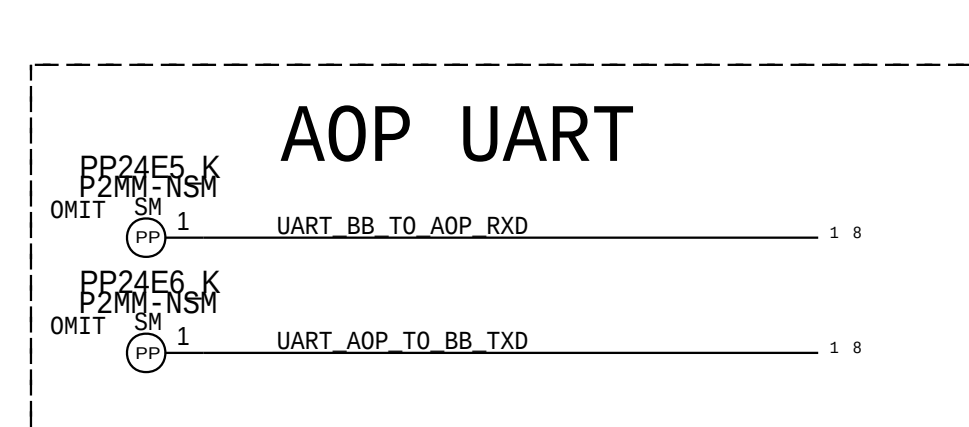
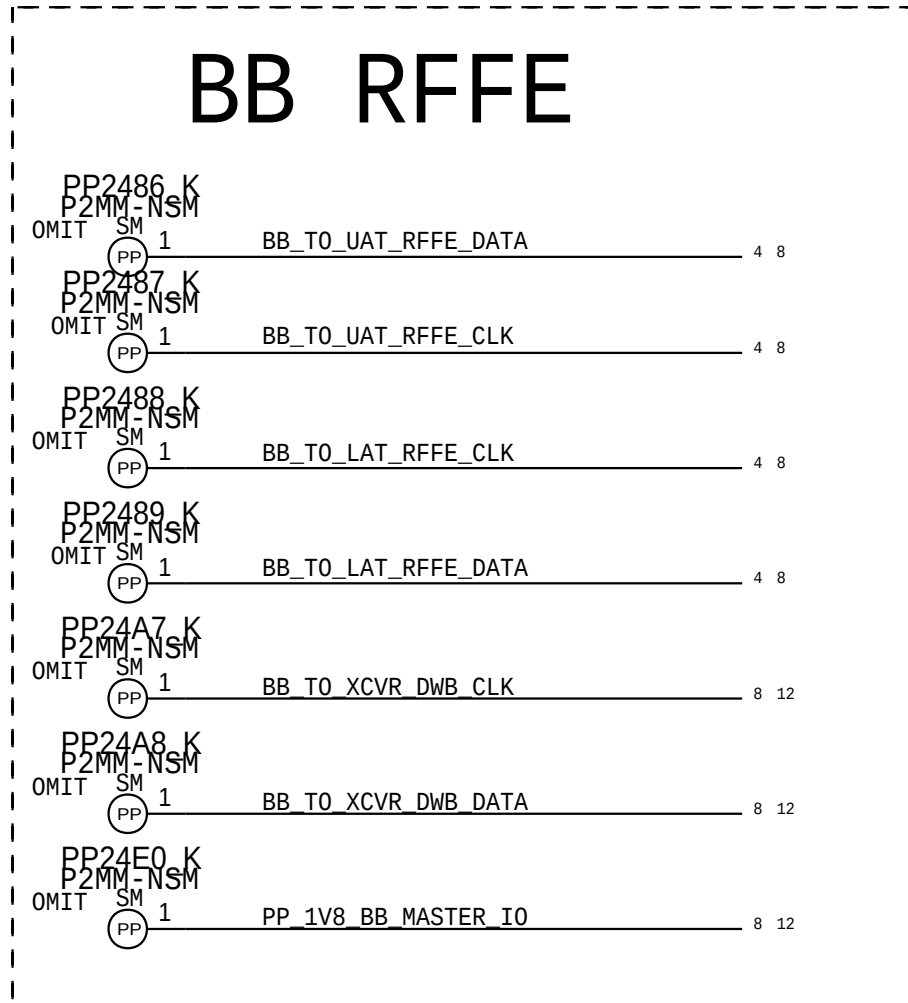
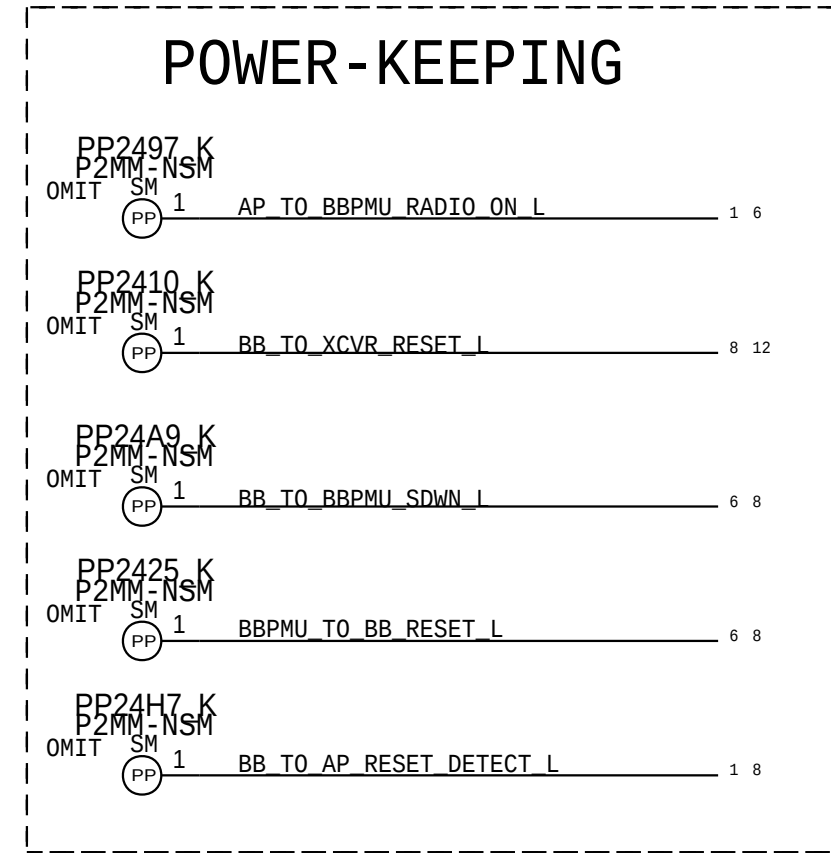
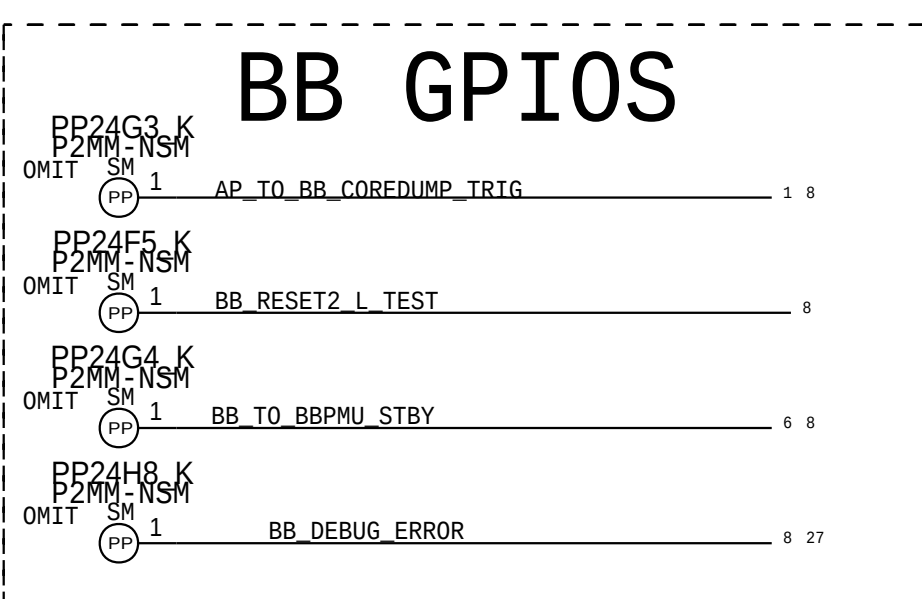
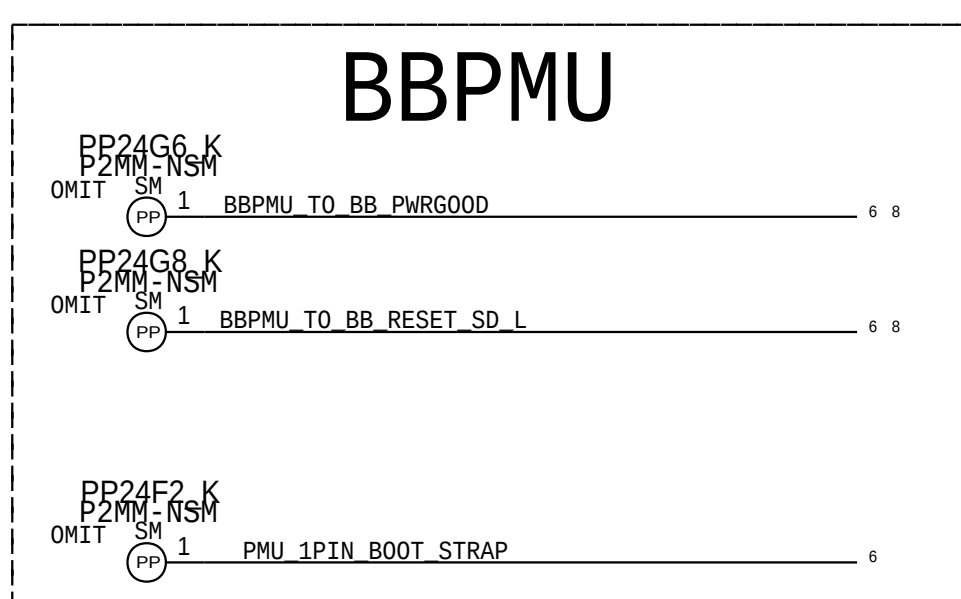
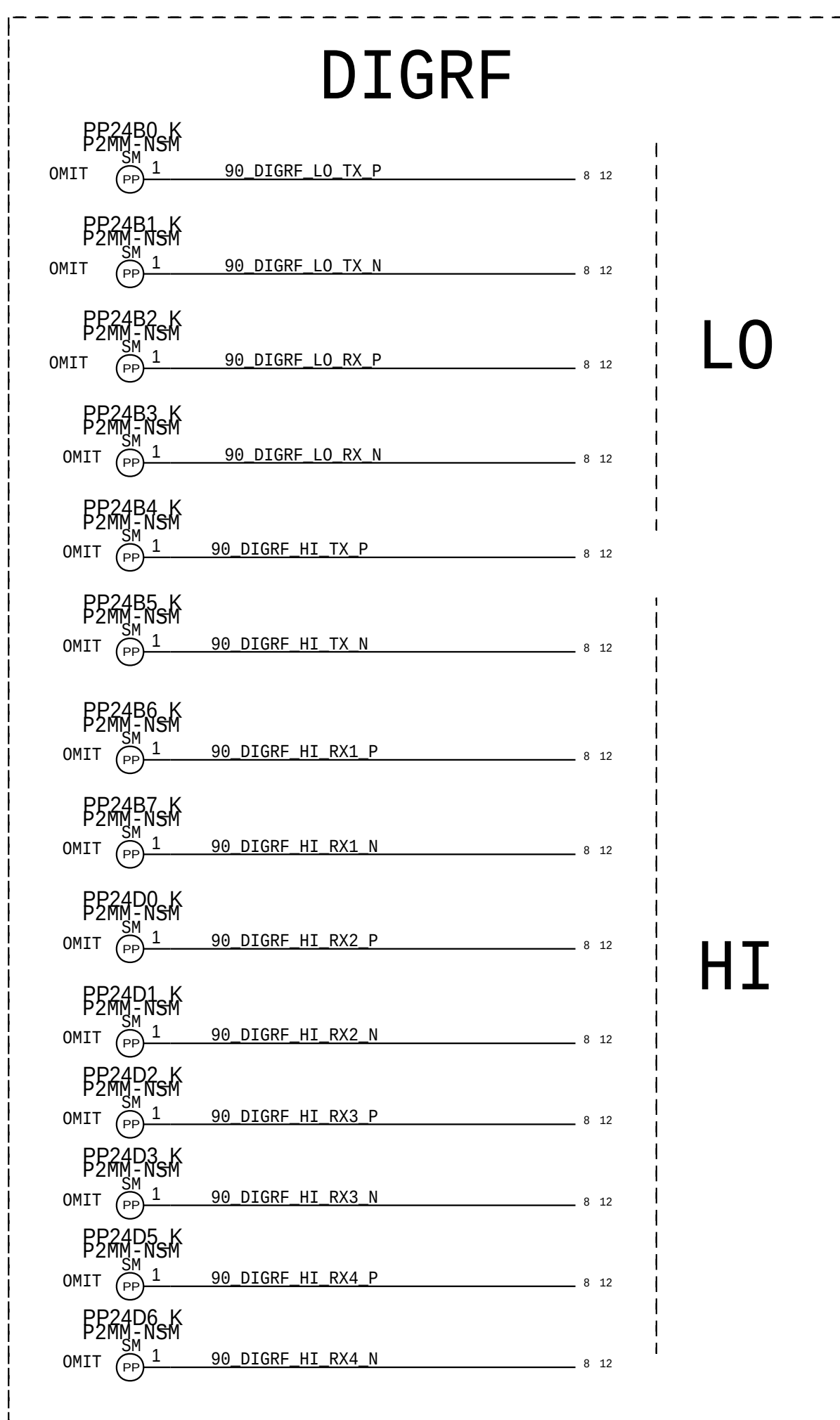
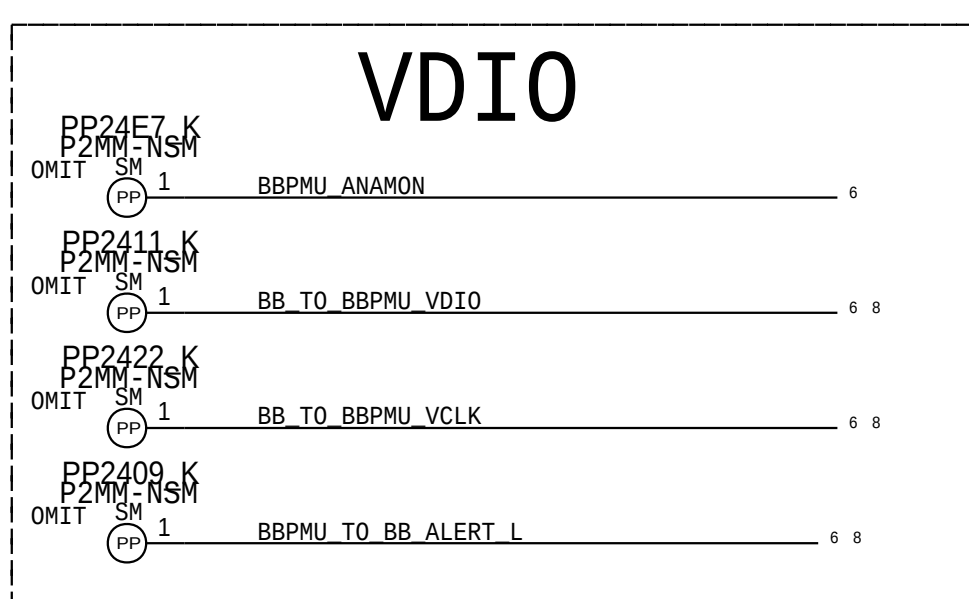
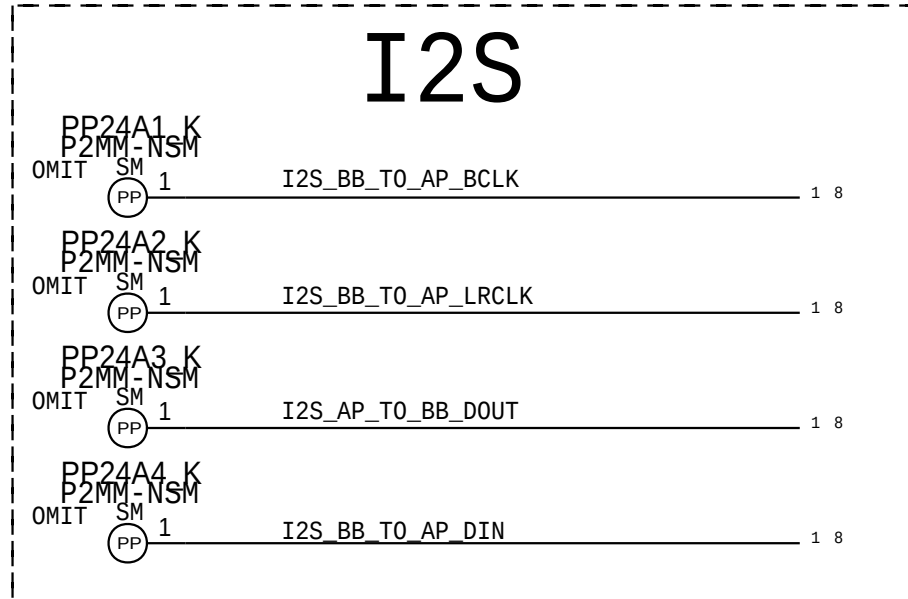
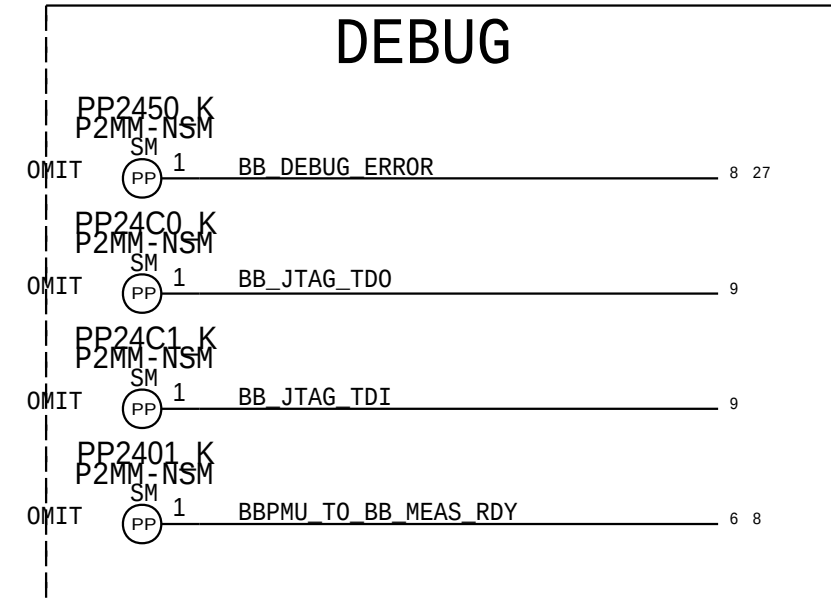


ESIM

27	8	SIM2_CLK	==	SIM2_CLK	26
		MAKE_BASE=TRUE			
27	8	SIM2_DATA	==	SIM2_DATA	26
		MAKE_BASE=TRUE			
27	8	SIM2_RST	==	SIM2_RST	26
		MAKE_BASE=TRUE			
27	8	SIM2_DETECT	==	SIM2_DETECT	26
		MAKE_BASE=TRUE			
27	10	PP_VDD_SIM2	==	PP_VDD_SIM2	26
		MAKE_BASE=TRUE			
27	7	SIM2_DATA_R	==	SIM2_DATA_R	26
		MAKE_BASE=TRUE			

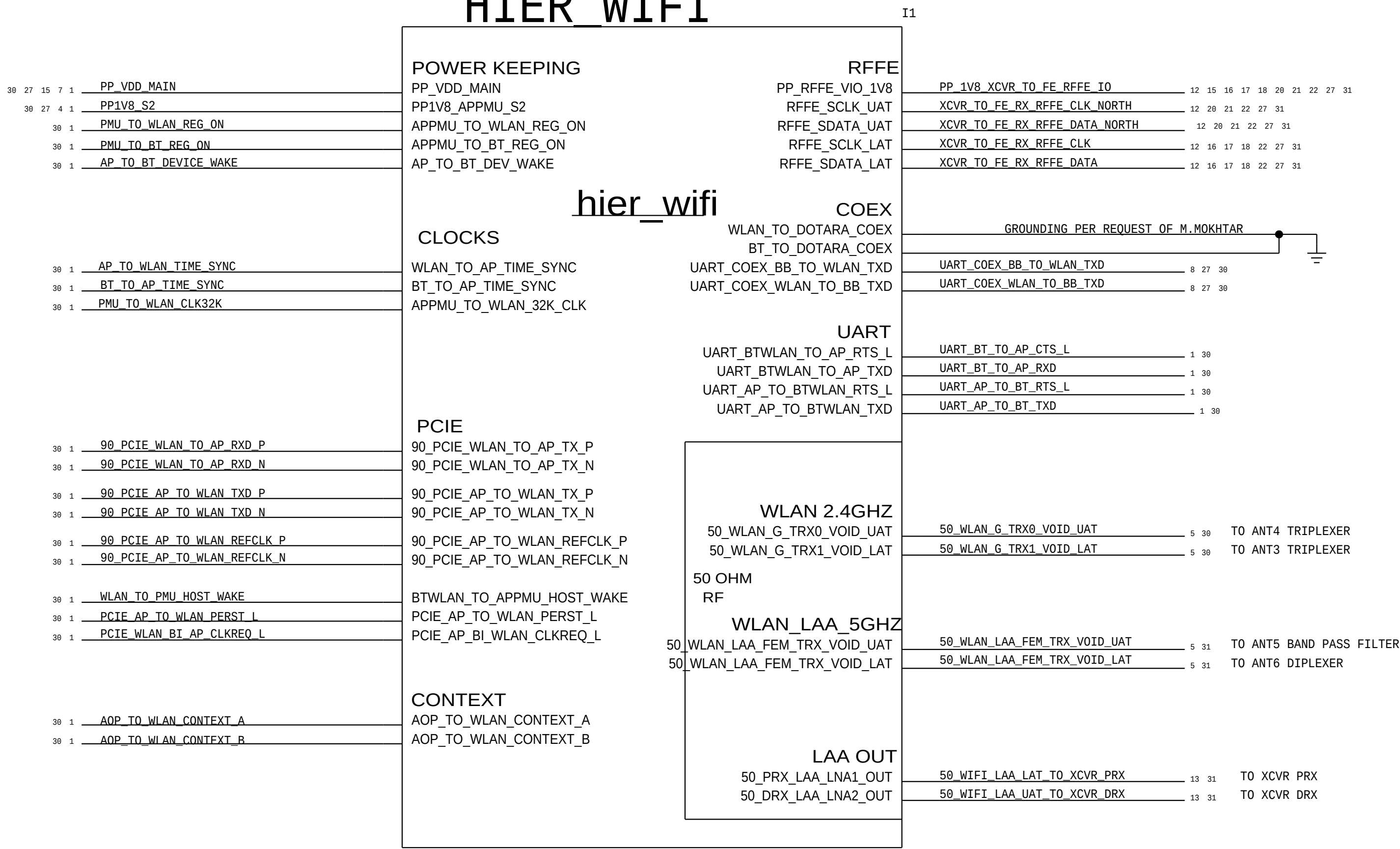


DEBUG: TEST POINTS



SYMBOL: WIFI

HIERARCHICAL SYMBOL FOR HIER_WIFI

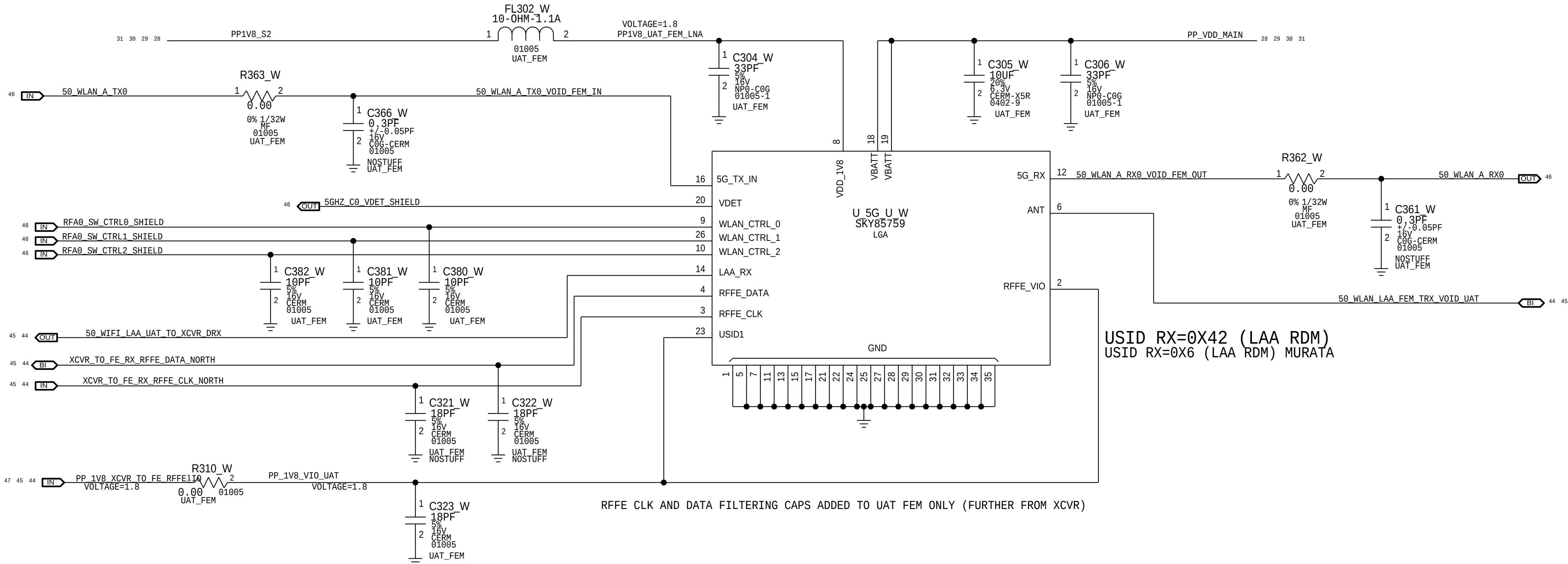


A

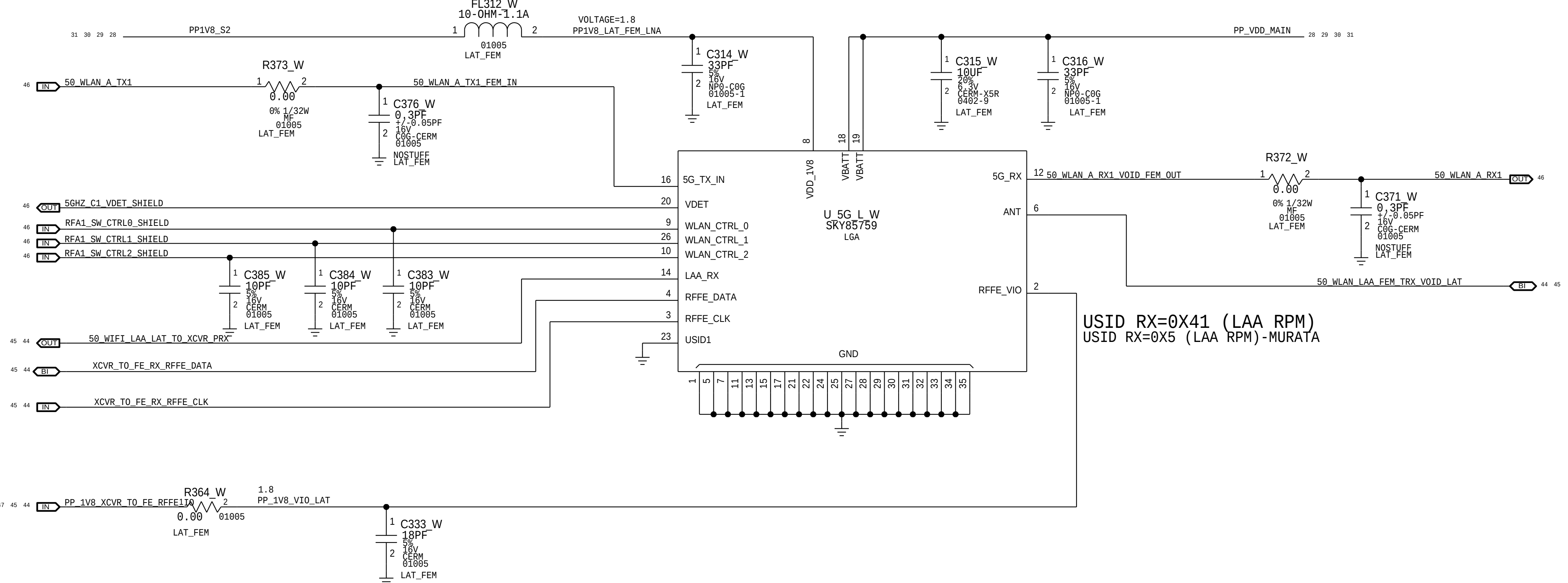


FEM MODULES

5GHZ UAT FEED



5GHZ LAT FEED



D

C



B

